MILITARY SPECIFICATION

COMPUTER, SONAR DATA; CP-1323/AQS-13E

AND MOUNTING BASE; MT-4948/AQS-13E

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all departments or agencies of the Department of Defense.

1. SCOPE

1.1 Scope - Computer, Sonar Data CP-1323/AQS-13E shall be a lightweight, airborne, digital acoustic processor designed for use onboard aircraft for real-time processing of acoustic signals from submerged objects.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Applicable Paragraph
Computer, Sonar Data	CP-1323/AQS-13E	3.5.1
Mounting Base, Sonar Data Computer	MT-4948/AQS-13E	3.5.2

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.7.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5845

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2. REFERENCED DOCUMENTS

2.1 Issue of Documents - The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-250/11	Aluminum Alloy 6061, Plate and Sheet

Military

MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For use with Electronic Equipment in Aircraft)
MIL-S-3786	Switch, Rotary, (Circuit Selector, Low Current Capacity) General Specification For
MIL-W-5088	Wiring; Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic and Associated Equipment
MIL-C-5809	Circuit Breakers, trip-free, Aircraft, General Specification for
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted (Asq)
MIL-P-7788	Panel, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-F-15733	Filter, Radio Interference, General Specification For

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SPECIFICATIONS

Military

MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance, And Life For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment including Ground Support Equipment
MIL-S-22710	Switches, Rotary (Printed Circuit) (Thumbwheel, Inline, and Pushbutton), General Specification For
MIL-S-22710/18	Switch, Rotary (Printed Circuit) (Pushbutton, Bidirectional, Sealed or Unsealed, Nonilluminated) Style SRPC18
MIL-T-23103	Thermal Performance Evaluation, Airborne Electronic Equipment and Systems, General Requirement for
MIL-P-81279	Power Supply, Miniature, General Specification for
MIL-C-85301	Classified Extract of Military Specification, Computer, Sonar Data; CP-3123/AQS-13E and Mounting Base; MT-4948/AQS-13E

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-471	Maintainability Verification/Demonstration/ Evaluation

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Military (Continued)

MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-790	Reliability Assurance Program for Electronic Parts Specifications
MIL-STD-794	Parts and Equipments Procedures for Packaging and Packing of
MIL-STD-2068	Reliability Development Tests
MIL-STD-2074	Failure Classification for Reliability Testing

HANDBOOKS

Military

MIL-HDBK-217	Reliability Prediction of Electronic Equipment

PUBLICATIONS

Naval Air Systems Command

WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections

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2.2 Source of documents - Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

b. Other parts and materials requirements shall conform to MIL-E-5400.

c. Subassemblies of high reliability or relatively low cost shall be constructed as nonrepairable as outlined in MIL-E-5400.

d. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including transistors, diodes, and thin and thick film hybrids) other than microelectronic devices shall be obtained as outlined in MIL-E-5400. (See 6.2.1.)

3.2.2 Microelectronic Modular Assemblies - Conformal coatings, encapsulate, embedments, or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removed without damage to the assembly and shall be in accordance with MIL-E-5400. (See 6.2.1.).

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3.2.3 Modules - The electronic portions of the equipment shall be functionally modularized at all levels of assembly/disassembly. The references in MIL-E-5400 to detailed mechanical and electrical design and to maintainability and to accessibility shall be applicable.

3.3 Design and Construction - The equipment shall conform to the requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 23.14 kilograms (51 pounds).

3.3.2 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously approved for use may be continued unless otherwise indicated in the contract or order. (See 6.2.1.)

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 425 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 5000 hours, with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean-Time-Between-Failure (MTBF) - The equipment, including any built-in test provisions, shall have 425 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The Sonar Data Computer, CP-1323/AQS-13E, shall contain a time totalizing meter in accordance with MIL-M-7793, Type MS 17322-10.

3.3.2.5 Reliability Program Specific Requirements - To prepare the necessary program consistency and program control visibility, the following specific requirements shall apply to contractor reliability programs.

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3.3.2.5.1 Reliability Design Analysis - Each contractor shall perform detailed reliability design analysis in accordance with MIL-STD-785. The analysis shall consider failure modes and effects, component loadings, environments, and failure rates, and mean-time-between-failures. Analysis shall continue throughout the program design, development, and test phases and shall progressively refine reliability predictions with the most current available design and environmental data. In general, early reliability projections shall be based upon estimated system configuration, component part ratings and stresses, and environmental conditions. Later predictions shall replace these estimates with actual design details, calculated loading and calculated local environmental exposures. Predictions in the final phase of the program shall reflect current design (i.e. include the latest design changes) and the component loadings and environmental exposures observed in the development test, adjusted where necessary to reflect maximum specified equipment ambients.

3.3.2.5.1.1 Failure Modes and Effects - The reliability analysis shall include all significant failure modes. Analysis shall document failure modes, effects, and critical analysis. Failure mode and effect analysis shall be documented in sufficient detail to enable correction of each distinct failure mode having significant impact on specified equipment performance.

3.3.2.5.1.2 Environment - Component failure rate predictions shall reflect the equipment's maximum specified environmental exposures, as defined in this specification and the estimated, calculated and test-observed, local component part environments present within the equipment during operation at the specified equipment ambients. Predictions shall acknowledge heat generated within the black box and the resulting local component temperature rises above the maximum specified black box ambient. Prediction shall also consider aircraft installation and operation and the resulting usual increase in failure rates above those observed in laboratory test. Component failure rate predictions, based upon MIL-HDBK-217, shall reflect the handbook's adjustment factors. Predictions based on other data shall be similarly adjusted as appropriate.

3.3.2.5.1.3 Statistics - Analysis of electronic components with wearout characteristics and age-dependent failure rates (e.g., lamps, batteries, etc.), and mechanical components shall consider the statistical implementations of this wearout. Analysis may neglect only that wearout precluded by the scheduled maintenance procedures as defined in the hardware specification.

3.3.2.5.1.4 MTBF Predictions - The reliability design analysis shall provide predictions of equipment and system MTBF. MTBF shall be based on all component parts in a series probability model. MTBF shall consider all redundancies which permit operation in accordance with specified requirements, despite component failure modes, and redundancies. This shall include the

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failure sensing elements and the mechanisms used to switch from a failed path to an alternate path. Analysis shall also consider testing done to insure all redundant paths are operable at the start of a mission. Degraded operation modes associated with redundancies shall be documented. It shall also consider component stress level variations with mode of operation.

3.3.2.5.2 Failure Analysis - The contractor shall prepare engineering analysis of all equipment failures occurring during:

a. Contract tests specified by the equipment specification.

b. Service use during the contractor support phase of the program.

Contractors shall employ failure analysis capabilities and facilities of the type necessary to define the cause of failure, including those described in MIL-STD-790. Positive corrective action shall be implemented to prevent recurrence of all failures. Program planning in response to MIL-STD-785, Management Tasks, shall include the necessary procedures for data collection and failed specimen control and routing.

3.3.2.5.2.1 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following: (See 4.5.)

a. Part type number
b. The circuit reference symbol number
c. The part function
d. Name of the manufacturer
e. Nature of the failure
f. The number of hours which the part operated prior to failure.

3.3.3 Cabling and Connections

3.3.3.1 Cables and Connectors - The equipment specified shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment specified shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when necessary to meet interference control requirements, and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cable shall not be supplied as part of the equipment.

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3.3.4 Control Panels - All rack or console-mounted control panels shall conform to the applicable requirements of MIL-C-6781, except the edge-lighted panels may be 0.350 inch maximum thickness. The configuration of all panels shall be approved by the procuring activity prior to preproduction testing. The lighting of the plastic panel shall conform to MIL-P-7788, Class 2R, Type V. (See 6.2.1.)

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of MIL-E-5400.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461. (See 4.2.2.2.)

3.3.7 Provisions for Maintainability - The Maintainability Program, built-in-test features, construction and packaging, provisions for test points, and other maintainability parameters shall conform to the requirements of MIL-E-5400.

3.3.7.1 Maintainability Program - Each contractor shall perform a maintainability program in accordance with MIL-STD-470 and the specific program requirements of paragraph 3.3.7.1.1 of this document, covering the complete design, development, and production phase.

3.3.7.1.1 Maintainability Program Specific Requirements - The following specific requirements needed to provide necessary program consistency and control visibility, shall apply to contractor maintainability programs.

3.3.7.1.1.1 Maintainability Design Analysis - Contractors shall perform a detailed maintainability design analysis in accordance with MIL-STD-470. The analysis shall also describe the equipment maintenance concept, as well as manhours and elapsed time consumption for each repair procedure at each level of maintenance. Maintenance studies shall detail mean-time-between scheduled and unscheduled maintenance actions, adjustment and calibration requirements and frequencies, built-in-test and failure diagnosis features, potential maintenance induced failure modes and special tool and test equipment requirements, and maintainability parameters specified in 3.3.7.1.1.3.

3.3.7.1.1.2 Maintenance Concept - The equipment shall contain Built-In Test (BIT) features which shall monitor performance and generate a failure indication when equipment performance falls below specified values. A visual indicator shall be provided for the Sonar Data Computer. The indicator shall be electrically resettable and shall reflect the status of the last test executed. The BIT function shall be used to locate the defective Weapon Replaceable Assembly (WRA) with a confidence level of 95 percent. Scheduled maintenance shall not be required.

a. Organizational ("O") Level - The BIT function will be used for fault detection and isolation to the Weapon Replaceable Assembly (WRA). This level of maintenance will remove the defective WRA and replace it with a Ready For Issue (RFI) unit. This system will then be checked after replacement. No Peculiar Ground Support Equipment (PGSE) shall be required.

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b. Intermediate ("I") Level - This level of maintenance will include WRA repair by correction of WRA discrepancies and Shop Replaceable Assembly (SRA) removal, repair and replacement, alignment if required, and repair verification.

c. Depot ("D") Level - This level of maintenance will include repair of SRA's and items that are beyond the capability of the "I" level. Maintenance action beyond the capability of the "I" level will be performed by the contractor until a capability is established at a Naval Air Rework Facility (NARF).

3.3.7.1.1.3 Repair - The Turn-Around-Time (TAT) at the "O" level includes all maintenance necessary to restore the SDC equipment to a ready status. BIT shall be used to fault isolate to the WRA with a 95 percent confidence factor. The TAT shall not exceed 30 minutes. The Mean-Time-To-Repair (MTTR) at the "I" level comprises all tasks necessary to restore the WRA to specified performance and includes necessary troubleshooting, fault verification and isolation, repair, and checkout. The MTTR at the "I" level shall not exceed four hours.

3.3.7.1.1.4 Service - The equipment shall contain BIT features that provide self-verification of operational readiness based upon an end-to-end GO/NO/GO checkout of all functions including redundancies. This test function shall obviate the need for special support equipment during preflight or post flight checks.

3.3.7.2 Compatibility with VAST - This requirement is not applicable.

3.3.8 Nomenclature, Nameplates, and Identification Marking - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307. (See 6.2.1.)

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature	Room ambient ($25^{o} \pm 10^{o}C$)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input power	115 ±1 Vac 400 Hz/3 phase voltage 27.5 ±0.5 Vdc

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 1B equipment, except as modified herein. (See 4.2.2.1, 4.2.2.1.3, 4.2.2.1.5, 4.2.2.1.6, 4.2.2.1.7.)

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3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves II and III of MIL-E-5400. (See 4.2.2.1.2.)

3.3.10.2 Temperature - The equipment shall meet the temperature requirements of Class 1B equipment except that the extreme operating temperature limits shall be -30 degrees C to +55 degrees C. (See 4.2.2.1.1.)

3.3.10.2.1 Overtemperature Shutoff - Overtemperature protection circuits shall be provided which shall interrupt input power under conditions of excessive internal temperature. The power shall automatically reset when the temperature returns to a safe operating level.

3.3.10.3 Humidity - The equipment shall withstand the effects of humidity up to 95 percent during operating and nonoperating conditions. During test, temperature shall be varied from +28 ±10 degrees C to +55 degrees C. Length of exposure shall be 240 hours. (See 4.2.2.1.4.)

3.3.11 Warmup Time - The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed 2 minutes under standard conditions and 3 minutes at extreme service conditions.

3.3.12 Input Electrical Power

3.3.12.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall be; ac (three phase), 115/200 V, Category B, 300 VA.

3.3.12.2 Lighting Power - Input power for lighting shall require not more than 2 amperes at 28 Vdc.

3.3.12.3 Transient Operation - The equipment shall continue to operate as specified, shall not be damaged, and shall not have any memory loss or alteration under the following transient conditions:

a. AC surge voltages as defined in Figure 3 of MIL-STD-704 (transient surge ac voltage step function loci limits for category B curve 2 equipment).

b. Rate of change of frequency due to frequency modulation, and duration of frequency transients in excess of MIL-STD-704 limits.

c. Spike transients of up to 600 volts of either polarity, duration of 5 microseconds, and with an effective source impedance of 50 ohms.

3.3.12.4 Overvoltage Protection - The equipment shall not be damaged by the abnormal high voltages specified in MIL-STD-704, Category B, and shall automatically resume specified operation when the voltage returns within limits.

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3.3.12.5 Undervoltage Protection - The equipment shall not be damaged by the abnormal low voltages specified in MIL-STD-704, Category B; it shall resume normal operation when the voltage returns within limits, with no memory loss or alteration. The processor shall commence an orderly shutdown when the voltage falls outside normal limits.

3.3.12.6 Under Frequency Protection - The equipment shall not be damaged by frequencies as low as 300 Hertz, occurring over indefinite periods of time.

3.3.12.7 Over Frequency Protection - The equipment shall not be damaged by frequencies as high as 500 Hertz, occurring over indefinite periods of time.

3.3.13 Cooling - The equipment shall meet the thermal design requirements of MIL-E-5400. The thermal evaluation shall be in accordance with MIL-T-23103.

3.3.13.1 Thermal Conditions - The equipment shall operate satisfactorily under the thermal conditions defined by the temperature/altitude requirements of MIL-E-5400 without exceeding the critical component maximum temperature limits as defined in MIL-T-23103. (See 4.2.2.1.1.)

3.3.14 Soldering - The soldering of printed circuit wiring boards shall comply with the procedures and requirements of WS-6536.

3.4 Performance - Unless otherwise specified herein, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances, or values setting forth acceptable variations from the performance under the standard conditions will be specified in the Acceptance Test Procedures.

3.4.1 Operation - Operation of the sonar data computer (SDC) shall require functional interfacing with associated avionic systems as shown in Figure 1. The SDC shall perform on-board, real-time processing of acoustic data. Data inputs shall consist of four channels of sonobuoy signals received on four channels from Receiver Set, AN/ARR-75 (sonobuoy receiver), or eight channels of dip sonar signals received on eight channels from Sonar Receiver, R-1695A/AQS-13B (sonar receiver). Multiple sonobuoy processing modes of operation shall be provided to process LOFAR, DIFAR, VLAD, CASS, DICASS, and RO sonobuoy types. An additional mode shall be provided to process time-shared multiplex signals from the eight dip sonar receiver channels during 200-millisecond COS^2 transmission mode. The dip sonar processing shall enhance the detectability of targets in reverberation and noise-limited conditions. In all sonobuoy and dip sonar modes

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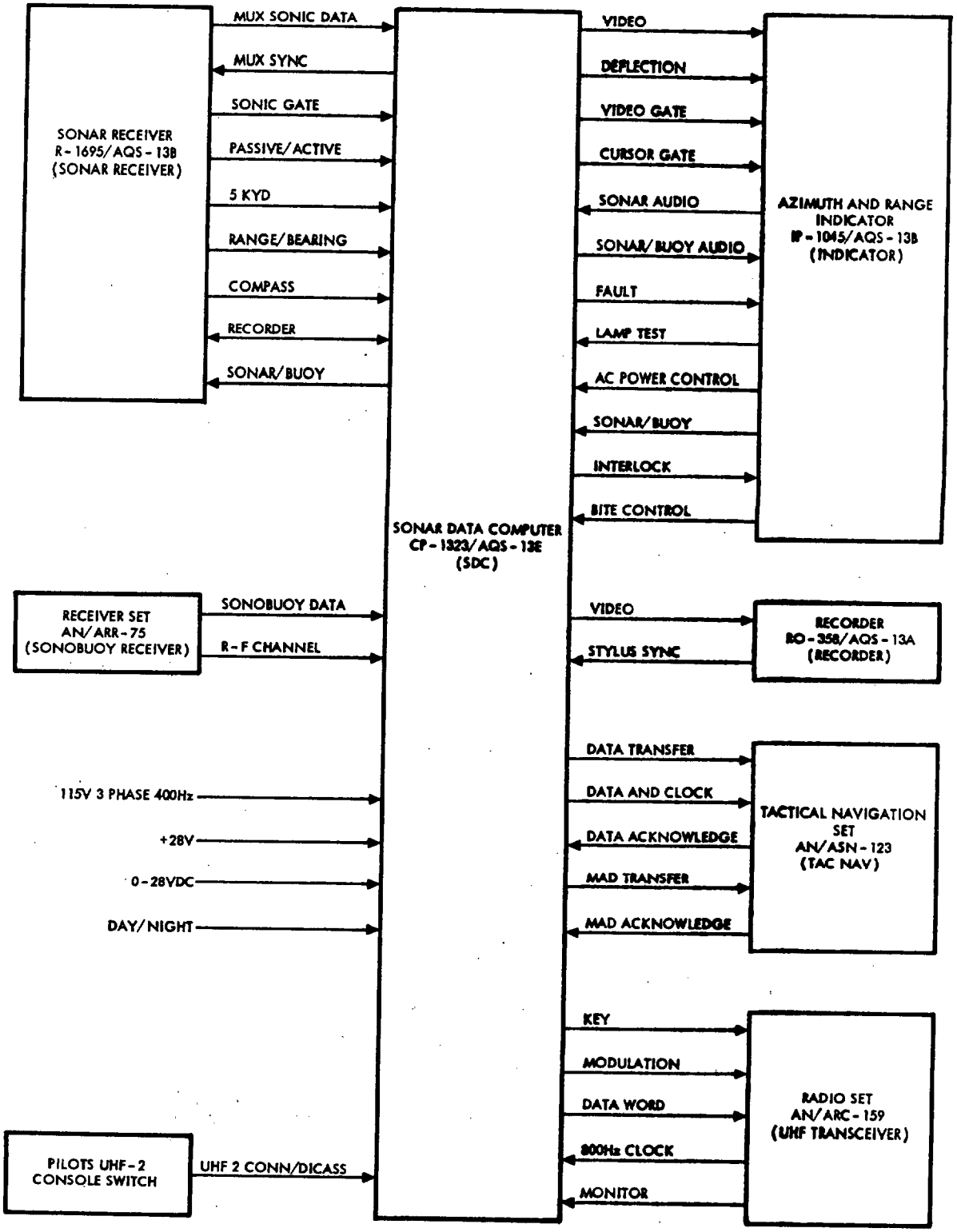


FIGURE 1. Computer, sonar data CP-1323/AQS-13E interface diagram.

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of operation, a digital process shall be performed to transform the source media from the time domain to the frequency domain by use of a Fast Fourier Transform (FFT) algorithm. Special characteristics of the resulting spectrum shall be detected, defined and noted in memory to establish a data bank. Data shall be digitally assembled with axes, cursors, and alphanumeric characters into selectable passive/active sonobuoy display formats (grams) or a refresh PPI (plan position indicator) dip sonar display format. The selected display format shall be converted into appropriate analog signals for presentation on Azimuth and Range Indicator, IP-1045/AQS-13B (indicator). Sonobuoy and dip sonar data shall be made available for transfer to Tactical Navigation Set, AN/ASN-123 (TACNAV). Provisions shall be made to modulate Radio Set, AN/ARC-159 (UHF transceiver) with downlink command functions for command activated sonobuoy types. Additional SDC outputs shall include sonobuoy audio and drive for Recorder, RO-358/AQS-13A (recorder), to provide hardcopy for passive sonobuoy waterfall and active sonobuoy or dip sonar range tracking data. A BIT capability shall be included within the SDC for the purpose of self testing each selectable processing mode of operation.

3.4.1.1 Input Information - The SDC shall control the acceptance and processing of specified input data from the following interface equipment:

3.4.1.1.1 Sonobuoy Receiver

a. Sonobuoy data composed of four channels of analog composite sonobuoy data.

b. 5-bit digital, r-f channel code for each of four receiver channels.

3.4.1.1.2 Sonar Receiver

a. Time-division multiplex of right and left half beam sum and left half beam in each of eight receiver channels.

b. Sonic gate pulse occurring at each COS^2 shaped ping time.

c. Passive/active mode, discrete to denote dip sonar mode.

d. 5 kyd discrete to denote dip sonar range scale of 5 kyd.

e. 28-bit serial, dip sonar range and bearing data with 6400 Hz clock.

f. Compass information composed of variable phase, 806 Hz sine wave and 806 Hz square wave reference.

g. Recorder range scale, video, temperature, and timing information for dip sonar hardcopy.

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3.4.1.1.3 Pilot's UHF-2 Console Switch - Enables downlink command modulation and data word to UHF transceiver in the DICASS position and inhibits the downlink command function when set to the UHF2 COMM position.

3.4.1.1.4 Indicator.

a. Left and right sonar audio to be returned to the indicator audio amplifiers during nonprocessed dip sonar operation.

b. Lamp test discrete to illuminate the SDC panel advisory indicators for test in accordance with TEST pushbutton switch on the indicator panel.

c. Ac power discrete to control SDC ac power on/off function in accordance with POWER ON pushbutton switch on the indicator panel.

d. Sonar (nonprocessed modes) or sonobuoy (including processed dip sonar) discrete to initiate SDC processing in the sonobuoy and SDC processed dip sonar sensor modes.

e. SDC BIT mode control consisting of two-digit BCD (10-1, 10-2, and 10-3) in accordance with TEST A and B step switches on the indicator panel.

3.4.1.1.5 Recorder - Optical trigger pulse and stylus rate clock for synchronizing video data to the recorder during sonobuoy and processed dip sonar sensor modes.

3.4.1.1.6 TACNAV

a. Data acknowledge discrete to verify insertion of dip sonar or sonobuoy data into TACNAV.

b. MAD acknowledge discrete to verify insertion of MAD contact into TACNAV.

3.4.1.1.7 UHF Transceiver

a. 800 Hz clock for data word transfer that automatically sets up the UHF transceiver for sonobuoy transmission.

b. Analog modulation monitor to verify downlink command tone interface during SDC BIT mode 10-3.

3.4.1.2 Output Information - The SDC shall be capable of providing output data and controlling the output data in accordance with the following:

3.4.1.2.1 Sonar Receiver

a. Multiplex synchronization pulse at each COS^2 ping and 19,728 Hz clock to control input of multiplex sonar data.

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b. Sonar (nonprocessed modes) or sonobuoy (including SDC processed dip sonar) digital discrete to enable COS^2 generator during SDC processed dip sonar operation.

3.4.1.2.2 Indicator

a. Sonobuoy and processed dip sonar video to brighten crt beam for processed sensor data.

b. Horizontal and vertical random deflection to position crt beam.

c. Video gate pulse allowing video to brighten crt beam during data point (target) deflection intervals.

d. Cursor gate pulse to allow internal azimuth and range indicator brightening during cursor, axes, and alphanumeric deflection intervals.

e. Sonar/sonobuoy audio providing separate left and right dip sonar audio or common left and right sonobuoy audio.

f. Fault discrete to denote SDC power fault.

g. Interlock ground return to enable azimuth and range indicator SDC data input circuitry.

3.4.1.2.3 Recorder -

a. Analog video for nonprocessed or processed dip sonar modes and for sonobuoy waterfall and range tracking hardcopy.

b. Range scale information for active sonobuoy range tracking hardcopy. Range scale information in passive sonobuoy modes and both processed and nonprocessed dip sonar modes shall feed through the SDC from the sonar receiver input connector to the recorder output connector.

c. Temperature, sound velocity correction, sweep triggering, recorder mode and ac power control information shall feed through the SDC chassis from the sonar receiver input connector to the recorder output connector.

3.4.1.2.4 TACNAV

a. Data transfer discrete to request acceptance of sonobuoy or dip sonar data.

b. The 28-bit serial data words containing dip sonar or sonobuoy information with 4 kHz to 1 MHz clock.

c. MAD transfer discrete to request acceptance of MAD contact.

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3.4.1.2.5 UHF Transceiver

a. Key pulse to enable radio set during downlink command transmission interval.

b. Downlink command modulation containing address, ping, and deep/scuttle tones for command activated sonobuoys.

c. 32-bit serial data word to automatically set up sonobuoy transmission mode.

3.4.1.3 SDC Processing Modes - The SDC shall provide seven selectable operating modes for processing acoustic data from passive/active sonobuoy sensor sets and active dip sonar. All sonobuoys in a sensor set shall be of identical nomenclature and shall be simultaneously processed. Only the dip sonar or one sonobuoy sensor set shall be processed at the same time. The processing of DIFAR or VLAD sonobuoys shall be identical in an SDC mode common to both sonobuoy types. All sonobuoys in a sensor set shall be processed over the same input band, bandwidth, and resolution. The SDC processing modes and related sonobuoy sensor sets shall be as follows:

	SDC MODE	SENSOR	MIL DESIGNATION	QUANTITY
a.	LOFAR (4)	LOFAR	AN/SSQ-41	4
b.	DIFAR/VLAD (2)	DIFAR	AN/SSQ-53	2
		VLAD	AN/SSQ-77	2

NOTE

Frequency resolution of processed output data in DIFAR/VLAD (3) mode is less than that for DIFAR/VLAD (2) mode. Frequency resolutions shall be as specified in the classified extract of this specification.

c.	DIFAR/VLAD (3)	DIFAR	AN/SSQ-53	3
		VLAD	AN/SSQ-77	3
d.	DICASS (2)	DICASS	AN/SSQ-62	2
e.	CASS (4)	CASS	AN/SSQ-50	4
f.	RO (4)	RO	AN/SSQ-47	4
g.	Q13 (PPI)	Dip Sonar	AN/AQS-13B	8 Channels

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3.4.1.3.1 Passive Mode Processor Parameters

3.4.1.3.1.1 Input Band and Passband - Refer to the classified extract of this specification.

3.4.1.3.1.2 Vernier Bandwidths, Transform Size, and Analysis Bandwidths - Provisions shall be made to process passive sonobuoys in four selectable vernier bandwidths of 1000, 200, 40, and 8 Hz. A means shall be provided to slew the selected vernier bandwidth to any position within the overall passband. Refer to the classified extract of this specification for transform size and analysis bandwidth.

3.4.1.3.1.3 Analyzer Transform Rate - Refer to the classified extract of this specification.

3.4.1.3.1.4 ALI Time Constants - Four selectable automatic line integration (ALI) time constants of reset, 0.5, 1.5, and 5 minutes shall be provided to exponentially integrate the amplitude data in each available frequency bin of successive Fast Fourier Transforms (FFT). When selected, the integration process shall cause the random background noise amplitude in each frequency bin to seek a time averaged level, while stable discrete frequencies in any frequency bin average out to a level proportional to the energy input. The ALI time constants shall apply to the directional channel processes and to the true omni channel process for DIFAR/VLAD sonobuoys, causing bearing indications to assume a time averaged indication. When processing DIFAR/VLAD sensors, the integrated frequency and bearing data shall not be lost when selecting the various display modes.

3.4.1.3.1.5 Bearing Correction - Bearing bias correction shall be selectable for application to all directional channels of DIFAR/VLAD. Bearing bias correction shall subtract local spectral noise from directional spectral lines prior to bearing computations.

3.4.1.3.2 Active Mode Processor Parameters

3.4.1.3.2.1 Bandwidths, Pulse Lengths, and Resolutions - Refer to the classified extract of this specification.

3.4.1.3.2.2 Range Scale - The SDC shall process active CASS and DICASS sonobuoy data on three selectable range scales of 5, 8, and 12 kyd, active RO sonobuoy data on 5 kyd, and shall process dip sonar data on five selectable range scales of 3, 5, 8, 12, and 20 kyd.

3.4.1.3.2.3 Analyzer Sweep Rates and Range Resolutions - Refer to the classified extract of this specification.

3.4.1.3.2.4 Range Zero Triggering - Range zero trigger for CASS and DICASS processing shall be provided by the SDC downlink function. For RO processing, range zero triggering in each process channel shall be a function of RO trigger detection. Range zero triggering in the processed dip sonar mode, shall be received from the sonar receiver.

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3.4.1.3.2.5 Analysis Enhancements - Reverberation whitening and automatic target candidate selection shall be applied to all active sonobuoy processing modes when displayed in any active buoy display mode except (2) DOP. Reverberation whitening and automatic target candidate selection shall also apply to all sectors of processed dip sonar data.

3.4.1.4 SDC Display Modes - The SDC shall provide four selectable passive and four selectable active sonobuoy display modes. Additionally, a refresh PPI display mode shall automatically be made available when the Q13 (PPI) dip sonar processing mode is selected. The selected sonobuoy or the dip sonar display mode shall produce video and deflection output components compatible with the display requirements of the indicator. Each sonobuoy and dip sonar display mode shall be comprised of appropriate data, axes, cursors, and alphanumeric information for the associated sensor set being processed. Controls shall be provided on the SDC front panel to actuate the buoy display mode cursors. The cursor appearing on the processed Q13 (PPI) display shall be positioned by means of the CURSOR controls on the sonar receiver. Data from sonobuoys being processed, but not selected for display, shall be retained in memory and be available for immediate recall and display in all passive sonobuoy display modes except 'FALL and 'FALLS. The SDC display modes related to applicable process modes shall be as follows:

DISPLAY MODE		PROCESS MODE
a. Passive		
(1)	(2) ALI	LOFAR (4), DIFAR/VLAD (2), (3)
(2)	'FALL	LOFAR (4), DIFAR/VLAD (2), (3)
(3)	'FALLS	LOFAR (4), DIFAR/VLAD (2), (3)
(4)	BRG/ALI	DIFAR/VLAD (2), (3)
b. Active		
(1)	(4) DOP	CASS (4), RO (4)
(2)	(2) BRG	DICASS (2)
(3)	(2) DOP	CASS (4), DICASS (2), RO (4)
(4)	BRG/DOP	DICASS (2)
(5)	Refresh PPI	Q13 (PPI)

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3.4.1.4.1 (2) ALI Display Mode - The (2) ALI display mode shall provide SDC output data for the simultaneous display of two frequency versus amplitude plots containing the data from any two LOFAR, or DIFAR/VLAD (true omni channel) sonobuoys being processed.

3.4.1.4.2 'FALL Display Mode - The 'FALL (waterfall) display mode shall provide SDC memory and output data for displaying a column of 20 frequency versus amplitude plots (suppressed amplitude) containing data from any one LOFAR or DIFAR/VLAD (true omni channel) sonobuoy being processed. The latest data shall be inserted at the bottom of the display, while the more historical data is moved up one line or deleted from the display to maintain a maximum of 20 lines of data.

3.4.1.4.3 'FALLS Display Mode - The 'FALLS (waterfalls) display mode shall provide SDC memory and output data for simultaneously displaying separate waterfall displays at a lower resolution, relative to 'FALL display mode, for any two LOFAR or DIFAR/VLAD (true omni channel) sonobuoys being processed.

3.4.1.4.4 BRG/ALI Display Mode - The BRG/ALI (bearing/automatic line integration) display mode shall provide SDC output data for simultaneously displaying a frequency versus bearing plot and a frequency versus amplitude plot, each containing data from any one DIFAR/VLAD sonobuoy being processed.

3.4.1.4.5 (4) DOP Display Mode - The (4) DOP (doppler) display mode shall provide output data for simultaneously displaying four doppler versus range plots, containing data from up to four CASS or RO sonobuoys being processed. A reverberation-whitening and target candidate select algorithm shall be employed to enhance the displayed data by minimizing reverberation and background noise clutter, while intensifying target candidates.

3.4.1.4.6 (2) BRG Display Mode - The (2) BRG (bearing) display mode shall provide SDC output data for simultaneously displaying two bearing versus range plots containing data from any two DICASS sonobuoys being processed. A reverberation-whitening and target candidate select algorithm shall be employed to enhance the displayed data by minimizing reverberation and background noise clutter, while intensifying target candidates.

3.4.1.4.7 (2) DOP Display Mode - The (2) DOP (doppler) display mode shall provide SDC output data for simultaneously displaying two doppler versus range plots containing all spectral data from any two CASS, RO, or DICASS (true omni channel) sonobuoys being processed.

3.4.1.4.8 BRG/DOP Display Mode - The BRG/DOP (bearing/doppler) display mode shall provide SDC output data for simultaneously displaying a range versus bearing plot and a range versus doppler plot containing data from any one DICASS sonobuoy being processed. A reverberation whitening and target candidate select algorithm shall be used to enhance the displayed data by minimizing reverberation and background noise clutter, while intensifying target candidates.

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3.4.1.4.9 Q13 (PPI) Display Mode - The Q13 (PPI) display mode shall provide SDC output for data simultaneously displaying all eight dip sonar sectors in a refresh PPI (plan position indicator) format. The display shall be stabilized with magnetic north maintained at top center of the displayed presentation. A reverberation-whitening and target candidate select algorithm shall be used enhance the displayed data in each sector by minimizing reverberation and background noise clutter, while intensifying target candidates.

3.4.1.5 Hardcopy Display - The SDC shall provide sonobuoy and dip sonar output data compatible with the recorder. Passive sonobuoy spectral data displayed in the 'FALL display mode, and active sonobuoy range tracking information displayed in the (4)DOP or BRG/DOP display mode, shall be made available for permanent recording. A means shall be provided to select the range track information to be recorded from any one plot in the (4)DOP display mode. The recorder range scale shall automatically be selected during active sonobuoy processing. Range track information shall also be made available from processed dip sonar data in each sector. The sector containing the cursor shall automatically be selected for permanent recording.

3.4.1.6 Aural - A means shall be provided to select the audio of one passive/active sonobuoy being processed for amplification on both channels of the indicator. Additionally, provisions shall be made to pass left and right dip sonar audio through the SDC during the normal dip sonar processing modes of operation.

3.4.1.7 Data Transfer - The SDC shall provide a means of manually and automatically transferring sonobuoy and dip sonar data to TACNAV. Sensor data, ID code and r-f channel assignment shall be assembled in word types I and II of the sonar input word format required by TACNAV. Data shall be manually transferable in all sonobuoy display modes except (2)ALI, 'FALL, and 'FALLS. In addition, a means shall be provided to initiate the automatic transfer of sensor data from one to four sonobuoys displayed in the (4)DOP sonobuoy display mode. Dip sonar data shall be manually transferable in both the SDC processed and standard sonar modes.

3.4.1.8 MAD Transfer - The SDC shall provide a means to manually notify TACNAV of a MAD contact.

3.4.1.9 Downlink - The SDC shall be capable of generating address, ping, deep, and scuttle downlink command tones compatible with command activated sonobuoy systems (CASS/DICASS). The downlink command tones shall modulate the UHF transceiver. Only sonobuoys being processed and displayed shall be pinged. Address and ping tone timing shall be such as to provide for the sequential pinging of one to four sonobuoys, depending on the number of sonobuoys being displayed. Appropriate address and ping tones shall be automatically selected in accordance with the sonobuoy receiver, r-f channel code for each sonobuoy. Provisions shall be made to select the address of an individual sonobuoy for the initiation of a deep or scuttle downlink function. When initiated, the deep or scuttle function shall automatically be executed at the completion of the ping sequence.

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3.4.1.9.1 Take Command - When the pilot's UHF-2 console switch is set to DICASS, the SDC shall present a 32-bit set up word (handshake) to the UHF transceiver. The downlink tones generated by the SDC shall modulate the UHF transceiver, only when the pilot's UHF-2 switch is set to DICASS.

3.4.1.9.2 Downlink Pulse and Range Parameters - The downlink command function shall provide three selectable ping pulse widths and three selectable range scales. The range scale of the active sonobuoy display mode shall coincide with the selected downlink range scale. See the classified extract for pulse width and range scales.

3.4.1.10 BIT - The SDC shall employ a means of: (1) self-testing each processing mode; (2) verifying the presence of downlink command tones at both the SDC and the UHF transceiver, modulator outputs; and (3) monitoring SDC power supply tolerances. The SDC BIT mode control, consisting of a two-digit input from the indicator, shall be used for BIT mode selection and activation. Test codes 10-1, 10-2, and 10-3 shall be used for SDC BIT functions. Power supplies shall be monitored continuously and if a fault occurs, a discrete signal shall be presented to the indicator for a fault indication.

3.4.1.11 Overall Passive Requirements

3.4.1.11.1 Dynamic Range - The dynamic range of the SDC shall be capable of handling outputs of the sonobuoy receiver, during passive modes of processing, without introduction of artifacts or degradation of detection thresholds (DT's). The SDC shall perform as specified herein when provided with an output from the sonobuoy receiver, having a linear response over a dynamic range of 46 dB for a high output level (80 mVrms to 16 Vrms) for the omni band, including DIFAR directional channel and reference frequencies.

3.4.1.11.2 Automatic Gain Control (AGC) - The passive mode processor shall utilize an 8-bit analog-to-digital converter with AGC. A 1 kHz frequency line, displayed in LOFAR mode in the 200 Hz vernier bandwidth, shall not increase more than 0.25 inch with a 1 kHz cw signal that is stepped in 6 dB increments every 4 seconds from -22 to +24 dBV at the SDC input. AGC shall also meet requirements in 3.4.1.11.1, 3.4.1.11.4, 3.4.1.11.5, and 3.4.1.11.9.

3.4.1.11.3 Presampling Filtering - Presampling analog filters shall be provided for each processing band selected. The filter response shall be such as to meet the requirements of 3.4.1.11.4 and 3.4.1.11.5.

3.4.1.11.4 Aliasing - Signals outside any band, with a signal to noise ratio (SNR) of 18 dB referenced to a bandwidth of 1 Hz in white Gaussian noise, shall have a probability of detection less than 0.5 percent at a probability of false alarm equal to 10^{-4} (P_D < 0.5 percent at $P_{FA} = 10^{-4}$) for a 5 minute ALI.

3.4.1.11.5 Artifacts - With a SNR of 18 dB referenced to a bandwidth of 1 Hz in white Gaussian noise at a level equivalent to sea state zero at the output of the sonobuoy receiver, any false spectral line shall have a P_D < 0.5 percent at $P_{FA} = 10^{-4}$ for a 5 minute ALI.

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3.4.1.11.6 Crosstalk - The channel being tested for crosstalk shall have input white Gaussian noise at the minimum broadband level (equivalent to sea state zero broadband noise at output of the sonobuoy receiver). All other channels shall have a signal applied to the input at a SNR of 18 dB referenced to a bandwidth of 1 Hz in white Gaussian noise at a broadband level (35 dB above sea state zero). For the channel being tested, spectral lines shall be detectable with P_D <0.5 percent at $P_{FA} = 10^{-4}$ for a 5-minute ALI.

3.4.1.11.7 Frequency Accuracy/Stability - The actual center frequency of an analysis filter bin shall be within ±1 frequency bin from theoretical for long term (eight hours) operation and within ±0.1 frequency bin for ALI time constants of 0.5, 1.5, and 5 minutes.

3.4.1.11.8 Passive Bearing Accuracy - Refer to the classified extract of this specification. Utilization of bearing bias correction shall not degrade bearing accuracy.

3.4.1.11.9 Passive Detection Thresholds - Refer to the classified extract of this specification.

3.4.1.12 Overall Active Requirements

3.4.1.12.1 Dynamic Range - The dynamic range of the SDC shall be capable of handling the outputs of the sonobuoy receiver, during active modes of processing, without introduction of artifacts or degradation of DT's. The SDC shall perform as specified herein when provided with an output from the sonobuoy receiver, having a linear response over a dynamic range of 46 dB for a high level (80 mVrms to 16 Vrms), for sine wave signals within the frequency bands associated with CASS, DICASS, and RO sonobuoys.

3.4.1.12.2 Distortion - Pulsed signals at a SNR of +25 dB referenced to a 1000 Hz bandwidth shall result in false detections due to signal distortions having a P_D < 0.05 percent and a $P_{FA} = 1.5 \times 10^{-6}$.

3.4.1.12.3 Crosstalk - On a channel processing white Gaussian noise only at the minimum input level (equivalent to sea state zero broadband noise at output of Receiver, AN/ARR-75), false detections due to crosstalk shall be detectable at P_D <0.05 percent at $P_{FA} = 1.5 \times 10^{-6}$ or less. All other channels shall be processing maximum input level discrete frequency signals within the operating band of the channel being tested for crosstalk.

3.4.1.12.4 Frequency Stability/Accuracy - The system shall exhibit single ping, short term (ping-to-ping), and long term (eight hours) stability of ±1 bin in doppler and range.

3.4.1.12.5 Active Bearing Accuracy - Refer to the classified extract of this specification.

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3.4.1.12.6 Active Detection Thresholds - Refer to the classified extract of this specification.

3.4.1.12.7 Dip Sonar Reverberation Limited Conditions - Refer to the classified extract of this specification.

3.5 Detail Requirements

3.5.1 Sonar Data Computer, CP-1323/AQS-13E - The sonar data computer shall meet the following requirements:

3.5.1.1 Function - The primary function of the SDC is to perform onboard, real-time acoustic data processing of sonobuoy and dip sonar signals received on four channels of the sonobuoy receiver, and eight channels of the sonar receiver, respectively. The SDC shall interface with the indicator, which shall display the processed sonobuoy or dip sonar data and amplify the sonobuoy audio. Provisions shall be made to modulate the UHF transceiver with downlink command tones directed to CASS and DICASS sonobuoys. Passive/active sonobuoy and processed dip sonar data shall be made available to the recorder for permanent hardcopy. Passive/active sonobuoy and dip sonar data shall also be available for transfer to TACNAV.

3.5.1.2 Form Factor - The SDC shall be enclosed in a maximum case size of 64.52 centimeters (25.40 inches) long by 30.56 centimeters (12.08 inches) wide by 15.90 centimeters (6.26 inches) high, enclosing less than one cubic foot. See Figure 2 for SDC outline drawing.

3.5.1.3 Weight - The weight of the SDC, excluding cables and mounting base, shall not exceed 21.32 kilograms (47 pounds).

3.5.1.4 Contents - The sonar data computer shall contain assemblies, subassemblies, and circuits to fulfill the following functions:

a. Primary Power Control and Overtemperature Protection

b. Panel Lamp and Advisory Indicator Control

c. Regulated DC Power Supplies and Fault Monitors

d. Sonobuoy Signal Conditioner

e. Digital Processor

f. Digital Processor Interfaces

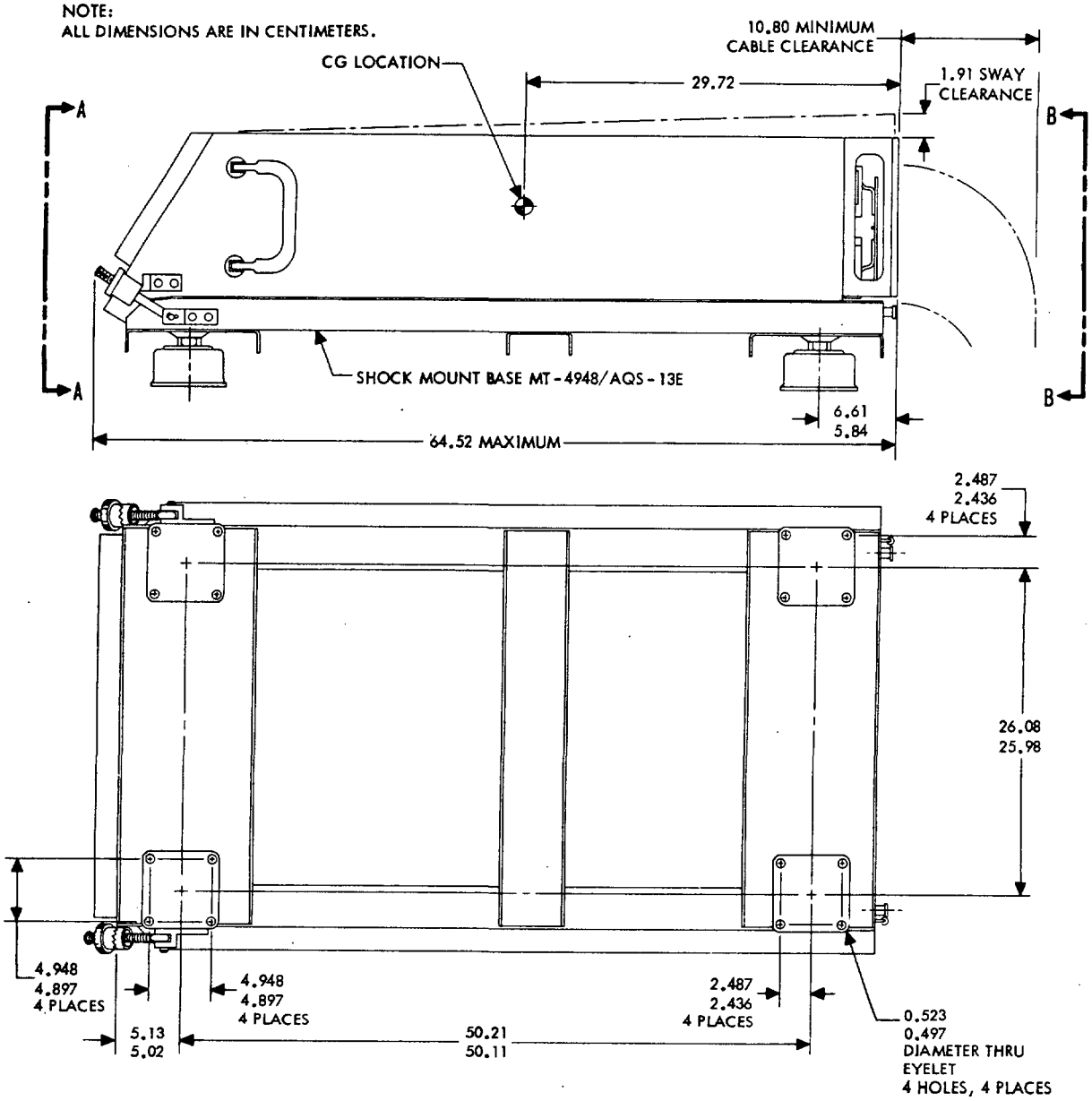
g. Equipment Interface

h. Downlink Command Generator

i. BIT Generator and Downlink Verification

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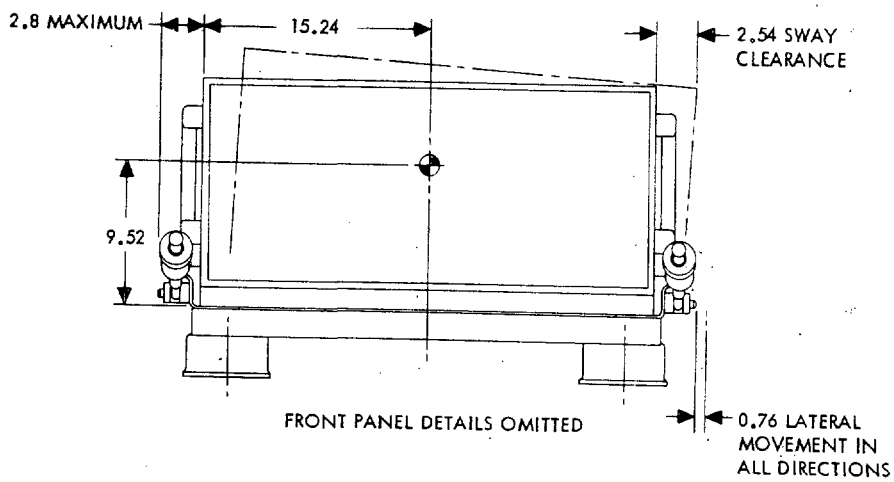
NOTE:
ALL DIMENSIONS ARE IN CENTIMETERS.



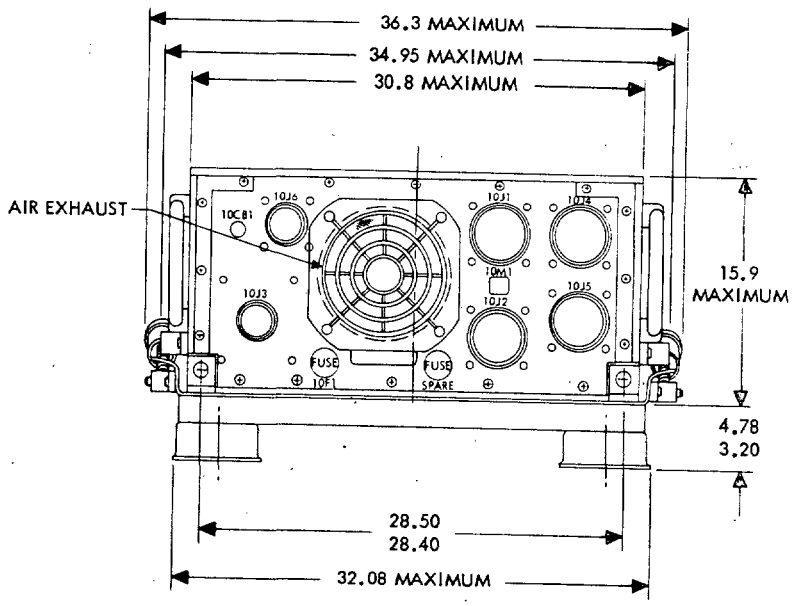
SONAR DATA COMPUTER CONNECTOR REF DESIGNATION	CONNECTOR PART NO.	MATING CONNECTOR	DESTINATION AND/OR FUNCTION	CONNECTOR REF DESIGNATION
10J1	M81511/41FJ01S1	M81511/46J01P1	TEST	———
10J2	MS3470L24-61P	MS3476L24-61S	IP-1045/AQS-13B	3J4
10J3	M83723-11H1608N	MS3476L16-8S	AIRCRAFT POWER	———
10J4	M83723/72R24616	M83723/75R24616	CP-1282/ASN-123	1J2
			ARC 159 (V)/ARR-75	1J2, 1J3/1J1, 1J2
10J5	M83723/72R24617	M83723/75R24617	R-1695/AQS-13B	4J2, 4J5, 4J6
10J6	MS3470L20-41SY	MS3476L20-41PY	RO-358/AQS-13A	2J1

FIGURE 2. SDC outline drawing (sheet 1 of 2).

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VIEW A-A



VIEW B-B

NOTE:
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FIGURE 2. SDC Outline Drawing

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3.5.1.4.1 Primary Power Control and Overtemperature Protection (See Figure 3) - The SDC shall be energized or deenergized by means of the POWER pushbutton switch/indicator located on the indicator front panel. An individual power switch or power indicator shall not be included as part of the SDC.

3.5.1.4.1.1 EMI Filter - A multiple circuit electromagnetic interference filter shall be employed by the SDC as an integral part of the power inlet connector 10J3. The filter shall be used to suppress conducted interference on all ac and dc power input and return lines. All applicable requirements of MIL-F-15733 shall apply.

3.5.1.4.1.2 Circuit Breaker - A three-phase circuit breaker shall be installed on the SDC rear panel for 115 V, three-phase, 400 Hz overload protection. All applicable requirements of MIL-C-5809 for the type MS21984 circuit breaker shall apply.

3.5.1.4.1.3 Power Relay - A dc relay shall be provided to control 115 V, 400 Hz, +28 Vdc aircraft, and 0 to +28 Vdc within the SDC. The relay shall be energized by -28 Vdc furnished by the indicator. Relay coil current shall be 120 milliamperes maximum, and coil transient suppression shall be provided.

3.5.1.4.1.4 Temperature Sensor - A temperature sensing device shall be installed within the SDC enclosure. When the internal temperature reaches 71 ±2.8 degrees Centigrade, the power relay shall automatically be deenergized. When the internal temperature of the SDC cools to 60 ±2.8 degrees Centigrade, the power relay shall automatically energize.

3.5.1.4.1.5 Overtemperature Indicator - A fault locator, flag indicator, type M83287/01-24, shall be mounted on the front panel and shall provide a means of indicating the occurrence of an overtemperature condition. When the power relay is deenergized due to excessive internal SDC temperature, the fault locator flag shall automatically change from black, to black and white. After the internal SDC temperature cools, the fault locator shall be manually reset. Coil transient suppression shall be provided.

3.5.1.4.1.6 Blower - A three-phase miniature fan shall be provided to furnish adequate cooling for the power supplies and SDC circuit cards. The blower shall operate only when the power relay is energized.

3.5.1.4.1.7 Time Meter - A miniature, digital, time-totalizing meter, type MS17322-10, shall be installed on the SDC rear panel and shall operate on the A phase of the 115 V, 400 Hz power when the power relay is energized.

3.5.1.4.2 Panel Lamp and Advisory Indicator Control (See Figure 3). The panel lamp and advisory indicator intensity shall be controlled by the aircraft PANEL LIGHTS control. The front panel illumination shall be varied from off to full bright by a 0 to +28 Vdc input from the PANEL LIGHTS control. When the variable 0 to +28 Vdc input is between 0 and +13 Vdc, all illuminated pushbutton switch/indicators shall remain illuminated at half intensity. The intensity of the illuminated pushbutton switch/indicators shall vary proportionally with the 0 to +28 Vdc when in the range of +13 to +28 Vdc. When the

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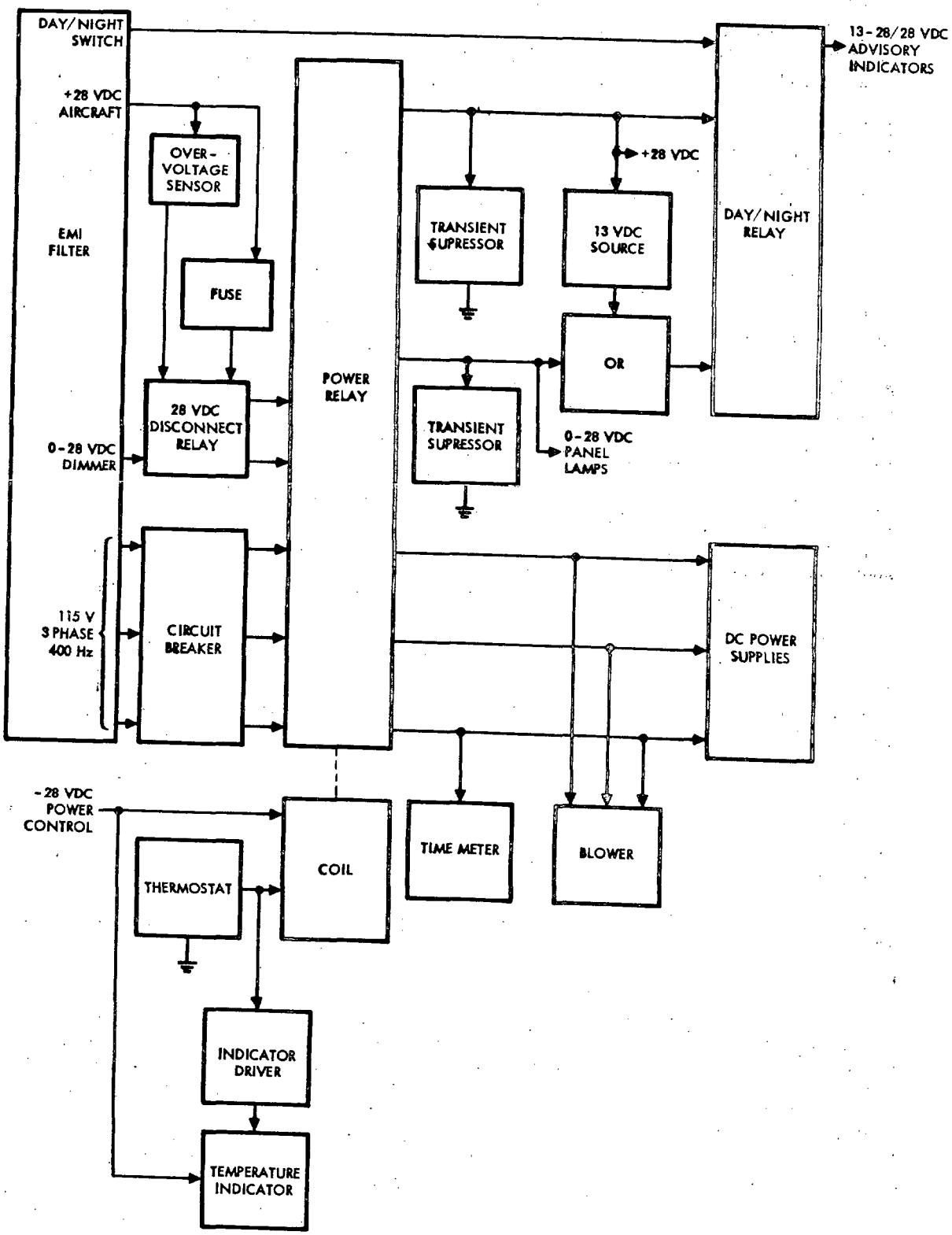


FIGURE 3. Power control block diagram.

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PANEL LIGHTS control is set to OFF, (day position), all panel lamps shall be extinguished and illuminated pushbutton switch/indicators shall be operated from +28 Vdc.

3.5.1.4.2.1 +28 Vdc Protection - A fuse, contained in a fuseholder mounted on the SDC rear panel, shall provide overload protection of the +28 Vdc aircraft input power. A spare fuse shall be housed in an unconnected fuseholder mounted on the rear panel. The protected +28 Vdc aircraft power shall be used as follows:

a. The +28 Vdc pushbutton switch/indicator lamp power when the PANEL LIGHTS control is set to OFF.

b. The source for deriving +13 Vdc to illuminate the pushbutton switch/indicator lamps when the PANEL LIGHTS control is set between 0 and +13 Vdc output.

c. The power supply fault monitor power supply voltage.

d. Range scale selection for the recorder when the SDC is operating in an active sonobuoy processing mode.

3.5.1.4.2.2 Day/Night Relay - A dc relay shall be provided to change the pushbutton switch/indicator lamp voltage from a +28 Vdc day source to a +13 to +28 Vdc variable night source. When the PANEL LIGHTS control is set to OFF, the day/night relay shall be deenergized and connect +28 Vdc to all pushbutton switch/indicator lamps. When the PANEL LIGHTS control is rotated away from the OFF detent, the day/night relay shall be energized by +28 Vdc furnished by the PANEL LIGHTS detent switch. When energized, the day/night relay shall connect a minimum of +13 Vdc to all pushbutton switch/indicator lamps and when the PANEL LIGHTS control variable output is between +13 and +28 Vdc, the pushbutton switch/indicator lamp voltage shall be varied proportionally. Coil transient suppression shall be provided.

3.5.1.4.2.3 Electrical Transient Suppression - A shunt-type, electrical transient suppressor device shall be used at both the +28 Vdc aircraft and variable 0 to +28 Vdc inputs.

3.5.1.4.3 Regulated DC Power Supplies and Fault Monitor - The SDC circuitry shall operate on regulated dc voltages produced by two solid state dc power supply modules contained within the SDC enclosure. The power supply modules shall meet the requirements of MIL-P-81279. Provisions shall be made to monitor all dc voltages for permanent or momentary over/under voltage levels. In the event of a permanent or momentary power supply failure, a power fault output shall be presented to the indicator which shall provide a fault indication.

a. Power supply module No. 1 shall produce:

+12.0 $\pm$0.54 Vdc
+15.0 $\pm$0.68 Vdc
-15.0 $\pm$0.68 Vdc

b. Power supply module No. 2 shall produce:

+5.125 $\pm$0.3 Vdc
-5.125 $\pm$0.3 Vdc

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3.5.1.4.3.1 Primary Input Power - Each power supply module shall require a primary input of 115-V, three-phase, 400-Hz, per MIL-STD-704 for Category B equipment.

3.5.1.4.3.2 Overcurrent Protection - Each power supply module shall provide a means of limiting the output current or shutting down the power supply to avoid power supply damage by excessive output current.

3.5.1.4.3.3 Output Voltage Protection - Overvoltage protection circuits shall be provided for each power supply output so that should any voltage regulator fail, the output voltage will be automatically clamped to a preset level.

3.5.1.4.3.4 Energy Storage - Energy storage shall be provided such that power supply outputs shall not be degraded if the 115 volt primary power drops out for a duration of 5 milliseconds or less. Power losses of less than 50 milliseconds shall not necessitate any manual reset of the power supply.

3.5.1.4.3.5 SDC Fault Output - The SDC shall provide a discrete output denoting power supply normal/fault conditions. The output of the driving device shall be TTL compatible. For a nonfault condition, the output driver shall produce an open collector. For a permanent or momentary power supply over/under voltage condition, the output driver shall latch to the low level output state and produce 0 to 400 millivolts at 4 milliamperes sink current.

3.5.1.4.3.6 SDC Fault Reset - When the power supply fault output device is latched in the low level output state as a result of a voltage surge or drop out, the output device shall be capable of being reset by manually cycling either the POWER or DISPLAY pushbutton switch/indicators located on the indicator front panel.

3.5.1.4.4 Sonobuoy Signal Conditioner (See Figure 4) - The SDC shall provide four input terminals (A, B, C, and D) to receive sonobuoy signals furnished by the sonobuoy receiver. Identical sonobuoy signal conditioners shall be provided at each sonobuoy input. Line receiver characteristics shall be similar to those provided by the device shown in Figure 5. The signal conditioners shall filter frequencies outside the designated input passbands, demodulate DIFAR/DICASS/ VLAD directional channels, heterodyne CASS, detect RO triggers, and provide AGC for all sonobuoy data. An analog-to-digital converter with a multi-shared multiplex input shall be used to sequentially digitize the sonobuoy data from each signal conditioner, for entry into processor memory. During dip sonar processing operation, the analog-to-digital converter shall digitize dip sonar signals furnished in time-shared multiplex from the receiver. The sonar receiver multiplexer shall be synchronized and clocked by SDC control outputs. Signal conditioner outputs shall also be made available as audio during sonobuoy and dip sonar processing operations.

3.5.1.4.4.1 Input Passband Filtering - Low pass filters shall be used to establish input passband limits for all sonobuoy inputs. Refer to the classified extract of this specification for input passbands.

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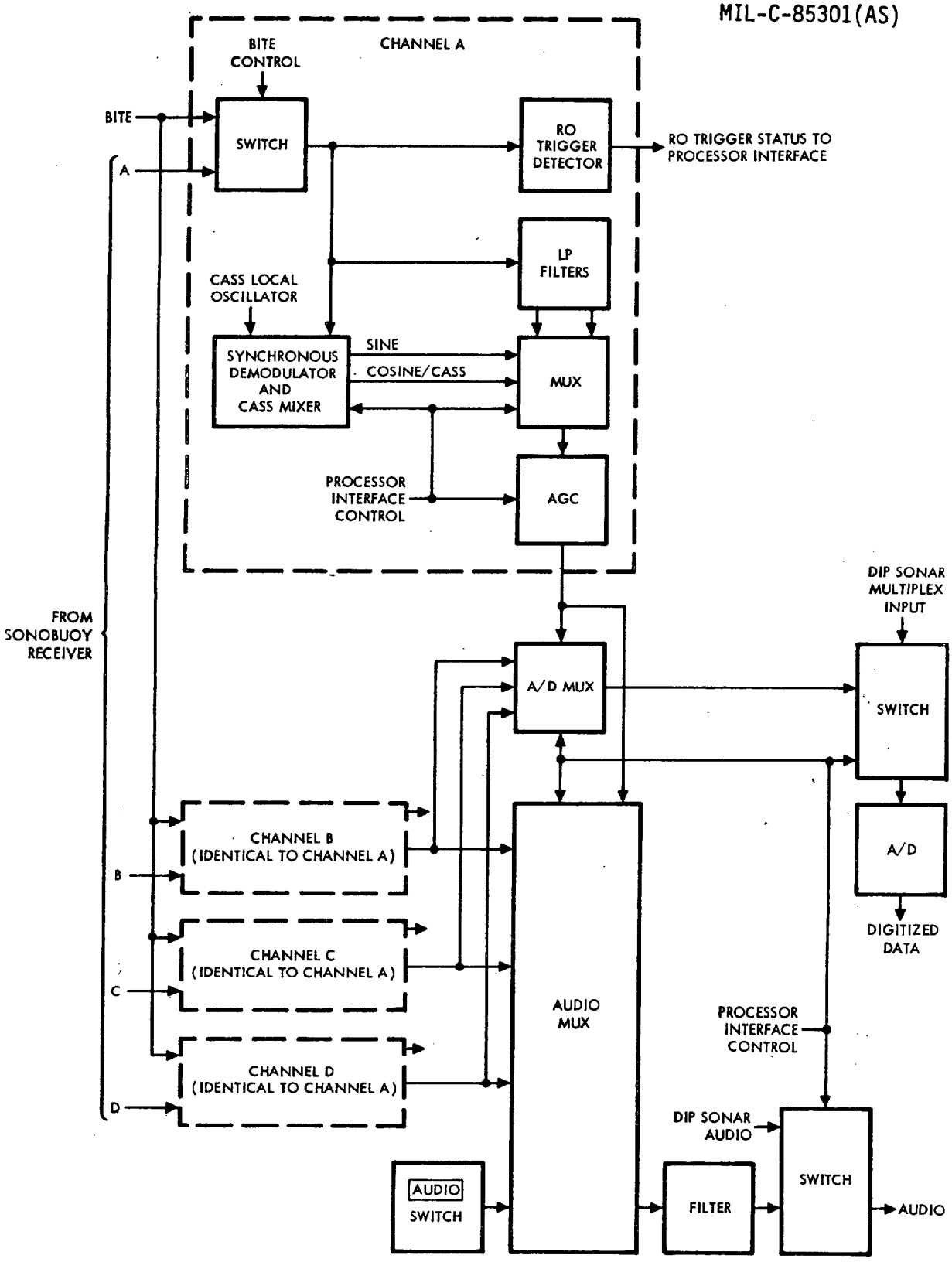


FIGURE 4. Signal conditioner block diagram.

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3.5.1.4.4.2 DIFAR/DICASS/VLAD Demodulator - Each signal conditioner shall provide a synchronous demodulation system to recover the sine and cosine information from the directional channel sidebands applicable to DIFAR, DICASS, and VLAD sonobuoys. Reference signals for the demodulators shall be taken from phase lock loops, captured by DIFAR, DICASS, or VLAD frequency and phase reference tones. The demodulated data shall be appropriately filtered to remove all frequencies other than those within the directional channel baseband. Phase lock status shall be periodically surveyed by the digital processor. The processor interface shall enter the phase lock status into processor memory.

3.5.1.4.4.2.1 Sync Loss Threshold - With a +5 dBV DIFAR/VLAD or DICASS frequency reference tone applied to the SDC input, an indication of synchronization loss (refer to 3.5.1.4.7.5.3) shall not be present for all signal-to-noise ratios above 14 dB related to a 2000 Hz bandwidth. At signal-to-noise ratios below 0 dB, a synchronization loss shall be present for more than 90 percent of the time. The synchronization loss indication may be intermittently present for signal-to-noise ratios between 0 and 14 dB.

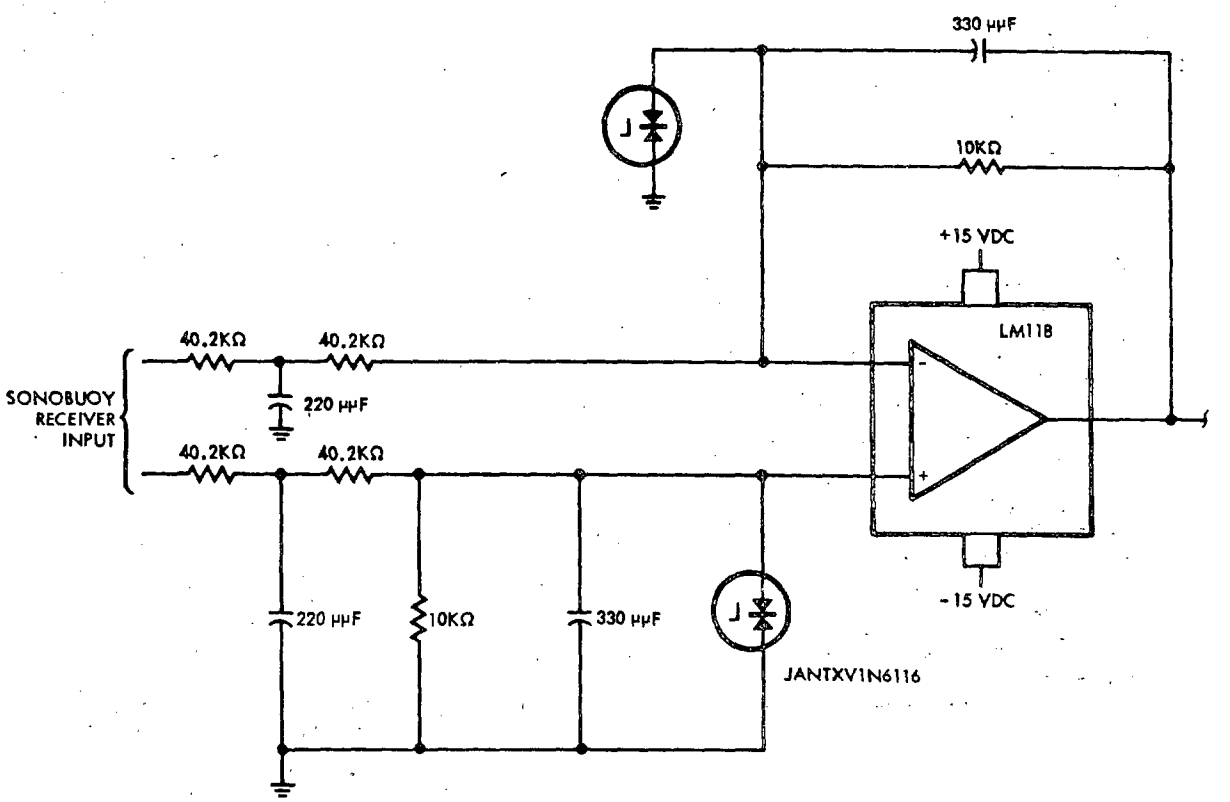


FIGURE 5. Differential line receiver, sonobuoy input.

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3.5.1.4.4.3 CASS Converter - CASS sonic frequencies received in any processing channel shall be heterodyned to a lower doppler passband that shall be common to all CASS sonic frequencies. Each time a CASS sonobuoy is pinged, one of four local oscillators shall be automatically selected for the heterodyne process in the signal conditioner receiving the sonobuoy being pinged. Local oscillator selection shall be governed by downlink command generator interrogation of the r-f channel codes furnished by the sonobuoy receiver (refer to 3.5.1.4.8.3). CASS r-f channel codes are related to CASS sonic frequencies. Filtering shall be used to remove unwanted frequencies from the converter output. Refer to the classified extract of this specification for the CASS heterodyned frequency band.

3.5.1.4.4.4 RO Trigger Detector - To establish range sweep zero during RO processing and display operation, the trigger tone transmitted by RO sonobuoys shall be detected by the signal conditioner. Refer to the classified extract of this specification for the RO trigger passband. RO trigger status shall be periodically surveyed by the digital processor. The processor interface shall enter the RO trigger status into processor memory.

3.5.1.4.4.5 AGC - An automatic gain control shall be provided to optimize sonobuoy signal quantization for the SDC input dynamic range. The AGC characteristics shall not degrade system detection thresholds (DT's) specified in the classified extract of this specification. For DIFAR and DICASS processing applications requiring the display of bearing data, the omni and two demodulated directional channels shall be time-share-multiplexed at the AGC input. Multiplex control shall be provided by the processor interface.

3.5.1.4.4.5.1 AGC Level - With a 0 dBV cw signal at a frequency of 840 Hz applied to an ADC channel input and with the SDC in the LOFAR mode, the AGC amplifier output level shall be 4.2 ±0.8 Vrms (after full recovery).

3.5.1.4.4.5.2 AGC Time Constant - AGC time constants shall be automatically selected in accordance with the sensor being processed. Time constants shall be controlled by the digital processor that enters the applicable time constant control information into interface registers. The time required for the AGC to recover 6.3 dB for a 10 dB step change shall be as follows:

a.	LOFAR/DIFAR/VLAD	2.56 ±1.0 seconds
b.	DICASS/CASS Medium and Long Pulse	1.01 ±0.4 seconds
c.	DICASS/CASS Short Pulse	0.17 ±0.07 second
d.	RO	0.17 ±0.07 second

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3.5.1.4.4.6 Analog to Digital Converter - Sonobuoy data at the AGC output of each signal conditioner shall be quantized by an 8-bit analog-to-digital converter. A 4:1 multiplexer at the analog-to-digital converter input shall sequentially sample all four AGC outputs under control of the processor interface. Synchronization and clocking operations required by the analog-to-digital converter shall also be controlled by the computer interface. During dip sonar processing, the analog-to-digital converter shall quantize sonar data furnished by the sonar receiver. The automatic switch-over between sonobuoy and dip sonar quantizing shall be controlled by mode information entered into interface registers by the digital processor. Sensor sample rates shall be as follows:

	SENSOR	SAMPLE CHANNEL (Hz)	A/D SAMPLE RATE (Hz)
a.	4 LOFAR	5120	20480
b.	2 DIFAR	5120	30720
c.	3 DIFAR	5120	46080
d.	4 CASS	2560	10240
e.	4 RO	3840	15360
f.	2 DICASS	2560	15360
g.	Q13 (PPI)	1233	19728

3.5.1.4.4.7 Sonobuoy Audio - The SDC shall provide left and right sonobuoy/dip sonar audio outputs that are compatible with the indicator audio input requirements. Using a front panel switch, sonobuoy audio shall be selected from any one sonobuoy signal conditioner and presented to both left and right audio outputs. Sonobuoy audio shall be derived at the AGC output and shall comprise all frequencies within the input passband of the SDC. The audio frequencies used for CASS shall be the intermediate frequencies resulting from the CASS conversion. For DIFAR and DICASS operations, the audio shall be sampled and filtered under control of the processor interface. The audio sampling shall be synchronized with the AGC input multiplexing, such that only the omni channel energy is contained in the audio output.

3.5.1.4.4.7.1 Audio Level - In LOFAR or DIFAR omni only modes, an 840 Hz cw signal at a level of 0 dBV applied to the SDC input shall produce an SDC audio output level of -21 ±3 dBV with no more than 10 percent total harmonic distortion into a 2400 ohm load.

3.5.1.4.4.8 Dip Sonar Audio - During SDC processed dip sonar operation, the SDC left and right audio outputs shall be automatically switched to left and right audio outputs from the sonar receiver. The gain of the dip sonar left and right audio channels shall be unity. Mode information entered into interface registers by the digital processor shall automatically switch the left and right sonar audio inputs to the SDC audio outputs. Audio line receiver and driver characteristics shall be similar to those provided by the device shown in Figure 6.

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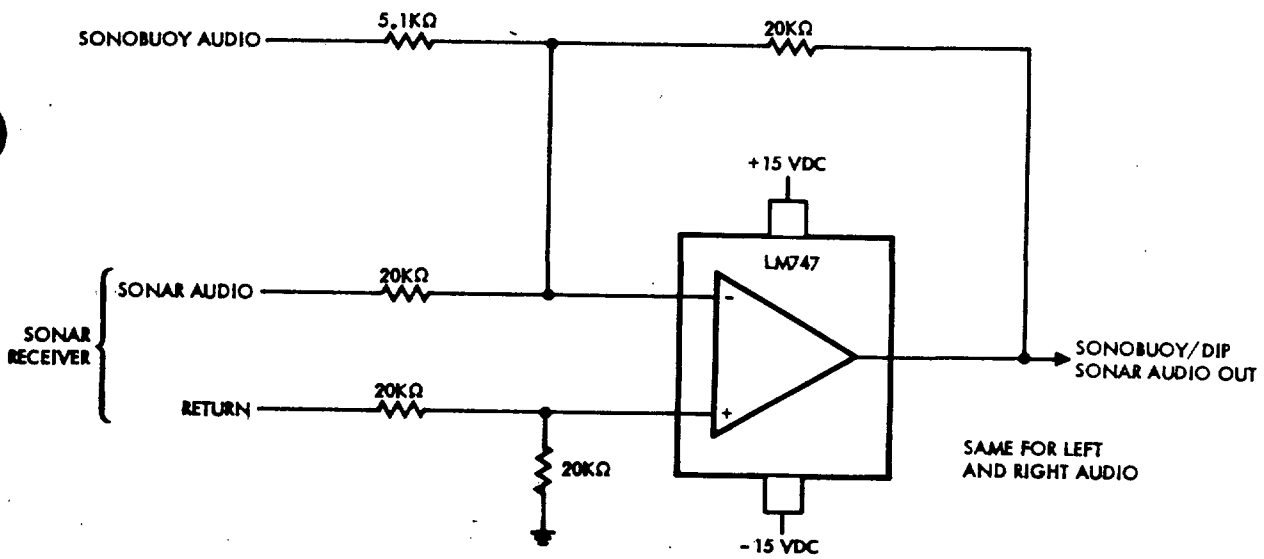


FIGURE 6. Audio line receiver/driver.

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3.5.1.4.5 Digital Processor - The digital processor shall contain two computational elements (CE's) which shall make common use of a semiconductor read-write random access memory (RAM). A memory control function shall be included to provide page addressing of the RAM and to assure timely access to RAM for the two CE's.

3.5.1.4.5.1 Computational Elements (CE's) - The two CE's shall be schematically identical, differing electrically only in their software (stored programs, constants, and microprograms). The CE's shall be high-speed microprogrammable, central processing units, capable of executing any microcommand in 0.156 microsecond. Each CE shall contain an arithmetic logic unit (ALU) and associated operand registers and multiplexers, capable of operation with parallel binary word widths of 12-bits, including sign, in two's complement notation. Each CE shall have a interrupt input port, and discrete input ports for direct input of flags or commands.

Each CE shall have the capacity for a repertoire of microprograms composed of four microcommands, 20-bits in width, accessible by any program instruction. Each CE shall be provided with program storage for 24-bit in width instructions, such storage having sufficient capacity to assure performance of the CE's respective functions as described in paragraphs 3.5.1.4.5.1.1 through 3.5.1.4.5.1.2.3.

3.5.1.4.5.1.1 Array Processing Computational Element Functions - The array processing computational element (ACE) shall perform functions which fall in two categories: Executive functions and sonic data processing functions.

a. The executive functions of the ACE shall include:

(1) The control of an 8-bit I/O data bus, including addressing status word data sources, and periodic read-in directly to RAM of their contents, reflecting front panel switch actuations and status inputs from associated interfaces.

(2) The generation of output control words via the I/O data bus to destination registers for the control of front panel indicators, command function generator, local oscillators, AGC's, and external interfaces.

(3) The computation of changes in analysis band center frequency in response to front panel switch actuation.

(4) The computation of changes in cursor position in response to front panel switch actuation of sonar receiver cursor controls.

(5) The computation of target characteristics such as range, bearing, doppler, and frequency for the target designated by the cursor.

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b. The sonic data processor functions of the ACE shall include:

(1) Providing direct memory access on interrupt, taking time samples of sonic data from as many as 16 time-share multiplexed input channels and storing them as ordered arrays in RAM.

(2) Performing digital heterodyning, filtering, and fast Fourier transform operations on the arrays from as many as eight of the input channels.

(3) Performing background whitening in the time domain and frequency domain to achieve automatic target candidate selection for active sensors.

(4) Performing bearing angle measurement on selected target candidate spectral components, by phase angle measurement or rectangular-polar conversion, as required for directional active sensors.

(5) Performing bearing angle measurement on all spectral components, as required for directional passive sensors.

(6) Performing integration of all passive spectral components with respect to time, using time constants selected from the front panel.

3.5.1.4.5.1.2 Display Interface Computational Element Functions - The display interface computational element (DICE) shall use processed sonic data and status data stored in RAM (by the ACE) to generate display data appropriately formatted for the crt display and recorder interfaces. In addition, the DICE shall provide the capability to automatically track as many as four active target echoes selected by the operator. It shall also provide target data output to the TACNAV interface in a serial format.

3.5.1.4.5.1.2.1 Crt Display Data - The DICE shall output to the crt display interface, in digital form and appropriately timed, all the information necessary to compose images on the crt screen as required for SDC data display. The specific data outputs are: horizontal and vertical beam deflection, video intensity, video unblanking (video gate) and cursor unblanking (cursor gate). The cursor gate shall be used to unblank the crt at fixed intensity during the writing of axes, cursors, and alphanumeric (axis and readout labels, cursor readouts, and error messages). The video gate shall be used for unblanking with variable intensity during the writing of processed sonic data. The DICE shall have, in the form of stored constants, the horizontal and vertical stroke sequences required to form each alphanumeric character or symbol. It shall also have, in the form of stored constants, the character string sequence codes to spell out the required axis and readout labels and error messages. The DICE shall refresh the crt display by repeatedly writing the entire image at intervals sufficiently frequent to maintain image visual continuity.

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3.5.1.4.5.1.2.2 Recorder Display Data - The DICE shall output to the recorder interface, in digital form and appropriately timed, the intensity data necessary for the hardcopy display requirements of 3.5.1.4.7.6. The DICE shall synchronize the output intensity data signal from the recorder, received via the recorder interface, at two of the DICE discrete input ports.

3.5.1.4.5.1.2.3 TACNAV Data Transfer - The DICE shall output to the TACNAV interface, in serial digital form and appropriately timed. It shall provide the target readout and sensor data necessary for the TACNAV data transfer requirements of 3.5.1.4.7.7. Data transfer shall be initiated or continued when a data transfer request signal from the TACNAV Interface is present at a DICE discrete input port.

3.5.1.4.5.1.3 Computational Element (CE) Hardware - Each CE shall contain an ALU working register, a scratch pad memory, address, and page registers for the various memories, and multiplexers to direct the flow of data between these devices. (See Figure 7.) Each CE shall also contain nonvolatile memory for microprograms, constants, and program instructions. Decoders shall be provided to develop control signals for the ALU, multiplexers, and registers, by decoding control data fields in each microcommand and instruction.

a. Each CE shall receive its input from the following three sources:

(1) RAM output data

(2) Direct or discrete input

(3) Interrupt

b. Each CE shall output the following functions:

(1) ALU buffered output

(2) RAM address

(3) RAM page change request

(4) RAM input data

(5) RAM access and read-write request

3.5.1.4.5.1.3.1 Arithmetic Logic Unit - The ALU shall perform arithmetical and logical functions upon two operand inputs, designated the "A" and "B" inputs, and a "carry" input. The function performed by the ALU shall be under microcommand software control.

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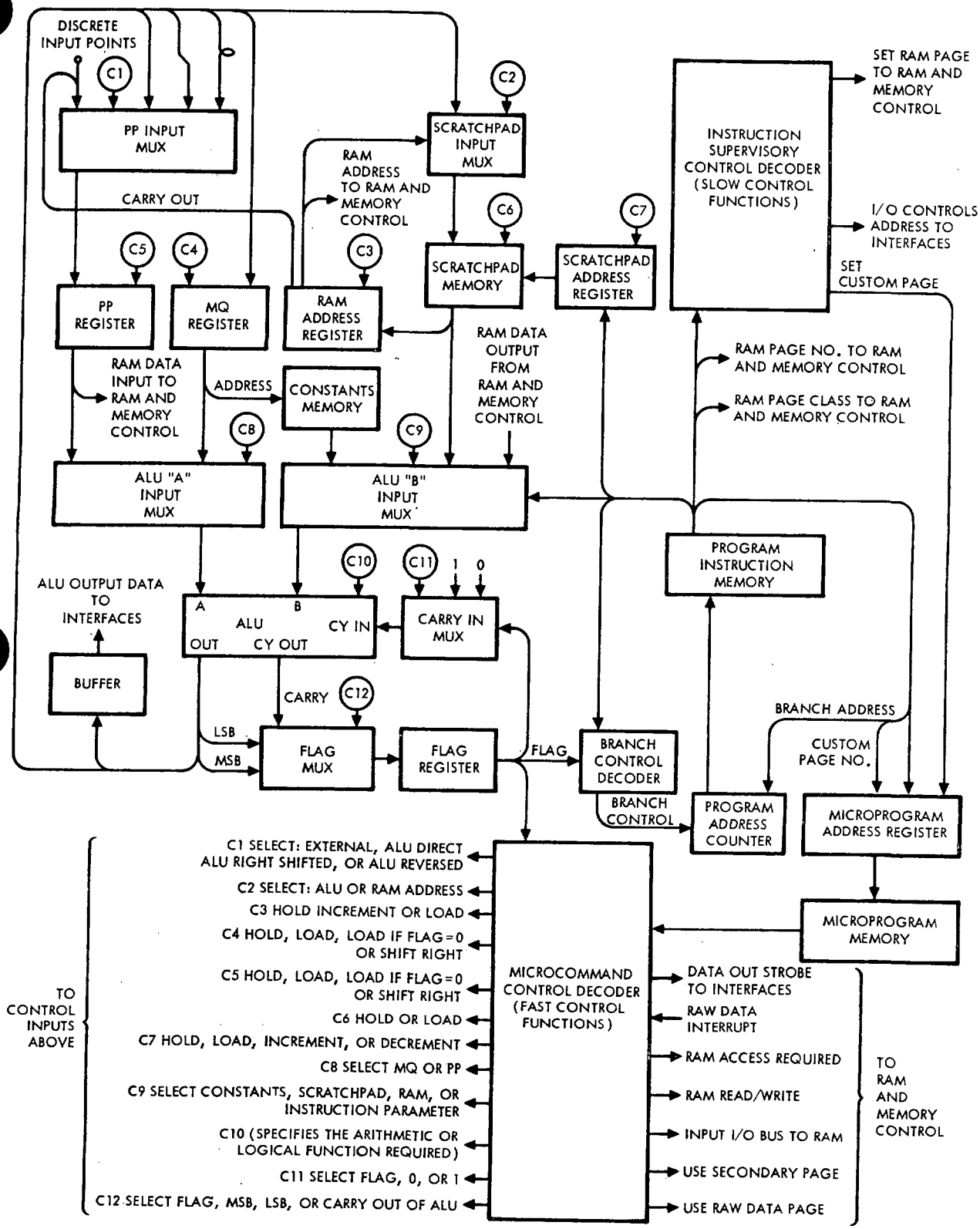


FIGURE 7. Computational element block diagram.

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a. The available arithmetic functions shall be:

(1) A + B + Carry

(2) A - B - 1 + Carry

(3) A + A + Carry

(4) A - 1 + Carry

(5) A + Carry

(6) -A + Carry

b. The available logic functions shall be:

(1) Copy A

(2) Invert A; Complement each bit of A

(3) Copy B

(4) Invert B; Complement each bit of B

(5) A·B; logical "AND" of corresponding bits

(6) A+B; logical "OR" of corresponding bits

(7) A⊕B; logical "EXCLUSIVE OR" of corresponding bits

(8) ZERO; output all bits logic low

The "A" input to the ALU shall be from one of the working registers, under software control. The "B" input shall be from RAM, scratchpad, constant memory, or the instruction parameter field, under microcommand software control. The "Carry" input shall be zero, one, or the FLAG bit, under microcommand software control.

3.5.1.4.5.1.3.2 Working Registers - Each CE shall contain two working registers for the storage of operands for the "A" input to the ALU. The multiplier quotient register (MQ) shall receive its input from the ALU and shall be capable of shifting its contents one bit to the right, under microinstruction software control. The MQ Register shall address the constants memory. The partial products register (PP) shall receive its input from a multiplexer that affords a selection of inputs, under microcommand software control.

3.5.1.4.5.1.3.3 Scratchpad Memory - Each CE shall contain a scratchpad read-write memory with a capacity of 16 words. It shall be loaded from a multiplexer that affords a selection of sources, under microcommand software control.

3.5.1.4.5.1.3.4 Flag Register - Each CE shall contain a register for conditional control storage. This register shall receive its input from a multiplexer that affords a selection of inputs, under microcommand software control.

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3.5.1.4.5.1.3.5 Address Registers - Each CE shall contain the necessary registers for addressing the various storage media used:

a. The microcommand address register is set at the beginning of each program instruction by the opcode field of the instruction. It shall then be incremented in three successive steps to address each of the four microcommands in the microprogram. The most significant bit of the opcode shall designate "base page" or "custom page" of the microprogram to addressed. The base page is always page zero. The custom page number shall be stored in a register under program control and used when designated by the opcode.

b. A program address counter shall be used to address the program instruction memory. It shall indicate the next address of program instruction memory to be executed. It shall be incremented after the execution of each program instruction unless branching conditions are met, in which case it shall be loaded with a new address taken from the parameter field of the instruction.

c. A scratchpad address register shall be provided to address one of 16 scratchpad locations. The scratchpad address register is loaded under microcommand control with an address derived from a field in the instruction. Alternatively, it may be held, incremented, or decremented in any microcommand.

d. A RAM address register shall be provided for each CE to address a location in RAM. The RAM address register shall be loaded with the output of scratchpad memory, incremented, or held, under microcommand control.

3.5.1.4.5.1.3.6 Multiplexers - Each CE shall be provided with multiplexers for the selection of data inputs for the ALU, scratchpad, PP register, and flag register. These multiplexers shall all be under microcommand control, and shall be connected so as to afford the data selection options listed on Figure 8, i.e., C1, C2, C8, C9, and C12.

3.5.1.4.5.1.3.7 Control Decoders - Each CE shall be provided with control decoders to convert the encoded instructions and microcommands into the discrete control functions required for the transfer, transformation, storage, and retrieval of data and addresses. The microcommand control decoder shall employ logic to decode the 20-bit microcommand into the fast control functions listed in Figure 7. The instruction supervisory control decoder shall employ logic to the instruction to derive the slow control functions shown in Figure 7. The branch control decoder shall use logic to the instruction and flag register to derive a branch control signal.

3.5.1.4.5.1.3.8 Program Instruction Memory - The CE's shall be provided with program storage capacity for a quantity of 24-bit width instructions, sufficient to fulfill the functional requirements of the digital processor. The program memory shall be nonvolatile (not subject to loss of memory due to loss of power). It shall have an access time no greater than .500 sec.

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3.5.1.4.5.1.3.9 Microprogram Memory - Each CE shall have a microprogram storage capacity of at least 128 microprograms. Each microprogram shall be composed of four microcommands, 20-bits in width. It shall be organized in pages, each containing a maximum of 32 microprograms. The microinstruction memory shall be nonvolatile. It shall have an access time no greater than .060 μsec.

3.5.1.4.5.1.3.10 Constants Memory - Each CE shall contain a memory with a capacity of 2048 words, 8-bits in width, for constants input to the ALU via the "B" input multiplexer with the MSB extended to a 12-bit width to preserve the two's complement notation. It shall be nonvolatile and have an access time no greater than .500 μsec.

3.5.1.4.5.1.4 Computational Element Software - Each CE shall have a repertoire of microprograms, a program of instructions, and a table of constants to fulfill its functional requirements as specified in 3.5.1.4.5.1.1 through 3.5.1.4.5.1.2.3. The instructions and microcommands shall be formatted for interpretation by the control decoders as specified in 3.5.1.4.5.1.3.7.

3.5.1.4.5.1.4.1 Program Instructions (See Figure 8) - Each instruction shall contain an opcode field to call a microprogram to be executed, which results in microcommands (controlled data transfers, transformations, storage, and retrieval peculiar to that microprogram). In addition, each instruction shall be capable of specifying scratchpad memory address, branching conditionality or supervisory command, and a 12-bit parameter field. The parameter field shall be formatted for interpretation in three conventions:

a. When an instruction specifies a program branch, it shall specify the branch destination address to be loaded into the program address counter if the branching condition is met.

b. When an instruction has supervisory content, it shall contain a supervisory command code, e.g. "set RAM page", and shall convey the appropriate numerical specification, e.g. "RAM page no." and "RAM page class".

c. When an instruction has neither branching nor supervisory control, it may be used to convey a 12-bit operand to the ALU "B" input for use as specified in the called microprogram.

3.5.1.4.5.1.4.2 Microprograms - Microprograms are composed of four microcommands which shall be used to control the transfer, manipulation, storage, and retrieval of data within the CE's, and to and from their interfaces. Each microcommand shall contain the information necessary to develop all the fast control functions listed in Figure 7. The microcommand format shall be subdivided into control code fields in such a way that the fields can be decoded, by discrete logic in the microcommand control decoder, to derive the fast control functions.

3.5.1.4.5.1.4.3 Constants - The constants memory shall provide up to 4096 words 8-bits in width for use by the CE as process control parameters, trigonometric function tables, and other quantities not suitable for efficient storage in the parameter field of the program instructions.

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3.5.1.4.5.2 Random Access Memory and Memory Control - The digital processor shall be provided with a random access read-write memory (RAM), accessible for data storage and retrieval by both ACE and DICE. The RAM shall be organized in a minimum of six pages, each containing 4096 locations. Three of the pages, intended for storage of sonic raw data sample arrays, shall have a minimum word size of 8-bits. The three remaining pages, used for storage of all other forms of data, shall have a word size of 12-bits. The RAM shall have an access time no greater than .450 μsec for reading or for writing of data. The RAM shall be addressed independently by each CE, such that interleaved or coincident access requests by ACE and DICE can be serviced on a priority basis without degrading the performance of either CE. A memory control function shall be provided for this purpose. The memory control shall contain logic circuitry to allow access for every request, even though deferred due to service of a higher priority access. The memory control shall also be capable of storing the RAM page numbers being used by CE, so that only the location address and the page class need be specified for proper access. There shall be three page classes for which a page number is stored in the memory control upon receipt of a "set RAM page" supervisory command from either CE. The classes are:

a. Raw data page (ACE)

b. Primary page (ACE, DICE)

c. Secondary page (ACE, DICE)

3.5.1.4.6 Digital Processor Interfaces - The digital processor shall interface with the other elements of the SDC by means of discrete input ports to each CE, a RAM input bus, and a buffered output bus for each CE. The RAM input bus and buffered output buses operate under supervisory command control.

3.5.1.4.6.1 Discrete Input Ports - Data can be input serially, via an asynchronous input to the PP register's mux (i.e. no storage of this on each CE board). DICE uses its discrete input ports to receive data from the SDC's associated equipment interfaces affecting display output formatting. These inputs are sampled by DICE as necessary during the routine execution of its programs.

3.5.1.4.6.2 RAM Input Bus - Data can be stored in RAM via interrupt. The input buffer (4096 words in length) is filled and reused continuously, with address 4095 followed by address 0. The last address loaded is shared in the ACE scratch pad memory. The input buffer page (4096 words per page) may be set or changed via a supervisory command. This memory storage is accomplished by a direct memory access (DMA) interrupt function and enters raw, also called sonic, data samples into RAM after each A/D conversion is complete. During an interrupt, the ACE disables all the input devices, switches the sonic data onto the RAM input bus, and enters the sonic samples into an ordered array in the specified page of RAM. The sonic data input is performed by interrupt retrievable ACE microprograms of the base page. This RAM input bus is the same RAM input bus which carries input data from up to 16 status word sources into RAM under supervisory command control. The status word sources may be in the

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form of tri-state registers, or buffers, or groups of switch contacts. The status word sources provide the digital processor with information defining front panel switch settings, input signal conditions, command function generator status, and signals from associated equipment. The input devices are enabled one at a time during an input operation, while the associated data is stored in RAM under ACE supervisory command.

3.5.1.4.6.3 Buffered Output Bus

3.5.1.4.6.3.1 ACE - The control word registers, e.g. output devices, are enabled one at a time for loading during an output operation under ACE supervisory command. The control word data is provided by the ACE via the buffered output bus. All control words governing A/D conversion rate, front panel lamps, input signal conditioner functions (including AGC's, multiplexers, and local oscillators), and command function generator are output in this manner.

3.5.1.4.6.3.2 DICE - The DICE buffered output bus is used by the DICE to feed registers which store the display drive data for the crt, the recorder, and the TACNAV external SDC interfaces. Register selection is controlled by supervisory command.

3.5.1.4.6.4 Supervisory Command Control - Supervisory commands are software executable commands to the hardware. These include:

ACE:
a. Increment Status Word (for input).
b. Set Status Word Address (for input). Allowable addresses are 0 thru F (hexidecimal) inclusive.
c. Increment Control Word Address (for output).
d. Set Control Word Address (for output). Allowable addresses are 0 thru F (hexidecimal) inclusive.
e. Reset Watchdog Timer.

DICE:
a. Set data output for crt vertical.
b. Set data output for crt horizontal.
c. Set data output for crt intensity.
d. Set data output for recorder.
e. Set data output for TACNAV data link.

ACE and DICE:
a. Set RAM page register.
b. Enable different custom microprogram page for this microprogram only.
c. Set custom microprogram page register.

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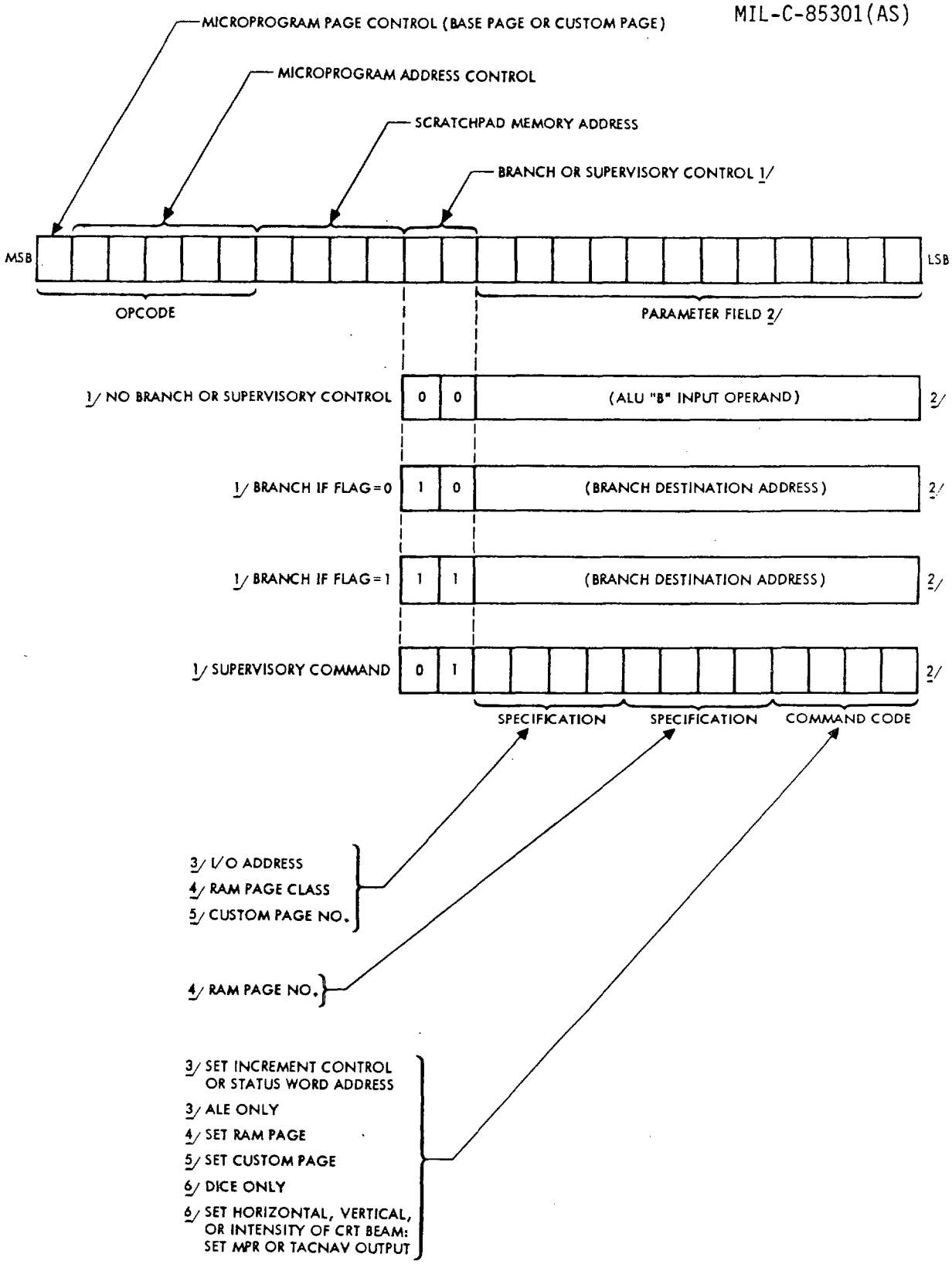


FIGURE 8. Program instruction format.

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3.5.1.4.7 Equipment Interfaces - The processor interfaces shall provide for the SDC input/output of data and control for the indicator crt displays, recorder hardcopy, TACNAV interface, and sonar receiver.

3.5.1.4.7.1 Display Interface - The SDC shall generate the necessary video, deflection, and control signals to provide sonobuoy and processed dip sonar displays on the indicator crt. The video and deflection signals shall be time shared by data and static format components. Time sharing shall be at a rate that is sufficient to produce a refresh display, in which visible fading and flicker is minimized. Output drivers having characteristics similar to those of the devices shown in Figure 9 shall be provided.

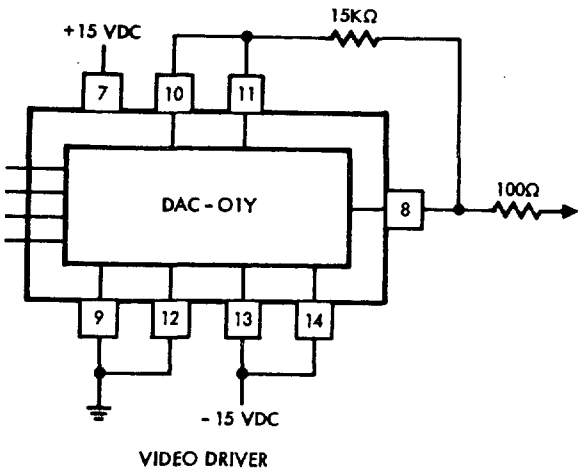
3.5.1.4.7.1.1 Video. - The SDC shall provide an output for an analog video signal that is compatible with the external video input requirements of the indicator. The video signals shall provide crt brightening (intensity modulation in the Z axis) for all passive and active data points (targets) displayed on the crt. The video signals shall be used for crt data brightening during intervals governed by a video gate which shall be generated by the SDC.

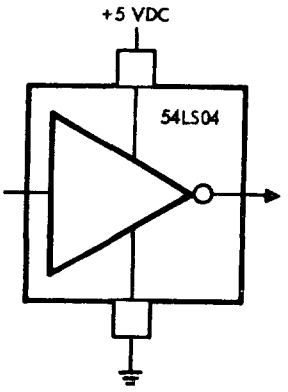
3.5.1.4.7.1.2 Video Gate - The SDC shall provide a digital output for a video gate signal. The video gate shall control the acceptance of video (intensity) information by the indicator. A high level video gate shall cause the indicator to accept the video signal for brightening of passive and active data points. Target data intensity shall be manually controlled by the VIDEO GAIN control located on the indicator front panel.

3.5.1.4.7.1.3 Cursor Gate - The SDC shall provide a digital output for a cursor gate signal. The cursor gate shall be time-shared with the video gate. A high level cursor gate shall cause the indicator to increase the crt beam intensity, while SDC generated deflection signals cause the beam to scribe axes, cursors, and alphanumerics (static format components). The static component intensity shall be manually controlled by the CURSOR INTENSITY control on the indicator front panel.

3.5.1.4.7.1.4 Deflection - The SDC shall provide outputs for analog horizontal and vertical deflection signals that are compatible with the external deflection requirements of the indicator. The deflection signals shall steer the crt beam during both the target data and static format brightening intervals as governed by the video and cursor gates. A position oriented deflection pattern, rather than a raster or spiral scan, shall be used to minimize refresh time limitations.

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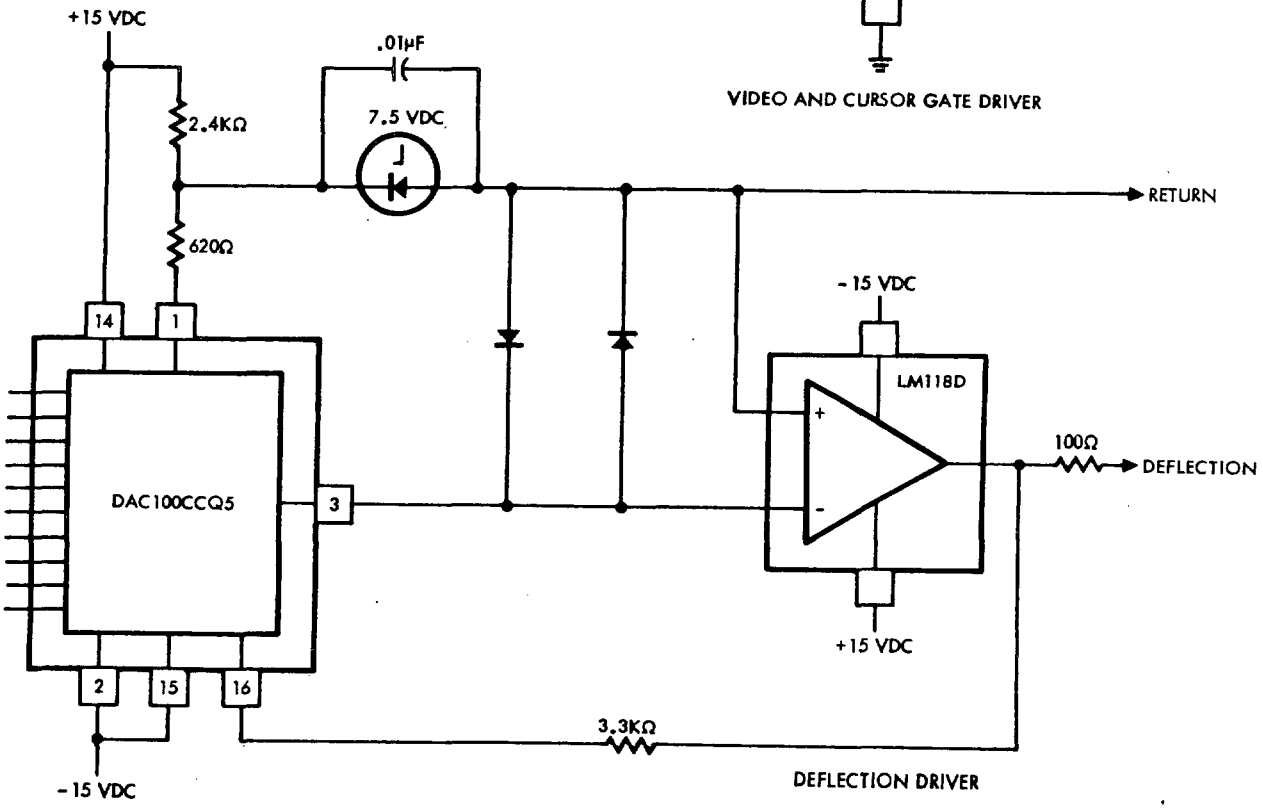


FIGURE 9. Display outputs.

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3.5.1.4.7.2 Passive Sonobuoy Display Characteristics - Passive sonobuoy display mode formats shall contain frequency versus amplitude or frequency versus bearing plots, as applicable to LOFAR and DIFAR/VLAD sonobuoys. The frequency data points shall be plotted along the horizontal axis, and amplitude or bearing data points shall be plotted along the vertical axis. All display formats shall include an analysis band scale at the bottom of the display and, for frequency versus bearing plots, a bearing scale shall be located at the left of the data plot. The amplitude scales shall not be marked. All display formats shall contain appropriate frequency or bearing alphanumeric readouts that are slaved to a manually operated cursor.

3.5.1.4.7.2.1 Analysis Band Scale - The scale shall be linear, having five equally spaced alphanumeric markers, with tick marks displayed horizontally on the crt along a four-inch axis. The alphanumeric markers shall indicate the frequency spectrum being processed, in accordance with the selected analysis band and the position of the analysis band within the input passband that is specified in the classified extract of this specification. The single analysis band scale shall apply to all plots in a display format. When a relocation of the analysis band within the input passband is initiated, the frequency scale alphanumerics shall slowly be incremented or decremented in 5 Hz steps. For large analysis band excursions, the step rate shall exponentially increase as a function of time. The center frequency of the scale shall always be an integral multiple of 5 Hz for each analysis band processed.

3.5.1.4.7.2.2 Bearing Scale - The bearing scale shall be linear and contain five equally spaced alphanumeric markers and tick marks. The bearing markers shall be 0, 90, 180, 270 and 360, with 0 at the bottom.

3.5.1.4.7.2.3 Cursors - When activated, each cursor shall step from data bin to data bin. Cursor step data shall initially be slow and then exponentially increase for prolonged cursor movement. The cursor accuracy shall be ± 1 frequency bin.

3.5.1.4.7.2.4 Alphanumeric Readouts - Appropriate alphanumeric frequency and bearing readouts shall be provided at the top of the display format and shall be slaved to the vertical cursor. The frequency readout shall increment or decrement with each frequency bin change of the cursor. Frequency readout resolution shall be equivalent to the frequency resolution of the selected analysis band. Frequency resolutions are given in the classified extract of this specification. The bearing readout shall automatically indicate the processed bearing of the frequency contained in the cursor selected data bin. The bearing readout shall be rounded off to the nearest degree.

3.5.1.4.7.2.5 Recall - Memory for display recall shall be provided for the (2) ALI and BRG/ALI display modes. When selected for display, sonobuoy data being processed but not displayed, shall be immediately presented to the display.

3.5.1.4.7.2.6 (2) ALI Display Format - The (2) ALI display format shall include two frequency spectrum plots, served by a single vertical cursor. The format shall provide for the display of the LOFAR, DIFAR, or VLAD sonobuoys. If only one sonobuoy is selected for display, the resulting spectrum shall appear in the upper plot. If two sonobuoys are selected for display,

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the spectrums shall be displayed in a top to bottom order, in accordance with the alphabetical order of the processed channels selected for display. A digital frequency indication shall be centered at the top of the display format and shall be slaved to the vertical cursor. The digital frequency readout shall indicate the frequency of a given data bin selected by the cursor and shall apply to both plots. The amplitude data contained in the upper and lower frequency bins selected by the cursor shall be automatically brightened. The brightness of all other frequency lines shall be uniform.

3.5.1.4.7.2.7 ALI Amplitude - Amplitude shall be displayed vertically on the crt with a dynamic range of 66 dB (least significant bit deleted). An 8-bit logarithmic conversion shall be used. The amplitude scale shall not be marked.

3.5.1.4.7.2.8 'FALL Display Format - The 'FALL display format shall provide memory and output data for displaying 20 horizontally oriented frequency analyses of one LOFAR or DIFAR/VLAD sonobuoy being processed. The 'FALL display shall be a time history of sensor data analyses, with earliest events at the top of the 20-line display. The vertical axis shall denote time and shall not be annotated. The latest data shall be inserted at the bottom of the display, while the more historical data is moved up one line or deleted from the display to maintain a maximum of 20-lines. Amplitude data in each frequency bin shall be proportionally brightened in accordance with the energy level of the displayed frequency component. The 'FALL display format shall be served by a single vertical cursor. A digital frequency indication shall be centered at the top of the display format and shall be slaved to the vertical cursor. The digital frequency readout shall indicate the frequency of a given data bin selected by the cursor.

3.5.1.4.7.2.9 'FALL Amplitude - The dynamic range of the 'FALL display shall be 4-bit log amplitude presentation.

3.5.1.4.7.2.10 'FALL Storage Capability - The storage capability of the 'FALL mode shall be limited to the sensor being displayed. Recall shall not be made available.

3.5.1.4.7.2.11 'FALL Reset - A means of manually resetting the 'FALL display (dumping the 20-line memory) shall be provided.

3.5.1.4.7.2.12 'FALLS Display Format - The 'FALLS display mode shall provide SDC memory and output data for simultaneously displaying two, 20 horizontally oriented frequency analyses of two LOFAR, DIFAR, or VLAD sonobuoys being processed. Both shall have similar characteristics to the 'FALL display, except that frequency resolution shall be reduced for selected analysis bands of 200, 40, and 8 Hz. Tandem vertical cursors shall be provided to serve the 'FALLS display format. A single digital frequency indication shall be centered at the top of the display format and shall be slaved to the tandem movements of the cursors. The digital frequency readout shall indicate the frequency of a given data bin that applies to both displays. An analog frequency indication shall be provided by a small arrow along the analysis band scale. The arrow shall travel with tandem cursor movement and shall rest at a position on the scale that corresponds to the digital frequency readout.

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3.5.1.4.7.2.13 BRG/ALI Display Format - The BRG/ALI display format shall include a frequency versus amplitude plot below a frequency versus bearing plot. The format shall provide a display for one DIFAR or VLAD sonobuoy being processed. The frequency spectrum displayed in the lower plot shall be similar to those displayed in the (2) ALI display format. Bearing information shall be displayed above and in alignment with the ALI data. A single dot in each frequency bin shall be positioned at the appropriate bearing. A manually operated horizontal cursor shall be provided within the upper plot for use as an aid in correlating common source bearing data. No readout information shall be derived from the horizontal cursor. A single vertical cursor shall serve both plots. A digital frequency and bearing indication shall be located at the top of the display format and both shall be slaved to the vertical cursor. The digital frequency readout shall indicate the frequency of a given data bin selected by the cursor and the bearing readout shall automatically indicate the bearing of the discrete frequency within the selected bin. The brightness of all frequency lines and bearing data shall be uniform, except that the amplitude line and bearing dot in the cursor selected data bin shall be automatically brightened. The bearing dot for any frequency whole computed bearing within ±5 degrees of the bearing of the cursored frequency shall also be brightened. The brightness of each bearing dot shall be proportional to signal amplitude.

3.5.1.4.7.3 Active Sonobuoy Display Characteristics - Active sonobuoy display formats shall contain range versus doppler or range versus bearing plots as applicable to CASS, DICASS, and RO sonobuoys. The range data points shall be plotted along the horizontal axis and doppler or bearing data points shall be plotted along the vertical axis. All display formats shall include a range scale at the bottom of the display, and either a doppler or bearing scale shall be located at the left of the data plot. All display formats shall contain appropriate range, doppler, or bearing alphanumeric readouts that are slaved to a manually operated cursor.

3.5.4.1.7.3.1 Range Scale - The scale shall be linear, having six equally spaced alphanumeric markers with tick marks for a selected 5-kiloyard range scale, and five equally spaced alphanumeric markers with tick marks for a selected 8 or 12-kiloyard range scale. The range scale shall be displayed horizontally on the crt, along a four-inch axis. The single range scale shall apply to all plots in a display format.

3.5.1.4.7.3.2 Doppler Scale - Knots (KTS) shall be displayed vertically, with 0 knots centered on the range axis. The doppler scale shall be linear, with two equally spaced tick marks at the ±10 and ±20 knot positions. Annotation shall be provided at the ±20 knots tick marks in the (4) DOP display format and at the ±10 and ±20 knots tick marks in the (2) DOP display format.

3.5.1.4.7.3.3 Bearing Scale - The bearing scale shall be linear and contain five equally spaced alphanumeric markers and tick marks. The bearing markers shall be 0, 90, 180, 270, and 360, with 0 at the bottom.

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3.5.1.4.7.3.4 Manual Cursors - Each active display format shall contain a vertical cursor common to all plots. When activated, the vertical cursor shall step from range bin to range bin. The (2) DOP display format shall contain a horizontal cursor to be used in either plot. When activated, the horizontal cursor shall step from doppler bin to doppler bin. Initially the step rate of all cursors shall be slow, and then exponentially increase for prolonged cursor movement. Additionally, a manually operated pointer shall be provided in the left margin of the (4) DOP and (2) BRG display formats. The pointer shall provide a means of keying the alphanumeric readouts at the top of the crt to a selected plot of interest.

3.5.1.4.7.3.5 Auto Tracking Cursors - The (4) DOP display mode shall provide for the manual insertion, or exclusion, of a small marker in any combination of plots, for the purpose of automatically transferring sonobuoy data to TACNAV. The auto tracking cursor shall appear over the target of interest, in the plot selected by the left margin pointer. The auto tracking cursor shall automatically slew along the range axis in a direction and at a rate proporational to the sampled doppler of the target in the auto track cursored data bin. On a ping-to-ping basis, the auto track slew rate shall be updated and doppler and range information from each plot containing an auto track cursor shall be sequentially transferred to TACNAV. Only one auto track cursor shall be allowed in each plot.

3.5.1.4.7.3.6 Alphanumeric Readouts - Alphanumeric range, doppler, and bearing readouts, as applicable, shall be provided at the top of the display format and shall be slaved to the manual cursor. The range readout shall increment or decrement with each range bin change of the vertical cursor. Range readout resolution shall be equivalent to the range resolution of the selected range scale, rounded off to the nearest 10 yards. (See the classified extract of this specification.) The range readout accuracy shall be ± 2 percent of full scale range. In the (4) DOP and BRG/DOP display formats, the doppler readout shall automatically indicate the processed range rate to the nearest knot of the target contained in the cursor selected range bin. In the (2) DOP display format, the doppler readout shall increment or decrement to the nearest knot, with each doppler bin change of the horizontal cursor. Doppler readout accuracy shall be ± 2 knots. In the (2) BRG and BRG/DOP display formats, the bearing readout shall automatically indicate the processed bearing to the nearest degree of the target contained in the vertical cursor selected range bin.

3.5.1.4.7.3.7 Active Display Enhancement - For each display mode plot, except the (2) DOP display format, processed data shall be subjected to a reverberation-whitening and target-candidate selection process at each range decision interval. The process shall involve the selection and display of only the most unusual frequency component (target candidate) in the active sonobuoy passband at each range bin interval. A brightened dot representing the target candidate, shall be displayed at the appropriate range and doppler bin coordinates. Dot intensity shall be a function of target candidate strength.

3.5.1.4.7.3.8 Range Zero Triggering - A new range sweep shall be initiated in an active display plot each time the associated CASS or DICASS sonobuoy is pinged by the SDC downlink command generator function or when an RO sonobuoy trigger pulse is detected by the SDC.

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3.5.1.4.7.3.9 (4) DOP Display Format - The (4) DOP display format shall include four range versus doppler plots, served by a single vertical cursor, cursor pointer and, if utilized, one auto track cursor for each plot. In the order of top-to-bottom, each plot shall be dedicated to SDC processing channels A through D, respectively. A digital knots and range indication shall be centered at the top of the display format. The pointer in the left margin shall be manually positioned opposite the plot for which the alphanumeric readouts are to be dedicated. Alphanumeric readouts shall be slaved to the vertical cursor at all times and shall not indicate auto track cursor data. The range readout shall indicate the range of the data bin selected by the vertical cursor. Range rate for the target candidate, in the range bin selected by the cursor and in the plot selected by the pointer, shall be automatically displayed by the knots readout.

3.5.1.4.7.3.10 (2) BRG Display Format - The (2) BRG display format shall include two range versus bearing plots, served by a single vertical cursor and a cursor pointer. A brightened dot shall indicate the bearing of the target candidate in each range bin. A digital bearing, doppler and range readout shall be centered at the top of the display format. The pointer in the left margin shall be manually positioned opposite the plot for which the alphanumeric doppler and bearing readouts are to be dedicated. The range readout shall indicate the range of the data bin selected by the vertical cursor. Range rate and bearing for the target candidate in the range bin selected by the cursor and in the plot selected by the pointer, shall be automatically displayed by the doppler and bearing readouts.

3.5.1.4.7.3.11 (2) DOP Display Format - The (2) DOP display format shall include two range versus doppler plots, served by a single vertical and horizontal cursor. All processed data, including reverberation at each range decision interval, shall be displayed. Displaying all data in the spectrum results in a brightened spot appearing in all doppler bins within a range bin. Dot brightness shall be a function of the energy level of the detected frequency component in each doppler bin. A digital range and doppler readout shall be centered at the top center of the display format. The range readout shall indicate the range of the data bin selected by the vertical cursor. The doppler readout shall be dedicated to the plot containing the horizontal cursor, and the doppler indication shall be dependent on the doppler bin selected by the horizontal cursor.

3.5.1.4.7.3.12 BRG/DOP Display Format - The BRG/DOP display format shall include one range versus doppler and one range versus bearing plot, both served by the same vertical cursor. The bearing plot shall be similar in size and display to the bearing plots in the (2) BRG display format. The range versus doppler plot shall be similar in size to the plots in the (2) DOP display format, but shall display data in a manner similar to that displayed in the (4) DOP display format. A digital range, doppler, and bearing readout shall be centered at the top of the display format. The range readout shall indicate the range of the data bin selected by the vertical cursor. Range rate and bearing for the target candidate in the range bin selected, and the cursor, shall be automatically displayed by the knots and degree readouts.

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3.5.1.4.7.4 Q13 (PPI) Display Characteristics - The SDC-processed, dip sonar Q13 (PPI) display shall provide for the simultaneous display of dip sonar targets at bearing and range coordinates in all eight sectors. The top of the display shall be magnetic north, and display stabilization shall be provided by sonar compass information.

3.5.1.4.7.4.1 Cursor - The Q13 (PPI) cursor shall be square and shall be manually positioned by the CURSOR controls on the receiver front panel. When activated, the cursor shall step from range bin to range bin or from bearing bin to bearing bin. Initially the range step rate of the cursor shall be slow, and then exponentially increase for prolonged cursor movement.

3.5.1.4.7.4.2 Alphanumerics - An alphanumeric, doppler readout shall be provided at the top of the display format and shall be slaved to the cursor. The doppler readout shall automatically indicate the processed doppler of the cursor selected target and shall be rounded off to the nearest knot.

3.5.1.4.7.4.3 Display Enhancement - Processed data in each sector shall be subjected to a reverberation-whitening and target-candidate selection process at each range decision interval. The process shall involve the selection of the most unusual frequency component (target candidate) in the sonar passband. A brightened dot representing the target candidate, shall be displayed at the appropriate range and bearing bin coordinates. Dot intensity shall be a function of target candidate strength.

3.5.1.4.7.5 Error Message - If the sensor operator sets up the sonobuoy receiver or the SDC for non-standard operation, an appropriate alphanumeric error message shall be generated and included within the output data format of the selected buoy display mode. An additional error message shall be generated if a DIFAR, VLAD, or DICASS sonobuoy being processed fails to provide a valid frequency reference tone used for directional channel demodulation. The error message characters shall be arranged in the display output data format such that, when viewed on the indicator, the error message appears at the bottom center of the crt presentation. An error message shall appear only in sonobuoy display modes and shall be inhibited from the Q13 (PPI) display output data. In the event that more than one error message applies, the message having the highest priority shall be generated. Error messages, in the order of priority, shall be as follows:

3.5.1.4.7.5.1 Too Many Processed Sonobuoys - The error message TOO MANY PRCS shall be generated when the number of sonobuoys simultaneously attempted to be processed exceeds the maximum allowable number of sonobuoys in a sensor set (as specified in 3.4.1.3). When the error message is generated, all sonobuoy information shall be deleted from the selected display mode data format (only axes, cursors and alphanumerics remain displayed).

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3.5.1.4.7.5.2 Too Many Displayed Sonobuoys - The error message TOO MANY DSPL shall be generated when the number of sonobuoys simultaneously attempted to be displayed exeeds the maximum allowable number to be displayed in the selected buoy display mode (as specified in 3.4.1.4). When the error message is generated, all sonobuoy information shall be deleted from the selected display output data format (only axes, cursor, and alphanumerics remain displayed).

3.5.1.4.7.5.3 Synchronization Loss - The error message SYNC LOSS shall be generated when the SDC is operating in the DIFAR/VLAD (2), DIFAR/VLAD (3), or DICASS (2) processing mode, and one of the processed sonobuoys fails to provide a valid frequency reference tone used for directional channel demodulation in the SDC.

3.5.1.4.7.5.4 Channel 12 Exceeded - The error message CHANNEL 12 MAX shall be generated when the SDC is operating in the RO processing mode and one of the sonobuoy receiver r-f channel code inputs to the SDC exceeds 12.

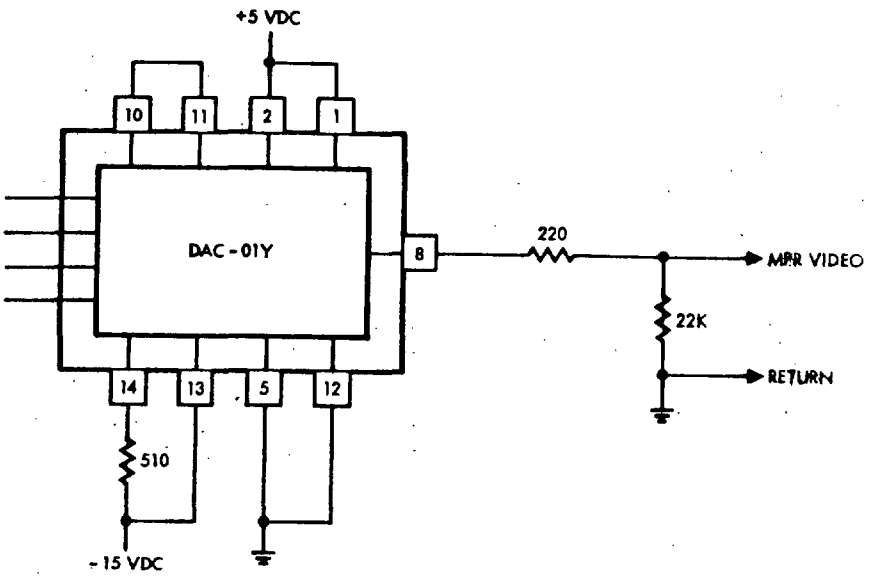
3.5.1.4.7.5.5 Audio Frequency Duplication - The error message AF DUPLICATION shall be generated when the SDC is processing two or more active sonobuoys, and the sonobuoy receiver r-f channel code inputs to the SDC channels being processed, relate to duplicate sonobuoy audio (ping) frequencies other than solely identical r-f frequencies.

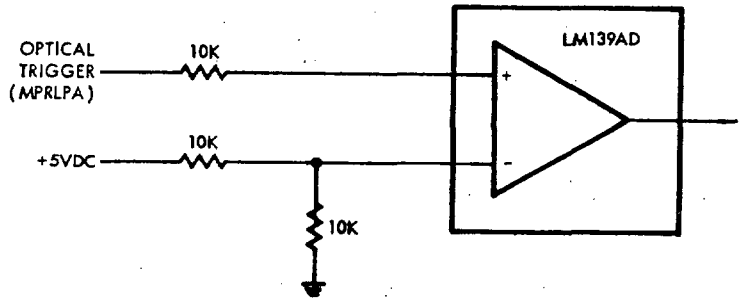
3.5.1.4.7.5.6 Radio Frequency Duplication - The error message RF DUPLICATION shall be generated when the SDC is processing two or more sonobuoys in the passive mode and the sonobuoy receiver r-f channel code inputs to the SDC channels being processed are duplicated.

3.5.1.4.7.6 Recorder Interface - The SDC shall provide a video output compatible with the recorder requirements. For sonobuoy or SDC-processed dip sonar sensor operations, recorder video shall be presented to the recorder from the processor interface. The video shall be derived from a digital-to-analog converter with characteristics similar to the device shown in Figure 10. Two SDC inputs shall be provided to receive control signals from the recorder to synchronize sonobuoy and processed dip sonar video with the recorder stylus position and sweep rate. Line receivers with characteristics similar to the devices shown in figure 10, shall be provided. During normal dip sonar operation, recorder video from the sonar receiver shall be channeled through the SDC to the recorder. The video switchover shall be controlled by the sonar/buoy discrete, originating at the DISPLAY pushbutton switch/indicator on the indicator front panel.

3.5.1.4.7.6.1 Active Sonobuoy Hardcopy - For active sonobuoy processing, the SDC shall automatically interrupt the range scale lines passing through the SDC to the recorder from the sonar receiver. The range scale line interrupt shall be controlled by mode information entered into interface registers by the digital processor. In accordance with the selected SDC range scale, the SDC shall apply +28 Vdc to the appropriate recorder range scale line. The recorder shall operate on the 8-kiloyard range scale for both the selected 5 and 8-kiloyard SDC range scale, and the 12-kiloyard recorder range scale shall be utilized for the 12-kiloyard SDC range scale. The 0 and 100-percent points on the chart paper shall correspond to 0 and full scale range for the selected SDC range scale. The stylus writing contrast shall vary in accordance with echo signal strength.

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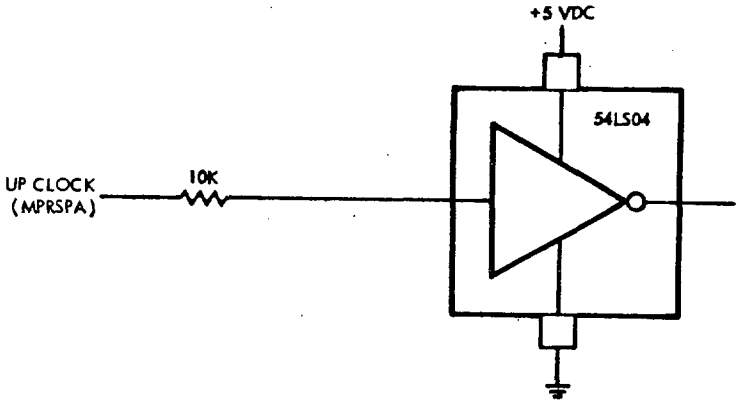


FIGURE 10. Recorder input/output.

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3.5.1.4.7.6.2 Passive Sonobuoy Hardcopy - Stylus sweep rate shall be determined by the RANGE SCALE-KYDS switch, located on the front panel of the sonar receiver. The 0 and 100 percent points on the chart paper shall correspond to the lower and upper frequency limits of the selected analysis band. The stylus writing contrast shall vary in accordance with frequency amplitude information.

3.5.1.4.7.6.3 SDC-Processed Dip Sonar Hardcopy - The hardcopy recording of SDC-processed dip sonar echo ranging shall be similar to the hardcopy recording of the standard echo ranging operation. The sector to be recorded shall be governed by cursor positioning.

3.5.1.4.7.7 TACNAV Interface - The SDC shall provide a means of transferring sonobuoy or dip sonar data to TACNAV. The SDC to TACNAV Interface shall utilize two successive data words with an accompanying shift clock to convey sonobuoy or dip sonar data to TACNAV. Necessary discrete signals shall be provided to perform the transfer and acknowledge functions. In addition, the SDC shall provide a manually operated MAD contact discrete with a means of indicating a TACNAV acknowledgement. Sonobuoy data shall be either manually or automatically transferred in selected process and display modes as follows:

	PROCESS MODE	DISPLAY MODE	TYPE OF TRANSFER
a.	DIFAR/VLAD	BRG/ALI	Manual
b.	DICASS	(2) BRG	Manual
		(2) DOP	Manual
		BRG/DOP	Manual
		(4) DOP	Automatic/Manual
c.	CASS and RO	(2) DOP	Manual

Dip sonar data shall be manually transferred in both the SDC processed and standard (non SDC enhanced) modes. Differential line receivers and drivers shall have characteristics similar to the devices shown in Figure 11.

3.5.1.4.7.7.1 Data - The SDC shall provide a differential output for data transmission by means of two, serial 28-bit words with an interval of 20 milliseconds minimum between the two. The exact format of the data words is shown in Figure 12. A type I word shall always be sent first and shall be followed by a type II word. Segments of words shall never be sent. Each data transfer shall be comprised of a single pair of words. The data line may have activity during the period between words.

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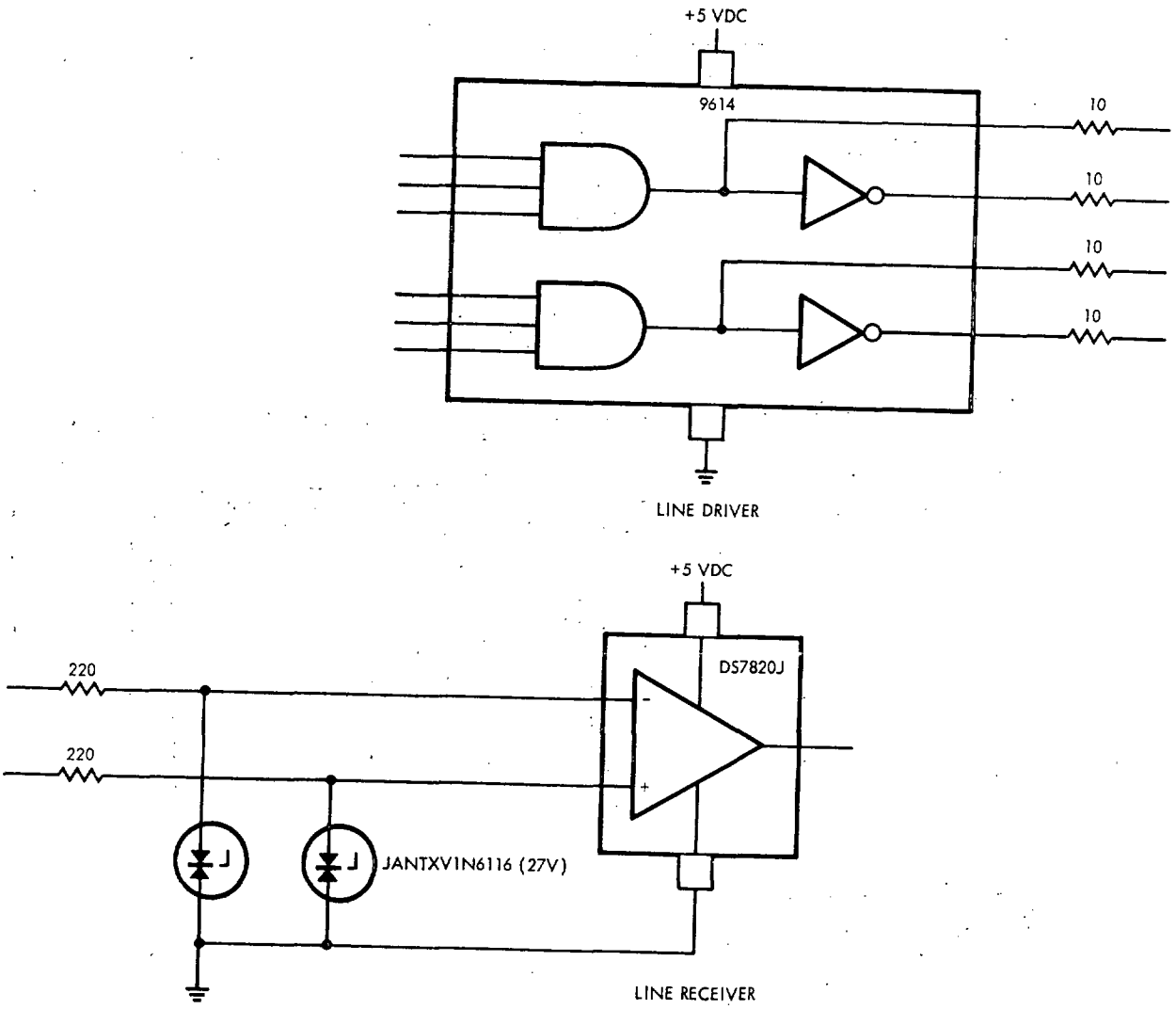
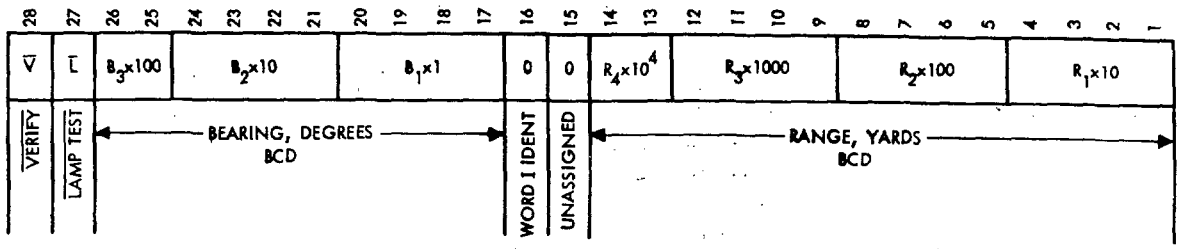


FIGURE 11. TACNAV input/output.

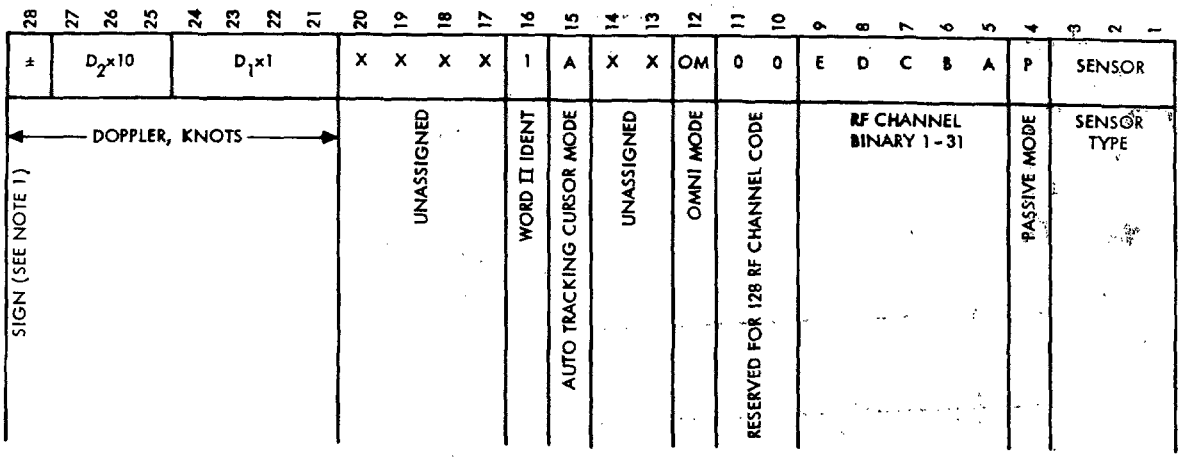
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TYPE I WORD:

FIRST BIT



TYPE II WORD:



SENSOR TYPE CODE

Q	AN/AQS-13E SONAR WITH SDC PROCESSING	0	0	0
A	RO -47	0	0	1
C	CASS -50	0	1	0
S	DICASS -62	0	1	1
D	DIFAR/VLAD (2) (SEE NOTE 2) -53	1	0	0
D	DIFAR/VLAD (3) (SEE NOTE 2) -53	1	0	1
L	LOFAR (SEE NOTE 3) -41	1	1	0
	AN/AQS-13E SONAR WITHOUT SDC PROCESSING	1	1	1

NOTES:
1. THE NUMERAL 1 IMPLIES CLOSING RANGE RATE.
   1 = HIGH LEVEL
   0 = LOW LEVEL
2. THE DIFAR ALSO IS USED FOR VLAD (SSQ-77).
   DATA IS TRANSFERRED ONLY IN BRG/ALI DISPLAY MODE.
3. DATA WORDS ARE NOT TRANSFERRED IN LOFAR SENSOR MODE.

FIGURE 12. TACNAV serial data format.

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3.5.1.4.7.7.2 Type I Word - The contents of a type I word shall be as follows:

a. Bits 1 through 14 are 3-1/2 BCD digits of range expressed in tens of yards. The greatest range is 20,000 yards and is represented by 2000.

b. Bit 15 shall always be low.

c. Bit 16 shall be low to identify a type I word.

d. Bits 17 through 26 shall be 2-1/2 BCD digits of magnetic bearing in degrees from 000 to 359.

e. Bit 27 is AQS-13E lamp test and bit 28 is AQS-13E remote bearing and range indicator verify. Bits 27 and 28 are low level when true and are ignored by TACNAV.

3.5.1.4.7.7.3 Type II Word - The contents of a type II word shall be as follows:

a. Bits 1 through 3 identify the type of buoy being processed, including both the 200-millisecond shaped pulse AQS-13E mode associated with Sonar Data Computer (SDC) processing and non-SDC processed rectangular pulse AQS-13E.

b. Bit 4 will be high to indicate a passive type sensor or passive operation of an active sonar system. This bit may be considered to be a prefix to the buoy-type code. This bit will be high for CASS or DICASS sensors not being pinged, and in all cases for LOFAR, DIFAR, and VLAD sensors. This bit will also be high if the AQS-13E sonar receiver is set for passive in both the non-processed and processed dip sonar modes.

c. Bits 5 through 11 designate the buoy up-link r-f channel. The ARR-75 code format will be converted to binary to represent r-f channels 1 to 31 in bits 5 through 9, and bits 10 and 11 will be set low and reserved for possible expansion of the r-f channel code to 128 channels. No other hardware provisions are provided at this time for future code expansion. R-f channel code bits 5 through 9 are low level when true.

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d. Bit 12 will be high to indicate an "omni" or nondirectional type sensor, or nondirectional operation of a directional type sensor. It signifies that the bearing data is invalid. This bit will be high for nondirectional display modes of DICASS, DIFAR, and VLAD sensors, and in all cases for RO, CASS, and LOFAR sensors. It will always be low for nonprocessed and processed dip sonar operation.

e. Bits 13 and 14 are unassigned and may be either high or low.

f. Bit 15 shall be high to indicate when an automatic tracking cursor is in use (available only with the (4) DOP display format selected).

g. Bit 16 shall be high to identify a type II word.

h. Bits 17 through 20 are unassigned and may be either high or low.

i. Bits 21 through 28 shall be 1-1/2 BCD digits plus sign of range rate data expressed in knots. The maximum range rate is ±79 knots (±40.6 m/s). Bit 28 high shall imply closing range rate.

3.5.1.4.7.7.4 Gated Shift Clock - The SDC shall provide a differential output for a gated shift clock, consisting of two bursts of 28 clock pulses which shall be used to shift the serial data into the input register of TACNAV. The clock rate shall be between 4.0 kHz and 1.0 MHz and may change during an individual word. The timing relationship between data and clock is shown in Figure 13.

3.5.1.4.7.7.5 Dip Sonar Range and Bearing Data - The SDC shall provide a differential input to receiver range and bearing data from the sonar receiver. The range and bearing data, which is received every 40-milliseconds, is contained in a 28-bit word, identical to type I word shown in Figure 12. Clocking activity occurs on the range and bearing data line between data words and the clock pulses shall not be transferred to TACNAV. During either processed or standard dip sonar operation, the data content of type I word set to TACNAV shall be a copy of the range and bearing data word from the sonar receiver. Range and bearing information contained in type I word shall be slaved to the CURSOR controls located on the sonar receiver front panel.

3.5.1.4.7.7.6 Dip Sonar Range and Bearing Data Clock - The SDC shall provide a differential input to receive a range and bearing data clock from the sonar receiver, for use as a gated shift clock to transfer the dip sonar range and bearing data word into SDC registers. The clock is a burst of 28 pulses, at a rate of 6400 Hz.

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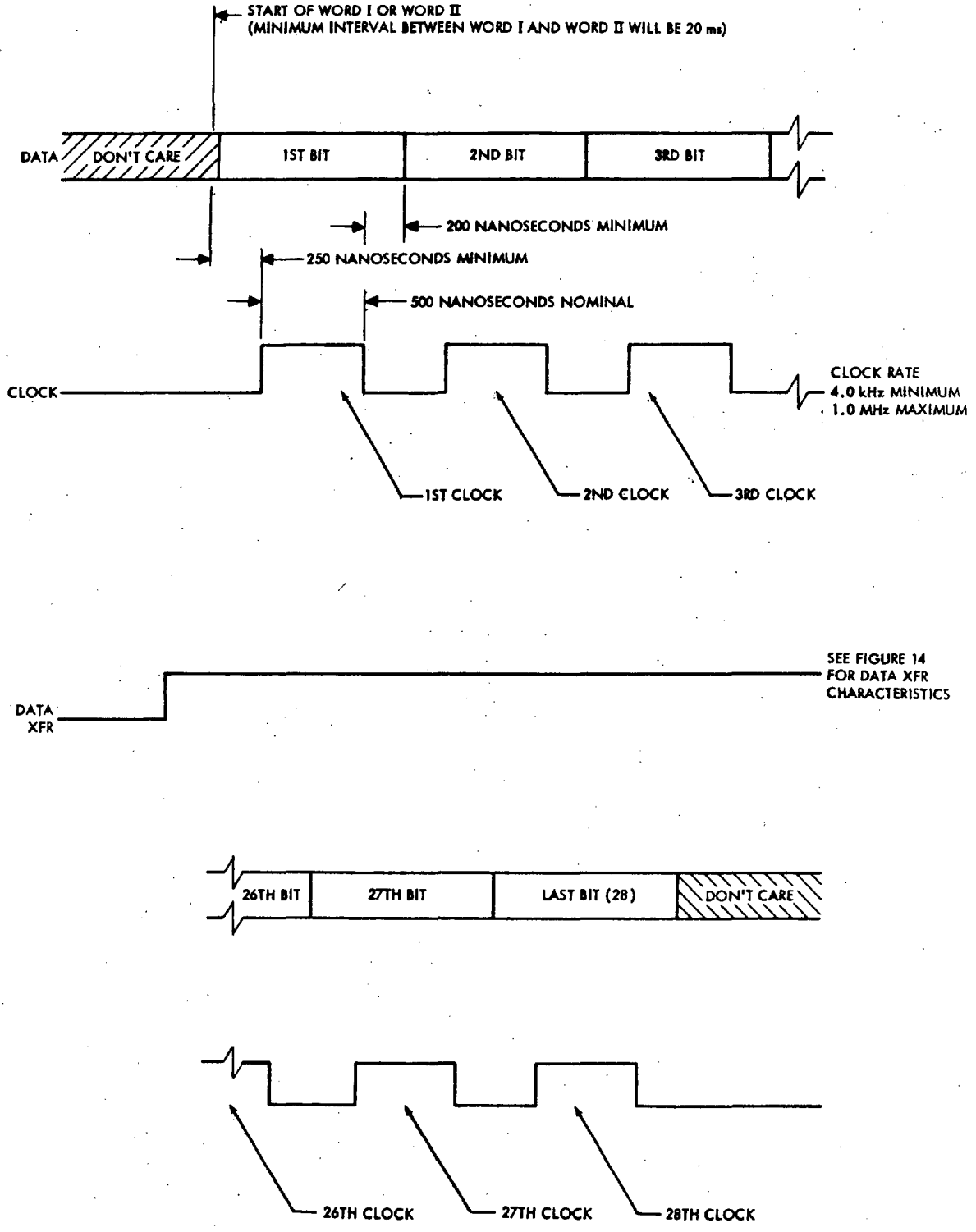


FIGURE 13. Data and shift clock timing.

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3.5.1.4.7.7.7 Data Transfer - The SDC shall provide a differential output for a data transfer digital signal. The data transfer signal shall go true whenever a data transfer to TACNAV is initiated, either manually or automatically, by the SDC. The data transfer signal shall remain true until the data acknowledge discrete from TACNAV goes true and returns to false, indicating that the data has been received and dispositioned by TACNAV. The data may be withdrawn by returning the data transfer signal to false if the data has not been dispositioned by TACNAV. The timing for both transfer and withdrawal is illustrated in Figure 14.

3.5.1.4.7.7.8 Data Acknowledge - The SDC shall provide a differential input for a data acknowledge discrete from TACNAV. The data acknowledge goes true within 100-milliseconds after the type II word has been received by TACNAV. A data acknowledge true shall inhibit the SDC from sending additional data for up to 60 seconds, as governed by TACNAV. When the data acknowledge signal returns to false, the TACNAV is ready to accept additional data. The front panel data transfer switch shall illuminate to indicate the true level of the data acknowledge signal.

3.5.1.4.7.7.9 MAD Contact - The SDC shall provide a differential output for a MAD transfer digital signal that notifies TACNAV of a MAD contact. The MAD transfer signal shall go true whenever a MAD transfer to TACNAV is initiated. The MAD transfer signal shall remain true for 100 milliseconds minimum. The timing for MAD transfer is illustrated in Figure 15.

3.5.1.4.7.7.10 MAD Acknowledge - The SDC shall provide a differential input for a MAD acknowledge discrete from TACNAV. The MAD acknowledge is expected to be true after MAD transfer has signalled a contact, and to remain true as long as the MAD contact is pending disposition by TACNAV. The front panel MAD transfer switch shall illuminate to indicate the true level of the MAD acknowledge signal.

3.5.1.4.7.7.11 Manual Transfer Characteristics - When a data transfer is initiated, a single word pair (consisting of a type I word followed by a type II word) shall be sent to TACNAV. The data transfer signal shall go true during the first gated clock pulse. When the data acknowledge received from TACNAV goes true, further data transfers shall be inhibited until data acknowledge signal returns to false. If the data transfer switch is depressed during the period that the data acknowledge is true, the data transfer signal shall return to false to indicate withdrawal of the data. The data acknowledge signal remains true until TACNAV responds and returns the data acknowledge signal to false, when further data transfers may be initiated.

3.5.1.4.7.7.12 Automatic Transfer Characteristics - The automatic transfer characteristic is available only with the (4) DOP display, which may be used with RO and CASS sonobuoys. The sequence of the data transfers will be the same in the automatic mode as in the manual mode, once the transfer is started by pressing the data transfer button. The only difference is that the data from each buoy will be sent in succession without further depression of the data transfer button. Data will be sent only for those buoys that have a tracking cursor activated. The maximum rate of data transfer is one word pair per ping, per buoy. Since the rate of data transfer may be limited by the TACNAV, using the data acknowledge discrete signal, the data transfer rate could fall behind the buoy ping rate. In this case, the most recent data would always be sent. Data from all buoys will be sent in succession, irrespective of the rate of data acknowledgement by TACNAV.

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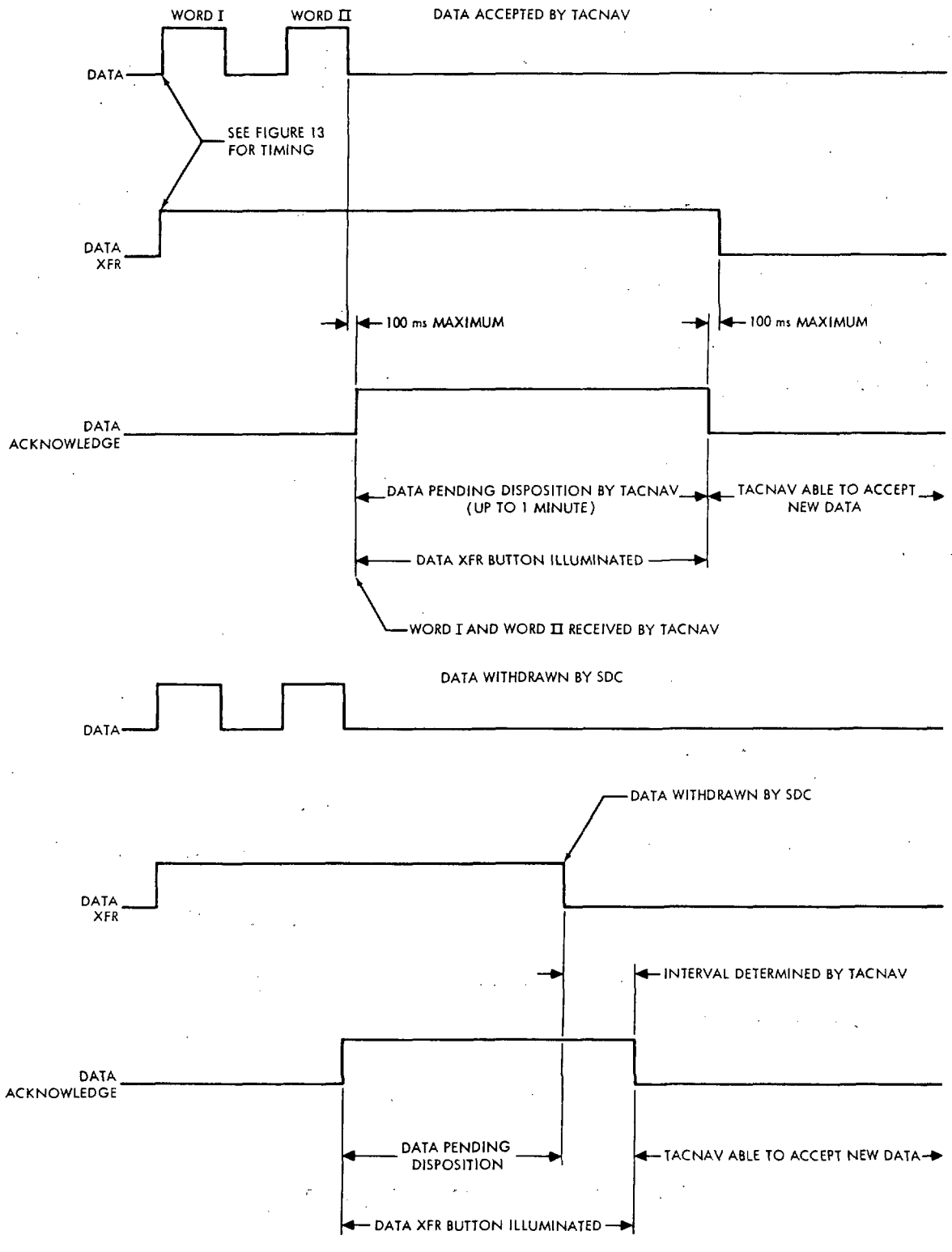


FIGURE 14. Timing of discrete control signals sonar data.

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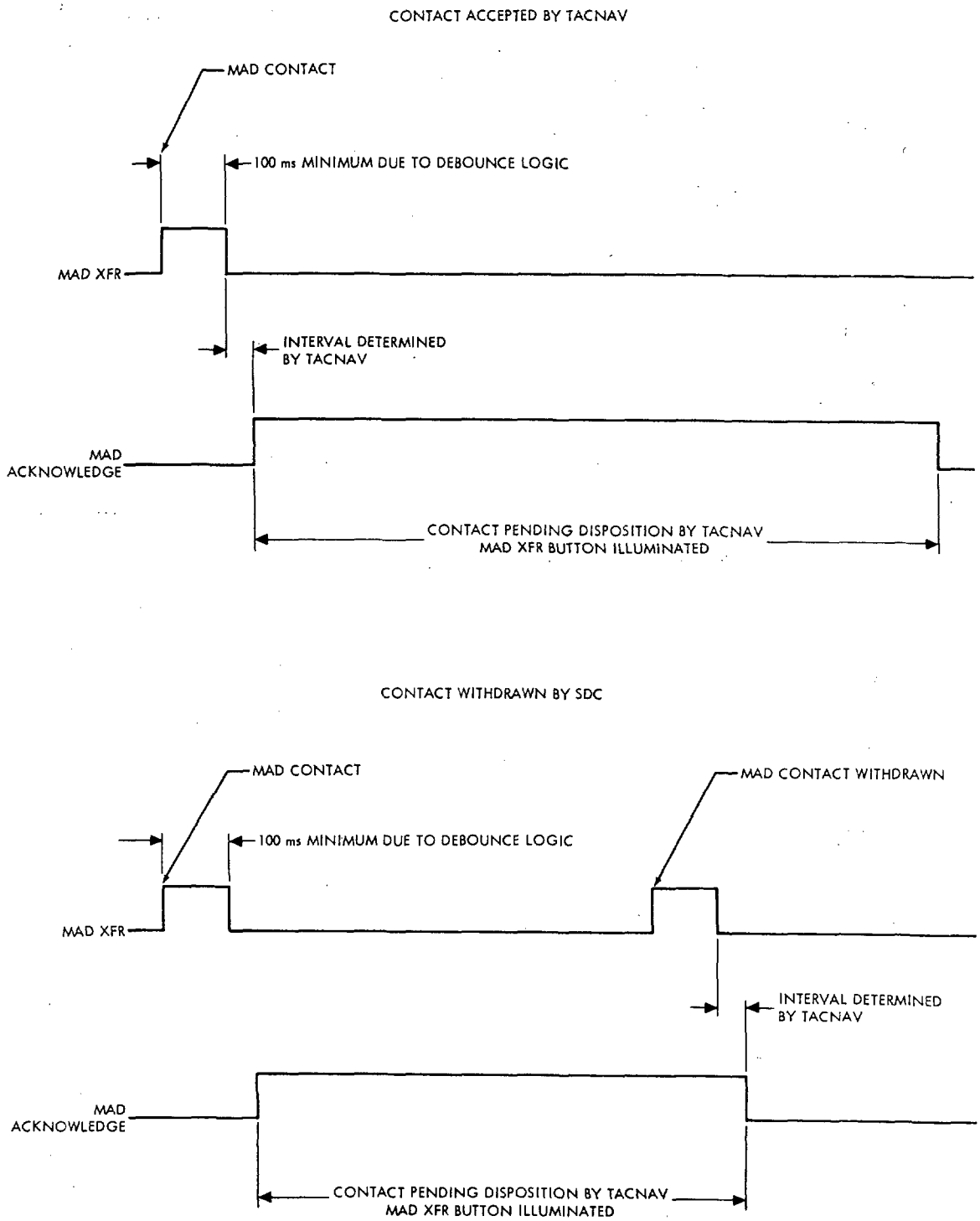


FIGURE 15. Timing of discrete control signals MAD contact.

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a. Operator feedback is provided by illuminating the data transfer button after the first word pair is acknowledged by TACNAV. The data transfer button shall remain continuously illuminated as long as automatic operation continues. If TACNAV ceases to acknowledge data transfers, the data transfer button shall be extinguished and automatic data transfer terminated. If TACNAV reduces the rate of data transfer by holding data acknowledge true for long periods, no indication of the reduced data rate is provided.

b. Automatic data transfer may be deselected by a second depression of the data transfer button.

3.5.1.4.7.7.13 SDC Processed and Conventional Dip Sonar Transfer Characteristics - Data transfer to TACNAV is initiated by the data transfer button and occurs in the same manner as sonobuoy data with manual transfer. Range and bearing data content of word I is controlled by the AQS-13E sonar receiver CURSOR controls. Doppler data is not made available to TACNAV.

3.5.1.4.7.8 Sonar Receiver Interface - The SDC shall provide inputs for receiving dip-sonar range scale, compass, start of transmission, and cursor data required for Q13 (PPI) processing and display. In addition, an input shall be provided for an active/passive advisory required in TACNAV transfer word II type. Sonar receiver multiplex control outputs shall be provided to synchronize and clock dip-sonar data into the SDC for quantization and entry into memory during dip-sonar processing operation. Line receivers and drivers shall have characterisitics similar to those of the devices shown in Figure 16.

3.5.1.4.7.8.1 Sonar Range Scale - The SDC shall provide inputs for 5, 8, 12, and 20-kiloyard discretes from the sonar receiver. The 5-kiloyard advisory is TTL, low level true. The 8, 12, and 20-kiloyard advisories are +30 Vdc true. With all range scale inputs false, a range scale of 3 kiloyards shall be assumed. Range scale discretes shall be entered into memory by the processor interface.

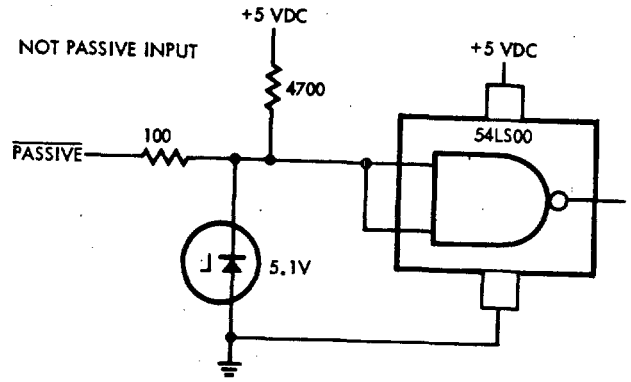
3.5.1.4.7.8.2 Sonic Gate - An input shall be provided to receive a TTL, high level true, 200-millisecond sonic gate pulse that denotes start and end of shaped pulse transmission during dip-sonar processing operation. The sonic gate shall be utilized as Q13 (PPI) range sweep zero. The sonic gate shall be entered into memory by the processor interface.

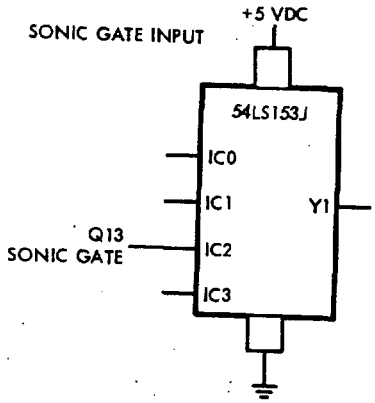
3.5.1.4.7.8.3 Q13 (PPI) Cursor Control - The SDC shall utilize the sonar bearing and range data and shift clock inputs specified in 3.5.1.4.7.7.5 and 3.5.1.4.7.7.6, to control the cursor position in the Q13 (PPI) display format. The bearing and range data shall be serially shifted into parallel registers and then entered into memory by the processor interface.

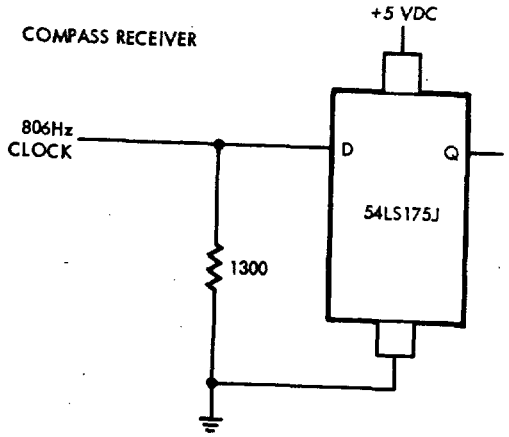
3.5.1.4.7.8.4 Q13 Compass - The SDC shall provide inputs for a variable phase, 806 Hz sine wave and an 806 Hz square wave, reference furnished by the sonar receiver. The two signals provide dip-sonar hydrophone orientation information, relative to magnetic north. The SDC shall monitor the phase difference between the two signals for the purpose of stabilizing the Q13 (PPI) display, regardless of rotational movement of the sonar hydrophone.

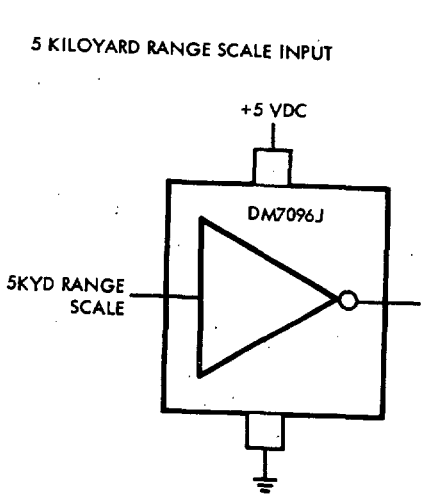
3.5.1.4.7.8.5 Not Passive - A TTL, low level true input shall be provided for entering a passive/active advisory into memory by the processor interface. The information shall be inserted into bit-4 of TACNAV transfer word II, during SDC-processed and normal dip-sonar operations.

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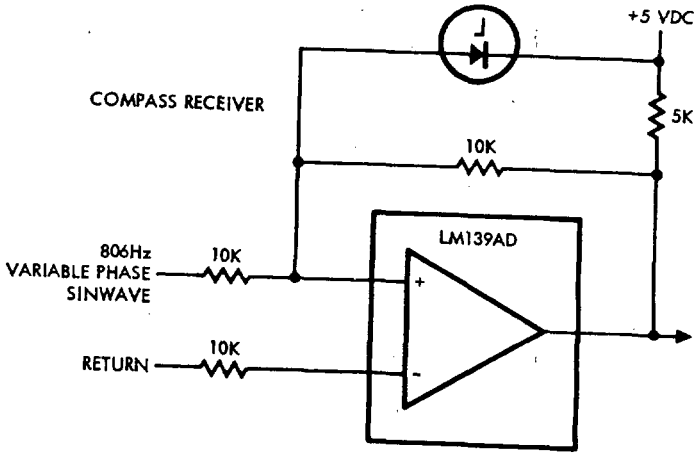


FIGURE 16. Sonar receiver advisory and compass inputs.

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3.5.1.4.7.8.6 Multiplexer Control - The SDC shall provide outputs for synchronizing and clocking the sonar receiver multiplexer (4A16) during Q13 processing operation. The multiplex clock shall be 19,728 Hz, TTL square wave that steps the multiplexer to the sum and and left half beams in each of eight sonar channels. A differential line driver shall be provided for the clock output and each leg shall have a TTL fan out of 10. A TTL, high level output, equal in duration to two clock periods, shall be provided once every ping to synchronize the sonar receiver multiplexer. The synchronizing output shall have a TTL fan out of 10. The multiplexer control outputs shall be inhibited in all sensor modes other than Q13 (PPI).

3.5.1.4.8 Downlink Command Generator (See Figure 17) - The SDC shall provide a means for modulating the uhf transceiver for the transmission of downlink commands to CASS and DICASS sonobuoys. The necessary modulation comprised of address, sonic, deep, and scuttle tones shall be generated by the SDC. Additionally, provisions shall be made for the SDC to automatically set up the UHF transceiver for the proper CASS or DICASS uhf downlink frequency. The SDC shall have the capability of producing a downlink ping command for each CASS or DICASS sonobuoy being processed and displayed. A means shall be provided by which an individual downlink deep or scuttle command may be directed to any CASS or DICASS sonobuoy, whether being processed and displayed or not.

3.5.1.4.8.1 Take Command - The SDC shall provide for an input from the pilot's UHF-2 console switch for controlling the output of downlink command tones and transmitter handshake (data word). An open supplied by the UHF-2 switch in the COMM position shall inhibit the output of all downlink command tones, transmit key, and transmitter handshake. When the UHF-2 switch is set to DICASS, a ground return furnished by the switch shall enable the downlink command output and cause the immediate transfer of data words to the UHF transceiver.

3.5.1.4.8.2 Transmitter Handshake - With the pilot's UHF-2 (take command) switch set to DICASS, the SDC shall generate and continuously transfer a frequency setup data word to the UHF transceiver. The data word shall be synchronous with a data clock furnished by the the UHF transceiver. The data word format and related data clock are shown in Figure 18. Handshake input/output characteristics shall be similar to those of devices shown in Figure 19.

3.5.1.4.8.2.1 Data Clock - The SDC shall provide a differential line input to receive the 800 Hz, square wave data clock furnished by the UHF transceiver. The data clock shall be used to shift the data word from the SDC to the UHF transceiver.

3.5.1.4.8.2.2 Data Word - The SDC shall generate and provide a serial 32-bit NRZ data word on a differential line output the UHF transceiver. The data word shall be shifted at a rate that is synchronous to the 800 Hz data clock furnished by the UHF transceiver.

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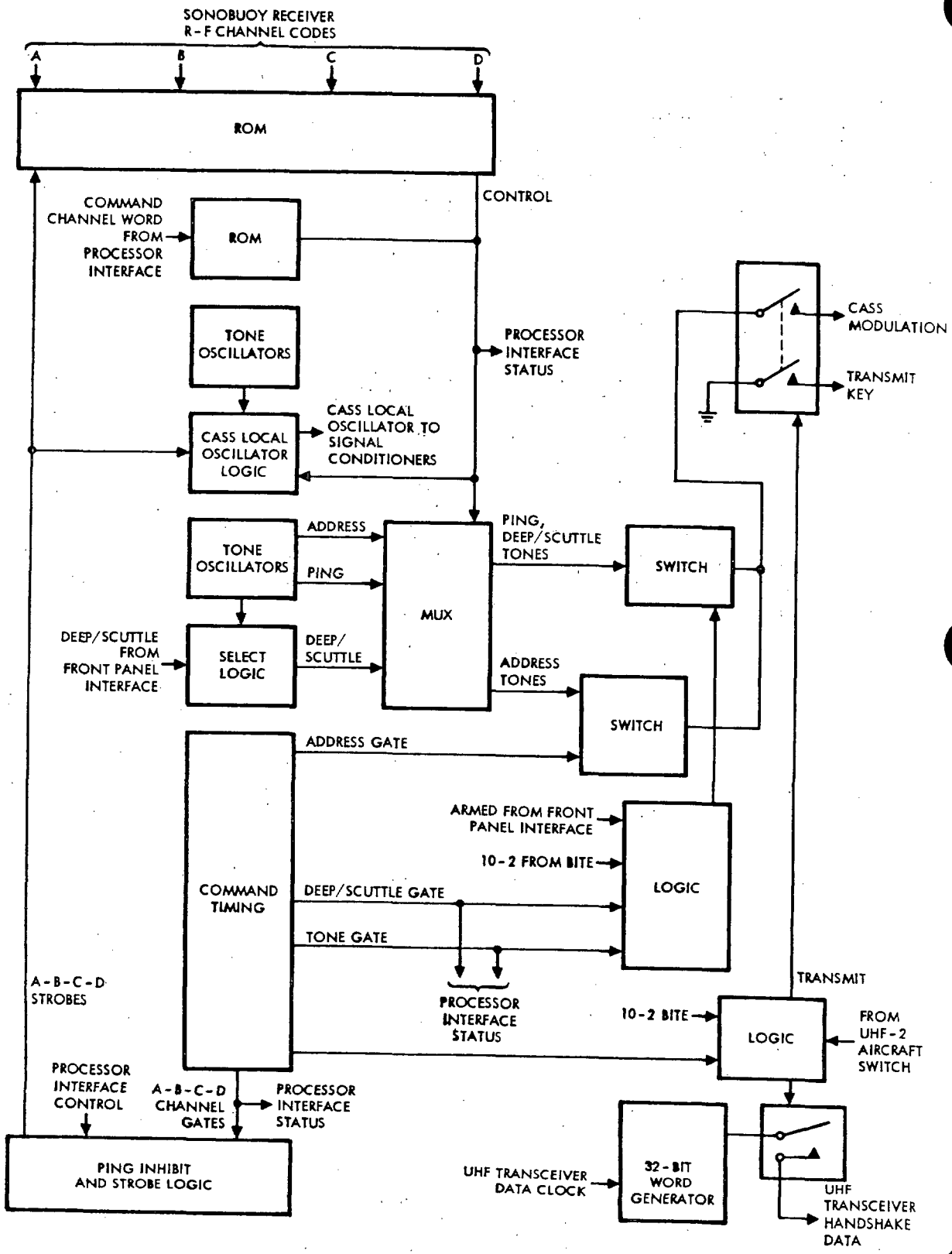


FIGURE 17. Downlink command generator, block diagram.

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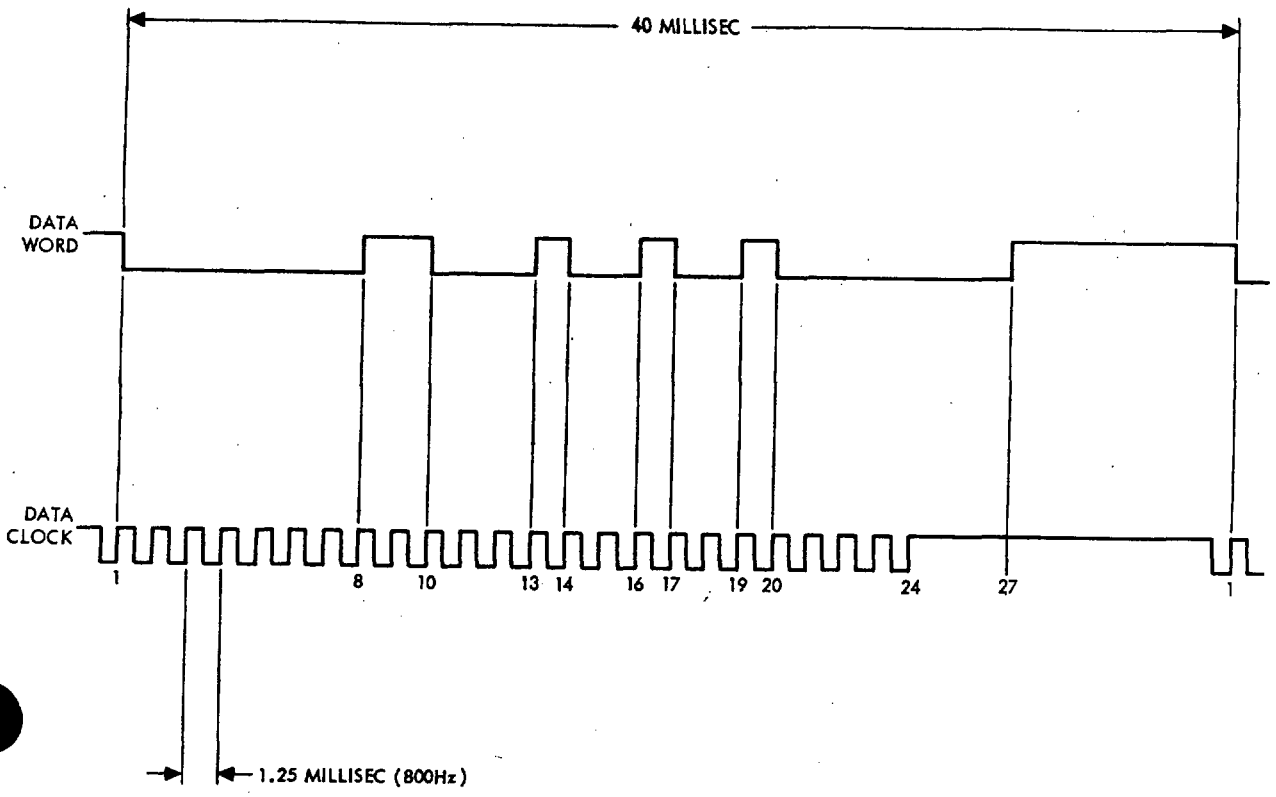


FIGURE 18. Transmitter handshake format.

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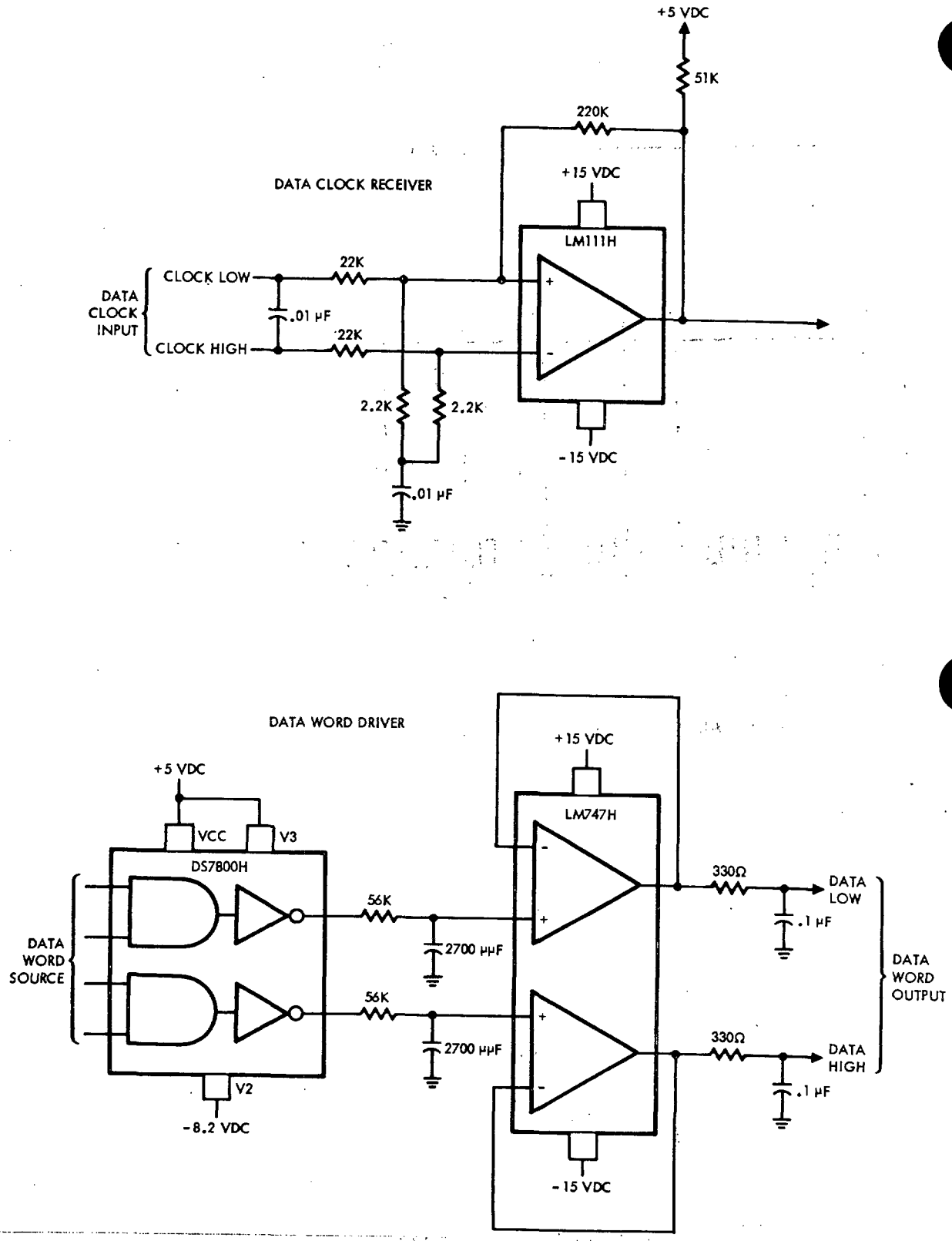


FIGURE 19. Handshake input/output.

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3.5.1.4.8.3 R-F Channel Codes - The SDC shall provide inputs to receive four separate 5-bit digital r-f channel codes furnished by the sonobuoy receiver. The digital code and logic levels for decimal 1 through 31 r-f channels are shown in Table I. Each r-f channel code shall provide the basis for automatic selection of downlink address and sonic tones in each process channel during CASS or DICASS active sonobuoy operation. R-F channel codes shall also be utilized for the automatic selection of CASS local oscillators, as specified in 3.5.1.4.4.3. The r-f channel code status for each process channel shall be entered into memory by the processor interface.

3.5.1.4.8.4 Transmitter Key - The SDC shall provide an output that presents a transmit key discrete to the UHF transceiver during an entire interval required to sequentially ping four sonobuoys. The transmit key discrete shall assume a ground potential 100-milliseconds minimum, prior to initiating the first downlink tone to the UHF transceiver. During the non-transmit interval, when the SDC is processing returning sonic data, the transmit key output shall produce an open line. The transmit key status shall be entered into memory by the processor interface.

3.5.1.4.8.5 Transmitter Modulation - The SDC shall provide an output for downlink command tones used to modulate the UHF transceiver during CASS and DICASS sonobuoy operations. All downlink commands shall be prefixed with an appropriate address tone pair, followed by a discrete sonic, deep, or scuttle tone. Downlink commands containing a sonic tone shall automatically be generated for each sonobuoy selected for pinging operation. When more than one sonobuoy in a processed and displayed sensor set is selected for pinging, a downlink command shall be sequentially generated for each sonobuoy to be pinged. The sonobuoys being processed in SDC channels A, B, C, and D shall be pinged in the alphabetic order of the process channel for which they are being processed and displayed. For each process channel not set for pinging operation, a downlink ping command time slot containing no tones shall form a part of the A, B, C, and D downlink ping command sequence. The SDC shall provide a means of manually initiating a downlink deep or scuttle command applicable to one CASS or DICASS sonobuoy. If a downlink deep or scuttle operation is initiated, the deep, or scuttle command shall be generated immediately following the next A, B, C, and D ping command sequence. Refer to Figure 20 for downlink command timing. The modulation output characteristics shall be similar to those provided by the driver shown in Figure 21.

3.5.1.4.8.5.1 Address Tones - Each downlink command generated by the SDC shall begin with a CASS or DICASS sonobuoy address composed of a composite, sine-wave tone pair. The frequency of each tone shall be automatically selected from eight tone frequencies that are compatible with the CASS/DICASS sonobuoy addressing requirements. The sonobuoy address tone pair automatically selected for sonobuoys to be pinged, shall correspond to the A, B, C, and D r-f channel codes furnished by the sonobuoy receiver. Provisions shall be made to manually select the desired sonobuoy address for use in a deep or scuttle downlink command. Deep or scuttle address information shall be entered into registers from memory by the processor interface. The tone frequencies for CASS/DICASS sonobuoy addresses and the duration of address tones are given in the classified extract of this specification.

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TABLE I. R-F Channel Codes.

R-F Channel Code	Code				
	16	8	4	2	1
1	0	1	1	1	1
2	0	1	1	1	0
3	0	1	1	0	1
4	0	1	1	0	0
5	0	1	0	1	1
6	0	1	0	1	0
7	0	1	0	0	1
8	0	1	0	0	0
9	0	0	1	1	1
10	0	0	1	1	0
11	0	0	1	0	1
12	0	0	1	0	0
13	0	0	0	1	1
14	0	0	0	1	0
15	0	0	0	0	1
16	0	0	0	0	0
17	1	1	1	1	0
18	1	1	1	0	1
19	1	1	1	0	0
20	1	1	0	1	1
21	1	1	0	1	0
22	1	1	0	0	1
23	1	1	0	0	0
24	1	0	1	1	1
25	1	0	1	1	0
26	1	0	1	0	1
27	1	0	1	0	0
28	1	0	0	1	1
29	1	0	0	1	0
30	1	0	0	0	1
31	1	0	0	0	0
0	1	1	1	1	1

Logic Level

0 = +1.2 Vdc

1 = -12 Vdc

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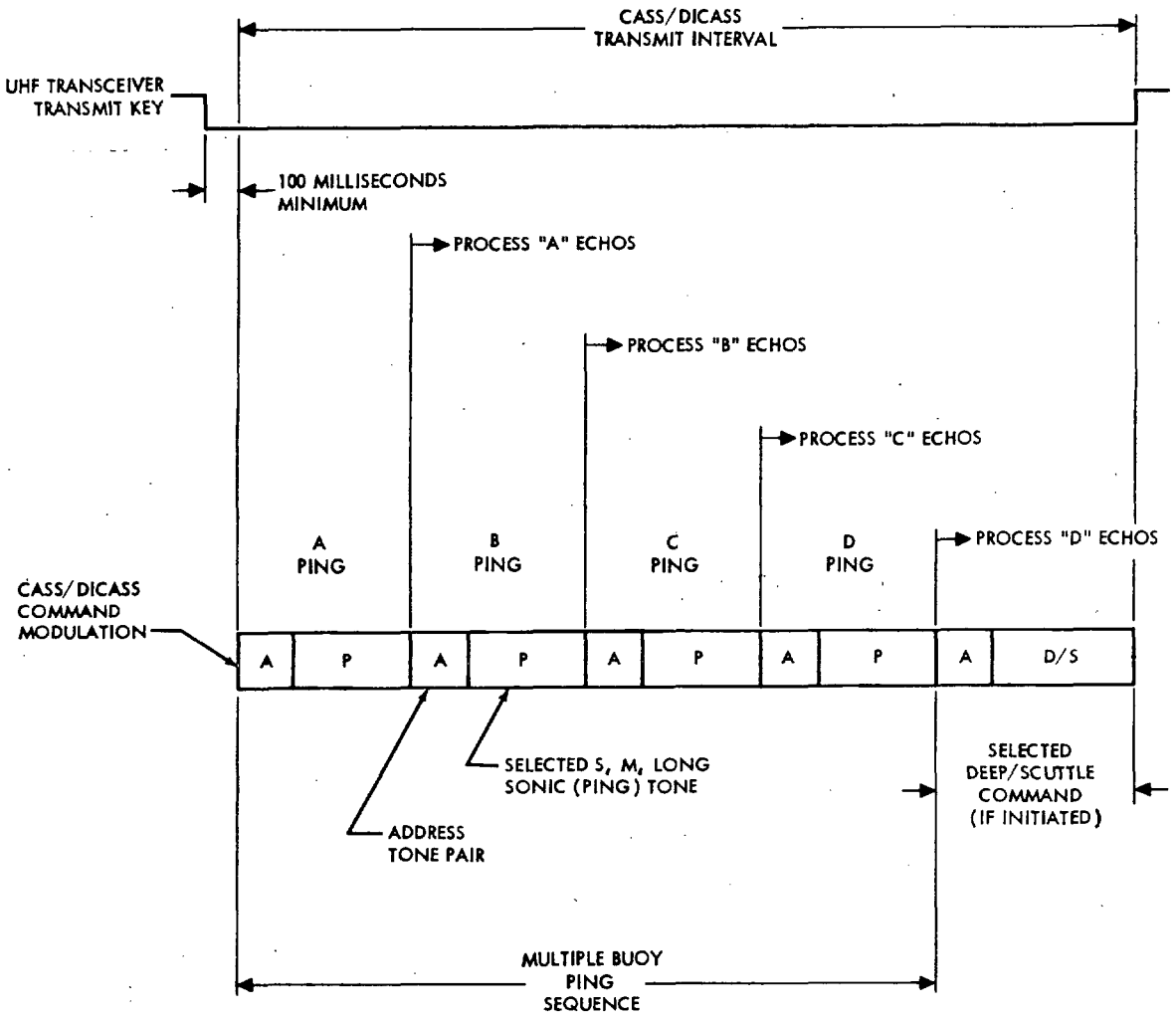


FIGURE 20. Downlink command timing.

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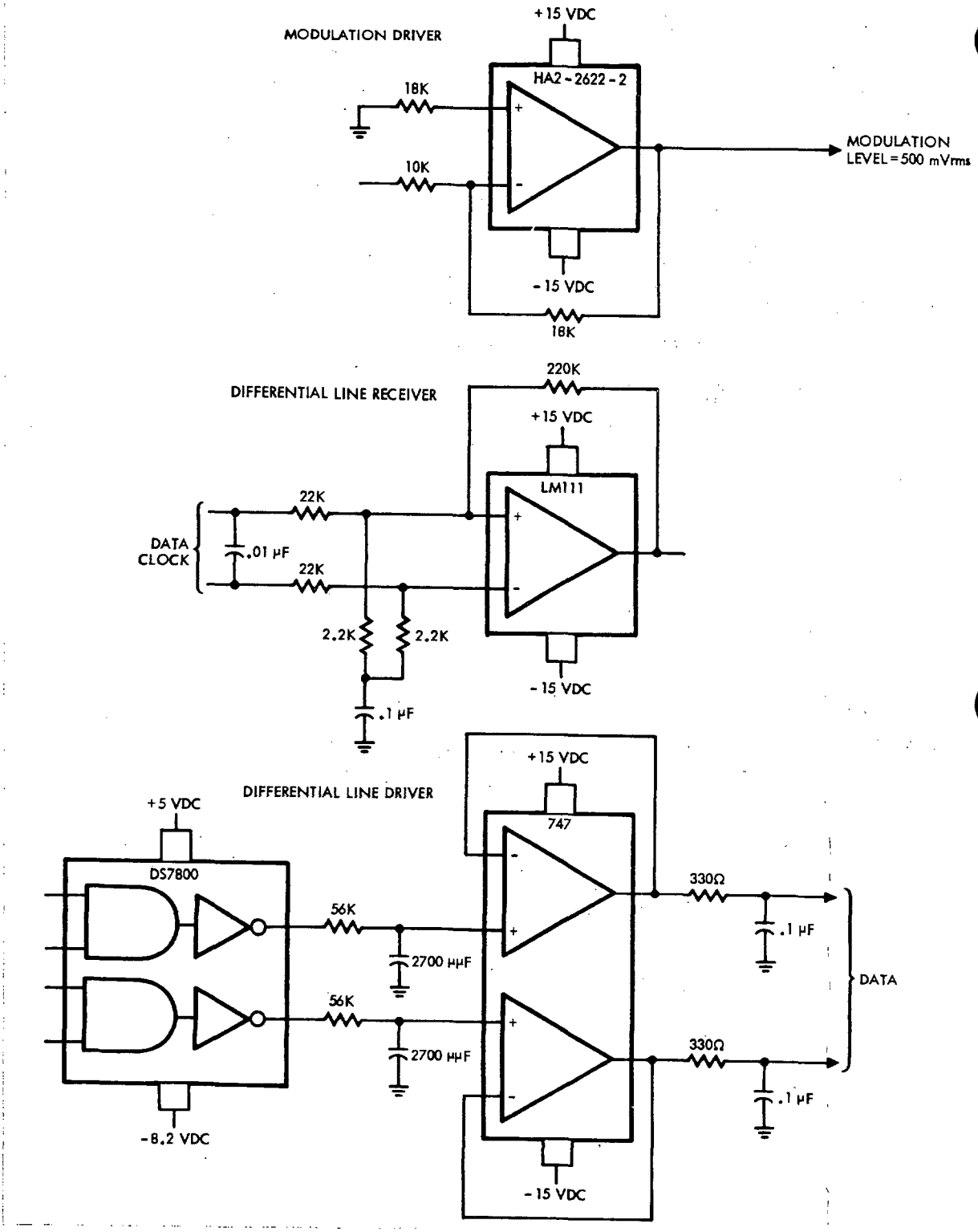


FIGURE 21. Downlink input/outputs.

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3.5.1.4.8.5.2 Ping Tones - One of four unique sonic, sine-wave tones shall be automatically selected for the downlink ping commands of each sonobuoy to be pinged. The sonic frequency for each sonobuoy to be pinged shall automatically be selected in accordance with the sonic frequency related to the appropriate A, B, C, and D r-f channel codes furnished by the sonobuoy receiver. The SDC shall provide three selectable sonic pulse lengths for CASS and DICASS active sonobuoy operations. Sonic frequencies and pulse lengths are given in the classified extract of this specification. The selected pulse shall simultaneously apply to sonobuoys in all process channels. Ping tone interval status for each process channel shall be entered into memory by processor interface.

3.5.1.4.8.5.3 Deep/Scuttle Tones - The SDC shall provide for two manually selected discrete sine-wave tones required for CASS or DICASS deep and scuttle downlink command operations. The SDC shall generate a deep or scuttle downlink command to cause a deep deployment of the hydrophone, or to flood and sink the sonobuoy. The deep/scuttle downlink command shall consist of an address tone pair, followed by the appropriate deep/scuttle tone. Deep/scuttle tone frequencies and durations are given in the classified extract of this specification.

3.5.1.4.8.5.4 Range Scales - The SDC shall provide three selectable range scales of 5, 8, and 12 kiloyards for CASS and DICASS operations. Downlink ping commands for sonobuoys selected for pinging operation, shall be recurrent at intervals dictated by the selected SDC range scale. The 5, 8, and 12 kiloyard range intervals for CASS and DICASS operations shall be 6.5, 10.5, and 15.4 seconds, respectively. The range sweep in the appropriate plot of the selected buoy display mode shall be synchronized by individual A, B, C, and D downlink ping commands during CASS and DICASS operations. The selected range scale shall simultaneously apply to sonobuoys in all process channels. Restricted use of range scales, in accordance with selected sonic pulse width, is given in the classified extract of this specification.

3.5.1.4.9 BIT Generator and Downlink Verification (See Figure 22) - BIT (built-in-test) capability shall be provided for the purpose of preflight and inflight performance verification of the SDC. SDC BIT functions shall be initiated by selecting the appropriate BIT code, using TEST A and B bi-directional pushbutton switches located on the indicator front panel. Test A and B codes and logic levels are given in Table II. The major SDC BIT functions selected by the TEST A and B pushbutton switches shall be as follows:

3.5.1.4.9.1 10-1 BIT Function - Process and display mode performance verification shall be initiated when a 10-1 BIT code is applied to the SDC. The SDC shall internally generate demodulated sensor set signals, similar to the type furnished by the sonobuoy receiver. Additionally, the SDC shall generate a dip sonar signal, similar to that produced by the sonar receiver, with center sector, or cw signals in all receiver channels. The particular sonobuoy or dip sonar BIT stimulus to be generated for a given process mode shall automatically be selected when the front panel SENSOR switch is set to the sensor mode of interest. The appropriate sonobuoy test stimulus shall be injected at the input of the sonobuoy signal conditioner to each SDC process channel. Sonobuoy receiver inputs to the SDC signal conditioners shall automatically be switched open during all process and display verification BIT operations. Downlink command modulation to the UHF transceiver shall be inhibited. The simulated data shall be presented to all display modes applicable to the processed sensor set.

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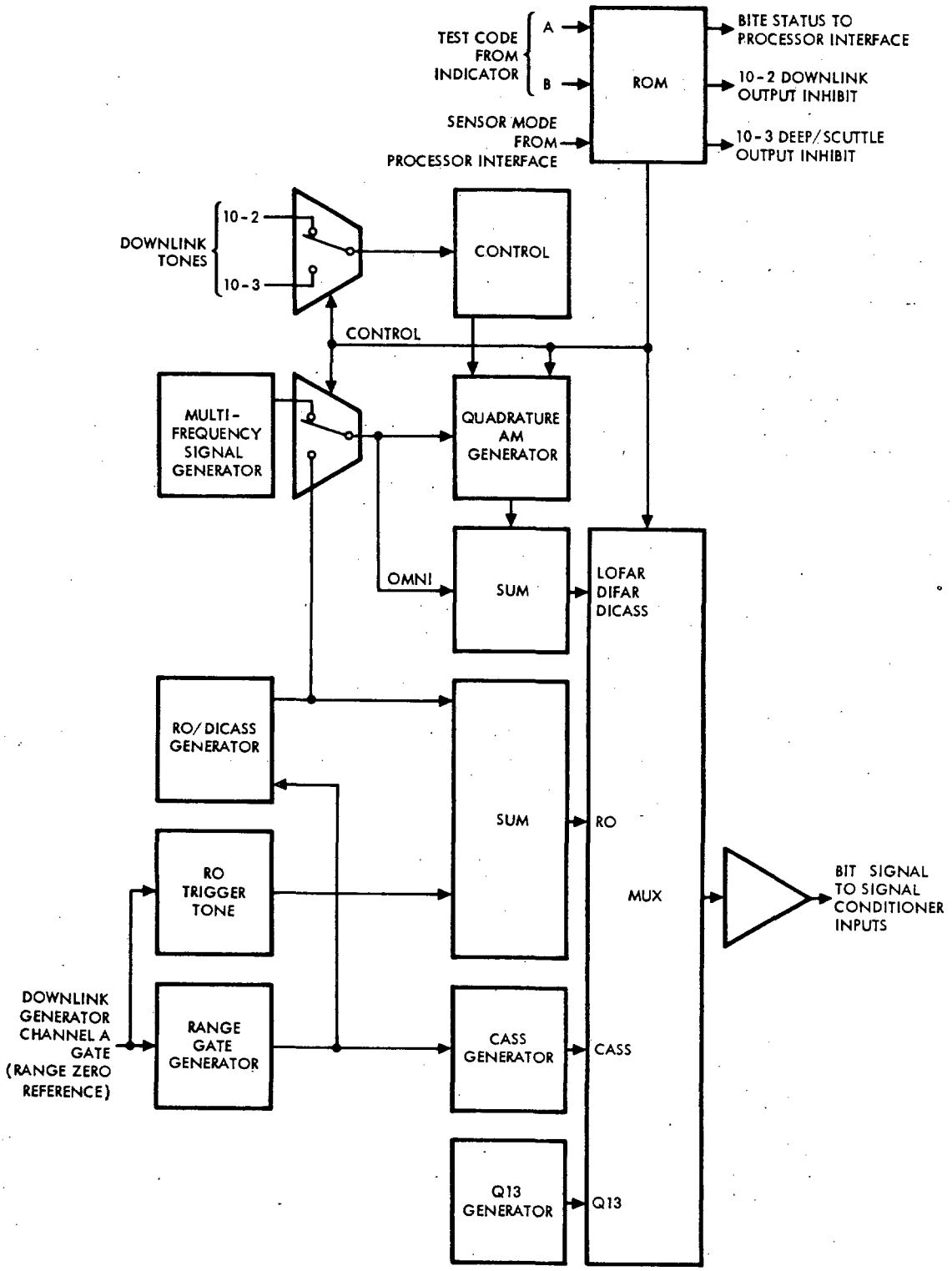


FIGURE 22. BIT, block diagram

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TABLE II. Test A and B Codes.

	TEST PUSHBUTTON SWITCHES							
	A				B			
	8	4	2	1	8	4	2	1
Process and Display Performance Verification 10-1	1	0	1	0	0	0	0	1
Downlink Tone Verification 10-2	1	0	1	0	0	0	1	0
Downlink Tone Interface Verification 10-3	1	0	1	0	0	0	1	1

Logic Levels

0 = 0 Vdc

1 = +5 Vdc

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3.5.1.4.9.1.1 LOFAR Simulation - A multifrequency signal generator shall be used to simulate a series of discrete frequencies across the input passband between 10 and 2010 Hz. The fundamental and harmonic progression of frequency lines in the spectrum, beginning at 10 Hz, shall be spaced at 5 Hz intervals throughout the passband. The energy contained in the frequency line at 320 Hz and each integral multiple thereof, shall have a magnitude greater than all other frequency lines. The energy contained in the frequency line at 40 Hz and each integral multiple thereof, shall have a magnitude between that of the 320 Hz spaced lines and all remaining 5 Hz spaced lines in the spectrum. All 5 Hz spaced lines, other than the 40 and 320 Hz spaced lines, shall be discernible in the (2) ALI display mode when processed in the 40 Hz analysis band with a five minute ALI.

3.5.1.4.9.1.2 DIFAR/VLAD Simulation - The same multifrequency signal generator used for LOFAR simulation shall be employed for DIFAR/VLAD simulation. The entire series of discrete frequencies shall form the omnichannel. In addition, each discrete frequency shall be used to derive directional channel frequency components within the suppressed carrier, double side band spectrum associated with DIFAR/VLAD sonobuoys. Frequency and phase reference signals shall be generated and mixed with the omni and directional channel data to form a complete frequency composite of a DIFAR/VLAD sonobuoy. With the simulated DIFAR/VLAD stimulus processed in the 1K analysis band with a five minute ALI, the 40 and 320 Hz spaced frequency lines shall provide a 45 ±10 degree bearing indication in the BRG/ALI display format.

3.5.1.4.9.1.3 DICASS Simulation - A cw tone (background) shall be generated within the heterodyned passband applicable to DICASS sonobuoys. The background tone shall affect the system AGC in a manner similar to sea noise which reduces system gain. A time zero range reference shall be established by the downlink command function timing, and shall be coincident with the end of ping time for a sonobuoy in process channel A. Between five and six seconds after time zero, the background tone shall be increased in frequency for 400 milliseconds to simulate an echo return from a closing target. Both the background and echo tones shall be used to form a complete frequency composite of a DICASS sonobuoy, in a manner similar to that specified for DIFAR/VLAD simulation. The echo frequency, range interval and simulated bearing shall be such that indications of 4920 ±240 yards, 24 ±2 knots, and 225 ±10 degrees shall be present under the following conditions:

a.	Pulse Length	Short
b.	Range Scale	5
c.	Buoy Display Mode	BRG/DOP
d.	Process Channel	A only set for ping
e.	R-F Channel	1

3.5.1.4.9.1.4 CASS Simulation - A cw tone (background) shall be generated within each of the four CASS sonobuoy passbands. Background tones shall affect the system AGC in a manner similar to sea noise which reduces system gain. A time zero range reference shall be established by the downlink command function timing and shall be coincident with the end of ping time for a sonobuoy in

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process channel A. Between five and six seconds after time zero, a tone in each of the four CASS passbands shall be simultaneously generated for 400 milliseconds to simulate a 9-knot echo in each passband. The echo frequencies and range interval (relative to end of ping time in process channel A) shall be such that 9 ±2 knot target indications at ranges of 5000 ±240, 4000 ±240, 2800 ±240, and 1700 ±240 yards shall be present in respective display mode plots A, B, C, and D under the following conditions:

a.	Pulse Length	Long
b.	Range Scale	12
c.	Buoy Display Mode	(4) DOP
d.	Process Channel	All channels set for ping
e.	R-F Channel	R-F channel 2 selected for each process channel.

3.5.1.4.9.1.5 RO Simulation - A cw tone (background/reverberation) shall be generated near the center of the heterodyned passband applicable to RO sonobuoys. The background tone shall affect the system AGC in a manner similar to sea noise which reduces system gain. In addition, the tone provides the necessary frequency reference for RO reverb centering. A time zero range reference for a 5-kiloyard range scale shall be established by generating a 100 to 350-millisecond tone at a frequency within the spectrum applicable to RO trigger frequencies. Between five and six seconds after time zero, the background tone shall be increased in frequency for 400 milliseconds to simulate an echo return from a closing target. The echo frequency and range interval shall be such that a 7 ±3 knot target indication at a range of 4480 ±440 yards shall be present in all display mode plots under the following conditions:

a.	Pulse Length	Short
b.	Range Scale	5
c.	Buoy Display Mode	(4) DOP
d.	Process Channel	All channels displayed
e.	R-F Channel	R-F channel 2 selected for each process channel.

3.5.1.4.9.1.6 Q13 (PPI) Simulation - A 2013 Hz cw tone shall be generated and injected into the analog-to-digital converter that normally digitizes sonobuoy and dip sonar data prior to digital processing. The simulated dip sonar stimulus shall be similar to that produced by the sonar receiver with center sector, cw signals in all receiver channels. The resulting PPI indications shall be a spoke near the center of each sector. The doppler indication provided by any spoke shall be 21 ±2 knots.

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3.5.1.4.9.2 10-2 BIT Function - Downlink command tone verification shall be initiated when a 10-2 BIT code is applied to the SDC. Downlink verification shall be performed with the SDC in the DICASS processing mode, utilizing the (2) BRG display format. The SDC shall simulate DICASS sonobuoy input stimulus to the sonobuoy signal conditioners in a similar manner as that specified in the 10-1 BIT function, except only the background tone is generated and the echos are inhibited. All downlink command tone modulation to the UHF transceiver shall be inhibited in the 10-2 BIT operation. When generated, downlink address, ping, deep, and scuttle tones shall cause a change in the simulated DICASS bearing so as to produce a visible bearing perturbation in one or both (2) BRG display plots.

3.5.1.4.9.3 10-3 BIT Function - Downlink modulation interface verification shall be initiated when a 10-3 BIT code is applied to the SDC. The downlink modulation interface verification shall be performed in a like manner to that specifed for the 10-2 BIT operation, except the downlink tones generated by the SDC shall modulate the UHF transceiver. Tones returned from the radio set shall be used to produce visible bearing perturbations in the (2) BRG display format. If selected and executed, deep and scuttle downlink command modulation shall be inhibited.

3.5.1.5 Controls - All controls and indicators necessary for normal operation shall be located on the SDC front panel. The layout of the controls shall be as shown in Figure 23. The SDC front panel controls shall have the following characteristics:

3.5.1.5.1 Pushbutton Switch/Indicators - Each pushbutton switch/indicator shall be a pushbutton switch and four lamp indicator, cap and base assembly with an engraved legend plate. Lamp type shall be 28 volts, T-1-based, in accordance with MS3338-6839. The cap assembly shall be removable from the front panel to facilitate lamp replacement. The cap and base assemblies shall be indexed to prevent accidental interchange. Cap details shall be as in Table III.

3.5.1.5.2 Bidirectional Pushbutton Switch - The item described herein shall be a bidirectional, binary coded, decimal pushbutton switch conforming to the requirements specified herein and the requirements of MIL-S-22710 and MIL-S-22710/18, as applicable. Each bi-directional pushbutton switch assembly shall contain a replaceable 28 volt, aviation red lamp, part no. 71744-CM6838R. RFI shielding and O-ring shaft seals are required. Dial details shall be as follows:

	Control	Dial Characters
a.	DOWNLINK, CHANNEL SELECT (tens)	0 through 3
b.	DOWNLINK, CHANNEL SELECT (units)	0 through 9
c.	AUDIO	A through D
d.	MPR	A through D

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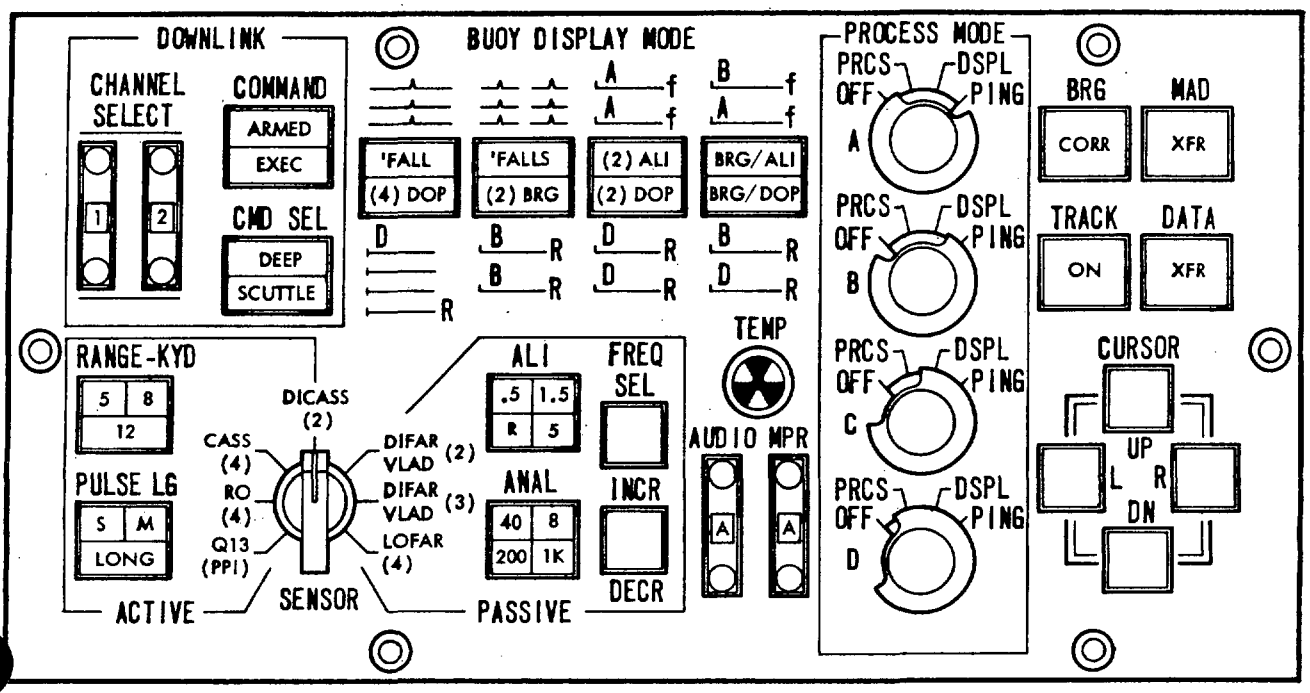


FIGURE 23. Sonar data computer, front panel controls.

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TABLE III. Pushbutton Switch/Indicator Cap Details.

LETTER COLOR		LETTER HEIGHT AND TYPE	LEGEND	BACKGROUND COLOR	
OFF	ON			OFF	ON
Visible White	Green --- Green	1/8 EC	'FALL (4) DOP	Grey Opaque	Grey Opaque
Visible White	Green --- Green	1/8 EC	'FALLS (2) BRG	Grey Opaque	Grey Opaque
Visible White	Green --- Green	1/8 EC	(2) ALI (2) DOP	Grey Opaque	Grey Opaque
Visible White	Green --- Green	1/8 EC	BRG/ALI BRG/DOP	Grey Opaque	Grey Opaque
Invisible	Black	1/8 N	ARMED EXEC	Black	Red --- White
Visible White	White	1/8 EC	DEEP SCUTTLE	Black	Amber --- Red
Visible White	Green	1/8 N	.5 1.5 R 5	Grey Opaque	Grey Opaque
Visible White	Green	1/8 N	40 8 200 1K	Grey Opaque	Grey Opaque
Visible White	Green	1/8 N	5 8 12	Grey Opaque	Grey Opaque
Visible White	Green	1/8 N	S M LONG	Grey Opaque	Grey Opaque
Visible White	Green	1/8 N	CORR	Grey Opaque	Grey Opaque
Visible White	Green	1/8 N	XFR	Grey Opaque	Grey Opaque
Visible White	Green	1/8 N	ON	Grey Opaque	Grey Opaque

NOTES: EC = Gorton Extra Condensed Gothic; N = Gorton Normal Gothic

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3.5.1.5.3 Pushbutton Switch - The item described herein shall be a modular momentary pushbutton switch. The pushbutton switch shall have no lamps, a clear filter lens, and a diffused white legend plate with no characters.

3.5.1.5.4 Miniature Rotary Switch - The item described herein shall be an explosion proof, rotary switch for low-level signal application, and conforming to the requirements of MIL-S-3786, as applicable. The PROCESS MODE switches shall have 30 degree detents, with stops ending at positions 4 and 1. The SENSOR switch shall have 45-degree detents, with stops ending at positions 7 and 1.

3.5.1.5.5 Front Panel Controls - The SDC shall have the following operational controls on the front panel:

3.5.1.5.5.1 DOWNLINK, CHANNEL SELECT - Bidirectional pushbutton switches (tens and units) shall provide a means of selecting an address of a CASS or DICASS sonobuoy that is to be issued a deep or scuttle downlink command tone.

3.5.1.5.5.2 DOWNLINK, CMD SEL - Pushbutton switch/indicator, which is normally extinguished, shall be depressed as required to select a deep or scuttle downlink command tone as applicable to a CASS or DICASS sonobuoy. Selection shall be indicated by illumination of DEEP or SCUTTLE legend.

3.5.1.5.5.3 DOWNLINK, COMMAND - Pushbutton switch/indicator, which is normally extinguished, shall be depressed once to enable the SDC to generate a deep or scuttle tone at end of CASS or DICASS sonobuoy ping sequence. Tone enabling shall be indicated by illumination of ARMED legend only if the pushbutton has been depressed before a downlink ping sequence has begun. EXEC legend shall illuminate at time downlink command is presented to the UHF transceiver. EXEC legend shall remain illuminated for duration of downlink address and tone, then both DOWNLINK, CMD SEL, and COMMAND pushbutton switch/indicators shall completely extinguish. DOWNLINK, COMMAND pushbutton switch/indicators shall be operable only after CMD SEL pushbutton switch/indicator has been operated. DOWNLINK command function shall be reset if DOWNLINK, COMMAND pushbutton switch/indicator is depressed after arming, but prior to the start of the downlink ping sequence.

3.5.1.5.5.4 BUOY DISPLAY MODE - Four pushbutton switch/indicators shall provide a means of selecting passive/active sonobuoy display modes related to the selected sensor processing mode, as specified in 3.4.1.4. Available passive or active buoy display modes shall be indicated by illumination of upper or lower legends, respectively, of the pushbutton switch/indicators. Available buoy display modes for a given sensor processing mode shall be indicated by the simultaneous illumination of all applicable legends on the pushbutton switch/ indicators when a sensor processing mode is selected. The desired buoy display mode shall be selected by depressing the appropriate pushbutton switch/indicator one time.

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The illuminated legends and colors of the pushbutton switch/indicators applicable to a given sensor processing mode shall remain constant, regardless of which buoy display mode is selected. The BUOY DISPLAY MODE pushbutton switch/-indicator legends to be illuminated for each sensor processing mode shall be as follows:

	SENSOR	BUOY DISPLAY MODE (filter plate color)
a.	LOFAR (4)	'FALL, 'FALLS, (2) ALI
b.	DIFAR/VLAD (3)	'FALL, 'FALLS, (2) ALI, BRG/ALI
c.	DIFAR/VLAD (2)	'FALL, 'FALLS, (2) ALI, BRG/ALI
d.	DICASS (2)	(2) BRG, (2) DOP, BRG/DOP
e.	CASS (4)	(4) DOP, (2) DOP
f.	RO (4)	(4) DOP, (2) DOP
g.	Q13 (PPI)	All pushbutton switch/indicators extinguished

3.5.1.5.5.5 PROCESS MODE - Four, four position, miniature rotary switches shall provide a means of selecting passive/active sonobuoys to be processed, displayed, and pinged in SDC processing channels A, B, C, and D. All PROCESSING MODE switches shall operate identically and each switch shall perform process mode functions in accordance with the switch setting as follows:

a. OFF - Process channel shall be inactive. The OFF position shall also provide a reset function for all passive sonobuoy, display mode data stored in memory.

c. PRCS - Shall process sensor data. Processed data shall be stored for immediate display recall, except in 'FALL or 'FALLS display mode. The maximum number of PROCESS MODE switches to be set to PRCS, without exceeding the processing capacity of the SDC, shall be the maximum number of sensors in a sensor set for a given sensor process mode, as specified in 3.4.1.3. If the processing capacity is exceeded, the appropriate error message shall be generated in the selected display mode.

c. DSPL - Shall process and display sensor data in the selected buoy display mode on the indicator. Maximum number of PROCESS MODE switches to be set to DSPL, without exceeding the display capability of the SDC, shall be the maximum number of display plots for a given buoy display mode, as specified in 3.4.1.4. If the display capacity is exceeded, the appropriate error message shall be generated in the selected display mode.

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d. PING - Shall process, display and ping CASS or DICASS sonobuoys. Downlink address and ping tones shall be presented to the UHF transceiver. The sonobuoy ping sequence shall always be in the alphabetic order of process channels A, B, C, and D. If the selected sensor processing mode is other than CASS or DICASS, the PING position shall duplicate the DSPL position.

3.5.1.5.5.6 BRG - The pushbutton switch/indicator shall provide a means of selecting a bearing bias correction function for DIFAR/VLAD sonobuoys being displayed in the BRG/ALI buoy display mode. This function is to reduce the effects of a broadband directional noise source on bearing calculations. The function shall be operational only in the BRG/ALI buoy display mode, and the CORR legend shall illuminate when the function is selected. If an alternate buoy processing mode or buoy display mode is selected, the BRG pushbutton switch/indicator shall automatically extinguish.

3.5.1.5.5.7 MAD - The pushbutton switch/indicator shall provide a means of notifying TACNAV of a MAD contact. XFR legend shall illuminate when the pushbutton switch/indicator is depressed and shall be automatically extinguished (within 60 seconds) by an acknowledgement from TACNAV.

3.5.1.5.5.8 DATA - The pushbutton switch/indicator shall provide a means of initiating a transfer of sonobuoy or dip sonar data to TACNAV. The data transfer function shall be applicable for sonobuoy data displayed in all buoy display modes except 'FALL, 'FALLS, and (2) ALI. Dip sonar data in both the SDC processed and conventional sonar modes shall be available for data transfer. Each time a data transfer is initiated, the displayed data that is selected by cursor positioning shall be presented to TACNAV. XFR legend shall be illuminated when TACNAV acknowledges the transfer and be automatically extinguished (within 60 seconds) when TACNAV accepts the data. By depressing the pushbutton switch/indicator prior to the acceptance of data, the intended transfer of data operation shall be cancelled and XFR shall be extinguished by TACNAV within 60 seconds.

3.5.1.5.5.9 TRACK - The pushbutton switch/indicator shall provide a means of selecting targets displayed in the (4) DOP buoy display mode for automatic transfer of data during each ranging interval. Each time the pushbutton switch/indicator is depressed, an auto track cursor (diamond) shall appear adjacent to a target previously selected by the display cursors. The auto track function shall be limited to only one target in each of the four display plots. Data from each auto track selected target shall be capable of automatically being transferred to TACNAV on a ping-by-ping basis. An auto track cursor appearing in any plot shall be erased by selecting the plot with the cursor pointer, then depressing the TRACK pushbutton switch/indicator. The ON legend shall be continuously illuminated as long as at least one target is selected for the auto track function. After targets have been selected by the TRACK pushbutton switch/indicator, the DATA pushbutton switch/indicator shall be depressed to initiate the auto transfer function. The XFR legend shall remain continuously illuminated until all auto track cursors have been erased, at which time both TRACK and DATA pushbutton switch/indicators shall extinguish.

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3.5.1.5.5.10 CURSOR, UP, DN, L, AND R - Four pushbutton switches shall provide a means of positioning buoy display mode cursors so as to select a displayed target for the formulation of data to be alphanumerically displayed and made available for transfer to TACNAV. The CURSOR UP and DN pushbutton switches shall position a horizontal cursor in the BRG/ALI and (2) DOP buoy display modes, and a left margin pointer in the (4) DOP and (2) BRG buoy display modes. The CURSOR L and R pushbutton switches shall position a vertical cursor in each buoy display mode.

3.5.1.5.5.11 MPR - The bidirectional pushbutton switch shall provide a means of selecting process channel A, B, C, or D for range tracking hardcopy by the recorder, only when the SDC is processing CASS or RO sonobuoys being displayed in the (4) DOP buoy display mode.

3.5.1.5.5.12 AUDIO - The bidirectional pushbutton switch shall provide a means of selecting the sonobuoy audio in process channel A, B, C, or D. The selected sonobuoy audio shall be presented to the indicator for amplification prior to aircraft ICS distribution.

3.5.1.5.5.13 PASSIVE, FREQ SEL INCR, DECR - Two pushbutton switches shall provide a means of incrementing or decrementing the displayed analysis band within the overall SDC input passband for all passive sonobuoys. A momentary depression of either pushbutton switch shall step the analysis band 5 Hz. Continuous depression of either pushbutton switch shall result in continuous 5 Hz stepping of the analysis band. Continuous stepping shall exponentially increase in rate as the pushbutton is held depressed.

3.5.1.5.5.14 PASSIVE ALI - The pushbutton switch/indicator shall provide a means of selecting three ALI time constants of .5, 1.5, and 5 minutes which are applicable to all passive sonobuoy display modes. A reset position shall be included to remove the ALI time constant from the display process. When a passive sensor processing mode is initially selected, the R (reset) legend shall automatically illuminate. Successive depressions of the ALI pushbutton switch/indicator shall be required to select a time constant. The selected time constant shall be indicated by the appropriately illuminated legend. Order of selection shall be R, .5, 1.5, 5, and return to R for the 1 kHz and 200 Hz analysis bands, and R, 1.5, 5, and return to R for the 40 Hz and 8 Hz analysis bands. The ALI pushbutton switch/indicator shall be completely extinguished when an active sensor processing mode is selected.

3.5.1.5.5.15 PASSIVE ANAL - The pushbutton switch/indicator shall provide a means of selecting four passive analysis bandwidths of 1 kHz, 200 Hz, 40 Hz, and 8 Hz. When a passive sensor processing mode is initially selected, the 1K legend shall automatically illuminate. Successive depressions of the ANAL pushbutton switch/indicator shall be required to select other analysis bandwidths. The selected analysis bandwidth shall be indicated by the appropriately illuminated legend. Order of selection shall be 1K, 200, 40, 8, and return to 1K. The ANAL pushbutton switch/indicator shall be completely extinguished when an active sensor processing mode is selected.

3.5.1.5.5.16 SENSOR - A seven position rotary switch shall provide a means of selecting the sensor set to be processed, as specified in 3.4.1.3. Preferred processing and display modes, as specified in 3.5, shall automatically be selected when a sensor set is initially selected for processing.

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3.5.1.5.5.17 ACTIVE, RANGE-KYD - The pushbutton switch/indicator shall provide a means of selecting three range scales of 5, 8, and 12 KYDS which are applicable to CASS and DICASS processing modes and hardcopy scale on chart paper of the recorder. Recorder chart shall indicate 8 kyd for both selected range scales of 5 and 8 kyd. When CASS or DICASS processing mode is initially selected, the 12 legend shall automatically illuminate. Successive depressions of the RANGE-KYD pushbutton switch/indicator shall be required to select a range scale for CASS or DICASS sonobuoys. The selected range scale shall be indicated by the appropriately illuminated legend. Order of selection shall be 12, 5, 8, and return to 12. When RO processing mode is initially selected, the 5 legend shall automatically illuminate and no further selection of range scale shall be possible. The RANGE-KYD pushbutton switch/-indicator shall be interactive with the PULSE LG pushbutton switch/indicator as specified in 3.5.1.5.5.18. When the Q13 (PPI) sensor processing mode is selected, the RANGE-KYD pushbutton switch/indicator shall be completely extinguished.

3.5.1.5.5.18 ACTIVE, PULSE LG - The pushbutton switch/indicator shall provide a means of selecting three ping pulse widths of S (short), M (medium), and LONG. Refer to the classified extract of this specification for pulse widths. When CASS or DICASS processing mode is initially selected, the LONG legend shall automatically illuminate. Successive depressions of the PULSE LG pushbutton switch/indicator shall be required to select a pulse length for CASS or DICASS sonobuoys. The selected pulse length shall be indicated by the appropriately illuminated legend. Order of selection shall be LONG, S, M, and return to LONG. When RO processing mode is initially selected, the S legend shall automatically illuminate and no further selection of pulse length shall be possible. When passive sonobuoy or the Q13 (PPI) sensor processing modes are selected, the PULSE LG pushbutton switch/indicator shall be completely extinguished. Selection of PULSE LG shall limit selection of RANGE-KYD for CASS and DICASS sonobuoy processing and display as follows:

	PULSE LG	RANGE-KYD
a.	S	5
b.	M	5, 8, 12
c.	LONG	8, 12

3.5.1.5.6 Preferred Mode - When the SDC is energized with the SENSOR switch at a given position, or when the SENSOR switch is rotated to a new setting, the following SDC functions shall automatically be selected:

a. PASSIVE:

	SENSOR	ANAL	ALI	BUOY DISPLAY MODE
(1)	LOFAR (4)	1K	R	(2) ALI
(2)	DIFAR VLAD (3)	1K	R	BRG/ALI
(3)	DIFAR VLAD (2)	1K	R	BRG/ALI

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b. ACTIVE:

	SENSOR	RANGE-KYD	PULSE LG	BUOY DISPLAY MODE
(1)	DICASS (2)	12	LONG	(2) BRG
(2)	CASS (4)	12	LONG	(4) DOP
(3)	RO (4)	5	S	(4) DOP
(4)	Q13 (PPI)	Not Applicable		

3.5.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

	Reference Designation	Receptacle Type	Function
a.	10J1	M81511/41FJ01S1	Test Point
b.	10J2	MS3470L24-61P	Azimuth and Range Indicator IP-1045/AQS-13B Inlet/Outlet
c.	10J3	M83723-11H1608N	Power Inlet
d.	10J4	M83723/72R24616	Receiver Set AN/ARR-75 Inlet; Radio Set AN/ARC-159 Outlet; Tactical Navigation Set AN/ASN-123 Inlet/Outlet
e.	10J5	M83723/72R24617	Sonar Receiver R-1695A/AQS-13B Inlet
f.	10J6	MS3470L20-41SY	Recorder RO-358/AQS-13A Outlet

3.5.1.7 Color - The Sonar Data Computer case and control panel shall be grey in accordance with No. 36231 FED-STD-595.

3.5.2 Base, Shock Mount MT-4948/AQS-13E - The shock mount base shall meet the following requirements:

3.5.2.1 Function - The shock mount shall provide a means of mounting the SDC on the sensor operator's console. The unit shall meet the applicable design and construction requirements of MIL-C-172.

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3.5.2.2 Form Factor - The shock mount base shall not exceed dimensions of 65.8 centimeters (26.0 inches) in length, 36.5 centimeters (14.4 inches) in width, and 10.0 centimeters (4.0 inches) in height. See Figure 2 for outline drawing of shock mount base.

3.5.2.2.1 Frame - The shock mount base shall be an aluminum alloy, anodized in accordance with MIL-A-8625, Type I or II, Class 1, Clear.

3.5.2.2.2 Materials - The shock mount base shall be constructed of 0.063 and 0.190 thick aluminum alloy sheet, 6061-16 or T651, in accordance with QQ-A-250/11. Extrusion number PAC055 (Pioneer Aluminum), 0.093 thick X 1.75 X 2.00 aluminum alloy 7075-T6, T6510, or T6511.

3.5.2.3 Weight - The weight of the shock mount base shall not exceed 1.8 kilograms (4.0 pounds).

3.5.2.4 Contents - The shock mount base shall contain assemblies and subassemblies to fulfill its functions as described herein. These shall include as a minimum:

a. Identification Plate

b. Serial Number Plate

c. Front Self Locking Retainers

d. Resilient Mounts

e. Rear Retainer Pin Assemblies

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

a. Preproduction (First Article) Tests (See 4.2.)

b. Initial Production Tests (See 4.3.)

c. Acceptance Tests (See 4.4.)

d. Life Tests (See 4.5.)

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a Government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review. (See 6.2.1.)

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications, and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of MIL-T-5422; electromagnetic interference tests in accordance with the procedures of MIL-STD-462; maintainability demonstration test in accordance with MIL-STD-471; and reliability tests in accordance with requirements of MIL-STD-2068.

4.2.2.1 Environmental Tests - Environmental tests shall be conducted in the sequence specified in MIL-T-5422. Environmental tests shall include, but not be limited to, the following: (See 3.3.10.)

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4.2.2.1.1 Temperature/Altitude Tests - A temperature/altitude test shall be performed in accordance with the MIL-T-5422 except the test conditions for class 1B equipment shall be as follows: (See 3.3.10.2, 3.3.13.1.)

	STEP	TEMP (^{o}C)	PRESSURE ALT. (FT)	EXPOSURE TIME
a.	1b	-62	ATM	2 Hours
b.	2	-30	ATM	Stabilization
c.	3	-30	15,000	Stabilization
d.	4	-10	ATM	Stabilization
e.	5	+85	ATM	16 Hours
f.	6	+55	ATM	4 Hours
g.	7	+55	ATM	30 minutes/cycle
h.	8	OMIT		
i.	9	+53	10,000	4 hours
j.	10	+55	10,000	30 minutes/cycle
k.	11	+51	15,000	4 hours
l.	12	+55	15,000	30 minutes/cycle
m.	13	OMIT		
n.	14	Standard Ambient Conditions		

4.2.2.1.2 Vibration Tests - A vibration test shall be performed in accordance with MIL-T-5422, using the applicable vibration level curves for helicopters. The following aircraft resonant frequencies shall be included during resonance dwell vibration tests: (See 3.3.10.1.)

a. 3.4 Hz
b. 17.0 Hz
c. 103.0 Hz

4.2.2.1.3 Shock Test - A shock test shall be performed in accordance with MIL-T-5422. (See 3.3.10.)

4.2.2.1.4 Humidity Test - A humidity test shall be performed in accordance with MIL-T-5422. (See 3.3.10.3.)

4.2.2.1.5 Salt Fog Test - A salt fog test shall be performed in accordance with MIL-T-5422. (See 3.3.10.)

4.2.2.1.6 Explosion Test - An explosion test shall be performed in accordance with MIL-T-5422, using 15,000 feet as the maximum operating altitude for class 1B equipment. (See 3.3.10.)

4.2.2.1.7 Sand and Dust Test - A sand and dust test shall be performed in accordance with MIL-T-5422. (See 3.3.10.)

4.2.2.2 Electromagnetic Interference Test - Electromagnetic Interference tests shall be conducted in accordance with MIL-STD-462. (See 3.3.6.)

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4.2.2.3 Maintainability Demonstration Tests - A maintainability demonstration test shall be conducted in accordance with MIL-STD-471, to demonstrate equipment compliance with the specified MTTR and turn-around time requirements. In addition, the test shall demonstrate that the BIT function and self-test features are capable of fault detection and isolation to a WRA; that repair is accomplished through removal and replacement; that a repair verification is accomplished exercising the stated equipment BIT and self-test features.

4.2.2.4 Reliability Test - A reliability development test (RDT) shall be conducted in accordance with MIL-STD-2068(AS). The equipment(s) under test will be representative of future production equipments in materials, configuration and workmanship. The test is intended to identify, analyze, and correct all deficiencies. All equipment failures shall require analysis and corrective action. (See 6.2.1.)

4.2.2.4.1 RDT Failure Reporting Analysis and Corrective Action - All RDT failures shall be reported to the procuring activity cognizant engineer or his appointed designate within 24 hours of the failure incident. All failures shall be analyzed. Positive corrective actions shall be developed for failures. Necessary modifications shall be incorporated into RDT equipments under test for the purpose of corrective action verification. The procuring activity may require verification of the corrective action modification by testing at a lower level of assembly or at the piece part level, as deemed necessary. (See 6.2.1.)

4.2.2.4.2 RDT Corrective Action - Verified corrective action(s) shall be incorporated in all systems submitted for acceptance and delivered systems shall be retrofitted.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

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4.3 Initial Production Test - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The procuring activity will provide shipping instructions. No other tests shall be conducted on equipment prior to starting the initial production tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality, and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for initial production tests, the contractor shall also prepare a listing of such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the initial production sample shall be by the procuring activity, upon satisfactory completion of all tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all limited life or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests

b. Sampling Tests

c. Reliability Assurance Tests

d. Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product

b. Operational Test

c. Manufacturing Run-in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated at room ambient temperature long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-In Test - As a minimum, each equipment shall be subjected to a manufacturing run-in test consisting of the following environments:

a. Random vibration.

b. Thermal cycling.

These environments may be applied sequentially in the order listed. Test levels, durations and tolerances shall be specified in MIL-STD-781 for optional nonstatistical production reliability test with sequential environments and as specified in the subparagraphs below.

Following random vibration, each equipment produced shall be tested for a minimum period of 40 operating hours in accordance with the conditions specified herein. All failures shall be repaired and the test continued except that the last 10 hours shall be failure free. A failure shall be defined as any out-of-specification performance or any condition which requires repair of the equipment under test. Adjustment of controls normally accessible to an operator shall be permitted.

During this 40-hour period of operation, the equipment shall be cycled periodically through its various phases of operation. Should a failure occur, it shall be repaired and the test resumed. However, if the failure occurs in the 10-hour failure-free period, the test shall be restarted at the at the beginning of the failure-free period. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures.

4.4.1.3.1 Random Vibration - Testing of more than one axis may be performed. If this option is authorized, the testing shall be in accordance with the test procedure approved by the procuring activity. (See 6.2.1.)

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4.4.1.3.2 Thermal Cycling - Each cycle shall consist of lowering the chamber temperature to -30°C, with equipment nonoperating until the equipment temperature stabilizes. During the cold period the equipment shall be turned on and tested, then the chamber temperature shall be raised to +55°C. The equipment shall be tested at the high temperature.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following:

	Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
a.	First 10	0 1/
b.	Next 50	1
c.	Next 75	1
d.	Next 100	1
	1 for each additional 200 or fraction thereof	

Sampling tests are not required when Reliability Assurance Tests are conducted.

NOTE 1/: If the Initial Production Test is deleted by contract action, a Sample Test shall also be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual, and special tests.

c. Manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

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4.4.3 Reliability Assurance Tests - A Reliability Assurance Test shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and MIL-STD-2074(AS). (See 6.2.)

4.4.3.1 Qualification Phase - When required by contract and prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781. The maximum number of equipments to be used shall be ten (10). For the qualification phase, a fixed-length test with a discrimination ratio of 2.0 and an 80 percent confidence level shall be performed. The specified MTBF is defined as 425 hours. For purposes of this test, θ_0 shall be 530 hours and θ_1 shall be 258 hours. Test conditions shall be in accordance with the requirements of MIL-STD-781 for helicopter-borne equipments, as limited by the equipment environmental specifications contained herein. (See 6.2.)

4.4.3.2 Reliability Production Acceptance (Sampling) Phase Test - The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein), except that all equipments produced shall be tested. (See 6.2.)

4.4.3.2.1 All Equipment Test - Each equipment produced shall be subjected to an all equipment test consisting of the following environments:

a. Random vibration.

b. Thermal cycling.

These environments may be applied sequentially in the order listed. Test levels, durations and tolerances shall be as specified in MIL-STD-781 for optional nonstatistical production acceptance reliability test and as specified in the subparagraphs below. Following random vibration, each equipment produced shall be tested for a minimum of 50 operating hours in accordance with the conditions specified herein. All failures shall be repaired and the test continued except that the last 10 hours of operation shall be failure free. A failure shall be defined as any out-of-specification performance or any condition which requires repair of the equipment under test. Adjustment of the controls normally accessible to an operator shall be permitted.

4.4.3.2.1.1 Random Vibration - Testing of more than one axis may be performed. If this option is authorized, the testing shall be in accordance with the test procedure approved by the procuring activity.

4.4.3.2.1.2 Thermal Cycling - Each cycle shall consist of lowering the chamber temperature to -30°C, with equipment nonoperating until the equipment temperature stabilizes. During the cold period the equipment shall be turned on and tested, then the chamber temperature shall be raised to +55°C. The equipment shall be tested at the high temperature.

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4.4.3.2.1.3 Reject-Accept Criteria - The accept-reject criteria shall conform to the requirements of the all-equipment production reliability acceptance test plan of MIL-STD-781. The lower test MTBF (θ_1) shall be as specified in 4.4.3.1 herein. During the all-equipment test, each equipment shall be exercised and tested in accordance with the approved test procedure. Should a failure occur, the equipment shall be repaired, proper operation verified and the test continued commencing at the beginning of the cycle in which the failure occurred. Corrective action shall be taken, should failures occur, as described in 3.3.2.5.2. Records shall be kept of all failures. To determine whether a reject condition has been reached at any time during the contract, the operating test hours and the failures therein shall be totaled and the results compared with the all-equipment production reliability acceptance test plan of MIL-STD-871. At the conclusion of each month, the test results shall be sent to the procuring activity. Any time that the current totals of test hours and test failures show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment any time that a reject situation exists, pending a review of the Contractor's efforts to improve the equipment. Any corrective actions made shall be incorporated in all units at the contractor's expense. All equipment shall be subjected to and shall pass the individual tests prior to the all-equipment test. When repair to an equipment, during or after all-equipment testing, is accomplished by card substitution, that card shall have been subjected to the burn-in requirements for spares as described in 4.4.3.4.

4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Reliability Assurance Tests.

4.4.3.4 Burn-In Requirement for Spares - The reliability testing procedures of this paragraph shall be applied to all assemblies (SRAs and sub-SRAs) that are procured separately from the CP-1323/AQS-13E computer.

a. As a minimum, in-process burn-in, screening, and other individual testing of separately procured assemblies shall be equivalent to that conducted on production assemblies. Following the burn-in, a complete functional performance test shall be conducted on each assembly offered for acceptance.

b. Included in this testing shall be at least 40 hours power-ON operation (signal voltages not required) of each assembly under the environmental test conditions specified in 4.4.1.3. Power supply assembly burn-in shall be conducted with the power supply loaded between 80 - 100 percent of rated full load.

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c. The tests specified in this paragraph shall be conducted by or for the supplier of the assemblies. Assemblies shall not be accepted by contractors for delivery to the procuring activity or a government activity, without objective evidence of having passed the above and other required tests. These test requirements shall be included in the manufacturing drawing, specification, or source controlled drawing for the assemblies.

4.4.4 Special Tests - When required by contract, special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment for special tests may be selected from equipment previously subjected to the sampling or reliability assurance tests. (See 6.2.)

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. (See 6.2.1.)

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance, or special tests, the following action shall be taken: (See 6.2.1.)

a. Determine the cause of the failure.

b. Determine if the failure is an isolated case or design defect.

c. Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

d. Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The tests shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions

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specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the initial production test, sampling tests, reliability assurance tests, or special tests may be selected for life tests.) (Test life accumulated during the reliability assurance test may be counted toward the life test, provided the entire 300 hours is accumulated on a single equipment and the equipment selected as follows.) (See 3.3.2.5.2.1.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
a. First 25	1
b. Next 175	1
c. Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

a.	Temperature	Normal Room
b.	Altitude	Normal ground (0-5000 ft)
c.	Humidity	Room ambient
d.	AC Voltage	115 ±5 volts (at lowest applicable frequency)
e.	DC Voltage	27.5 ±2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At appropriate intervals (6 to 16 hours) during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity. (See 6.2.1.)

4.5.4 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A failure report shall be prepared upon completion of test. In this report, the Contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity shall review such proposals and determine whether they are acceptable. (See 6.2.1.)

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4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests, and life tests shall be prepared by the Contractor. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests when deemed necessary. (See 6.2.1.)

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production, acceptance, and life tests shall be reconditioned by the Contractor by replacing all worn, life limited, or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects. Before resubmittal for aceptance, a failure report shall be prepared which describes the failure or trouble observed and the corrective action taken. This report shall be reviewed by the Contracting Officer. (See 6.2.1.)

5. PACKAGING

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

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6. NOTES

6.1 Intended Use - The equipment covered by this specification is intended for installation and operation in an ASW aircraft and shall be utilized while airborne to detect and process acoustical signals from submerged objects.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of testing (see 4. Quality Assurance Provisions).

c. Selection of applicable levels of packaging and packing (see 5.1).

d. Title, number, and revision of each end item data list.

6.2.1 Data Requirements - When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.2.1	Nonstandard Part Approval Requests	DI-E-7028
3.2.2	Microelectronic Devices	UDI-E-21337
3.3.2	Plan, Reliability Program	DI-R-2113
3.3.4	Drawing Requirements For Rack Or Console	UDI-E-21339
3.3.8	Nomenclature And Identification Assignment And Approval Request	UDI-E-21336A

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Paragraph	Data Requirement	Applicable DID
4.2.1	First Article Test Report	UDI-T-23790
4.2.2.4	Reliability Test Procedure	UDI-T-21347
4.2.2.4.1	Reliability Test And Demonstration Reports	DI-R-7034
4.4.1.3.1	Test Procedures	UDI-T-21347
4.4.4.2	Test Procedures	UDI-T-21347
4.4.5	Corrective Action Report	DI-R-1704
4.5.3	Test Procedures	UDI-T-21347
4.5.4	Report, Failure	UDI-R-21141A
4.6	Test Procedures	UDI-T-21347
4.9	Report, Failure	UDI-R-21141A

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply.

a. Contract - The contract shall have precedence over any specification.

b. This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

c. Reference Specifications - Any referenced specification (Section 2.) shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.5 Nonrepairable Subassemblies - As a general rule, nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the nonrepairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.6 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Associated Equipment

Item	Type Designation
Radio Set	AN/ARC-159(V)
Receiver Set	AN/ARR-75
Tactical Navigation Set	AN/ASN-123
Detecting Ranging Set, Sonar	AN/AQS-13E

Preparing activity
NAVY-AS
(Project 5848-N066)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-85301(AS) Computer, Sonar Data; CP-1323/AQS-13E and Mounting Base; MT-4948/AQS-13E

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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